Event tree และ Guard tree

แบบจำลอง 198 ตัว · มีเงื่อนไขจากแผนภาพ 146 ตัว · ไม่มีเงื่อนไข 52 ตัว

1. ที่มาของ tree

tree ทั้งสองแบบสร้างจาก guard ในแบบจำลอง Event-B ที่ได้จาก ChronoSeq-B โดยไม่แก้ไขแบบจำลอง

guard ในแบบจำลองแบ่งเป็น 2 กลุ่ม ดังนี้

กลุ่ม	ตัวอย่าง	ใช้สร้าง
guard ลำดับข้อความ (สูตรแปลงสร้างให้ทุกข้อความ)	`m ∉ sentMessages`, `p ∈ receivedMessages`, `currentMessage = ∅`	Event tree
guard เงื่อนไขจากแผนภาพ (loop, alt, opt)	`cycleNo < 8`, `mode = GOLD`, `cycleNo mod 3 = 0`	Guard tree

2. การตั้งรหัส

รหัส guard

- **ตัวอักษร** คือจุดตัดสินใจ เงื่อนไขที่นิพจน์ด้านซ้ายเหมือนกันใช้ตัวอักษรเดียวกัน เช่น `mode = GOLD` และ `mode = SILVER` เป็นตัว b ทั้งคู่
- **ตัวเลข** คือทางเลือกของจุดตัดสินใจนั้น เช่น b1, b2, b3
- ตัวอักษรและตัวเลขเรียงตามลำดับที่พบในแบบจำลอง

รหัส event (send และ receive นับเป็นคนละ event)

- **ตัวอักษร** คือลำดับขั้น event ที่ต่อกันเป็นเส้นตรงได้ตัวอักษรใหม่ทุกขั้น (a1 → b1 → c1)
- **ตัวเลข** คือทางเลือกที่จุดแยก event ขั้นเดียวกันของแต่ละกิ่งใช้ตัวอักษรเดียวกัน เช่น กิ่ง GOLD, SILVER, BRONZE ได้ m1, m2, m3 และขั้นถัดไปได้ n1, n2, n3
- ท้าย loop เป็นจุดแยกเช่นกัน คือทำต่อ (o1) หรือวนรอบใหม่ (o2)
- event ที่กิ่งมารวมกันได้ตัวอักษรใหม่ (q1)

3. การสร้าง tree

- **Event tree** — receive ตามหลัง send ของข้อความเดียวกัน และ send ของข้อความ m ตามหลัง receive ของข้อความ p เมื่อ send m มี guard `p ∈ receivedMessages`
- alt ทำให้ tree แยกกิ่ง และกิ่งกลับมารวมที่ event ที่มี guard `p ∈ receivedMessages ∨ q ∈ receivedMessages`
- event นับรอบ loop อยู่ท้าย loop เครื่องหมาย ↺ ชี้กลับไปยัง event แรกของ loop
- **Guard tree** — ไล่ตามแต่ละเส้นทางของ event tree แล้วบันทึกรหัส guard ตามลำดับที่พบครั้งแรก จึงแสดงได้ทั้งเงื่อนไขที่ซ้อนกัน (alt ใน loop) และเงื่อนไขที่ต่อกัน (opt ตามด้วย alt)
- เอกสารนี้แสดง event tree แบบย่อ คือ event ที่ต่อกันเป็นเส้นตรงเขียนในบรรทัดเดียว และขึ้นบรรทัดใหม่เฉพาะที่จุดแยก รูปเต็มอยู่ใน `models/<id>/tree.txt`

4. ตัวอย่าง

ตัวอย่างพื้นฐาน เงื่อนไขแรกคือ `age > 50` จากนั้นมี 2 ทางเลือก คือ `salary ≥ 500` และ `salary < 500`

a1	age > 50
b1	salary ≥ 500
b2	salary < 500

```
(root)
└── a1  age > 50
    ├── b1  salary ≥ 500
    └── b2  salary < 500

เส้นทาง:  a1 -> b1
          a1 -> b2
```

---

model_141 PharmacyDelivery — opt ตามด้วย alt 3 ทาง (เงื่อนไขต่อกัน)

Guard tree

a1	needAck = 1
b1	mode = GOLD
b2	mode = SILVER
b3	mode = BRONZE

```
(root)
└── a1  needAck = 1
    ├── b1  mode = GOLD
    ├── b2  mode = SILVER
    └── b3  mode = BRONZE
```

เส้นทาง

```
a1 -> b1
a1 -> b2
a1 -> b3
```

Event tree

รหัส	event	guard
a1	sendsyncSignature_1	
b1	receivesyncSignature_1	
c1	sendvalidateSignature_2	
d1	receivevalidateSignature_2	
e1	sendfetchRoute_3_opt1	a1
f1	receivefetchRoute_3_opt1	a1
g1	sendqueryRoute_4_opt1	a1
h1	receivequeryRoute_4_opt1	a1
i1	sendconfirmRoute_5	
j1	receiveconfirmRoute_5	
k1	sendissuePackage_6	
l1	receiveissuePackage_6	
m1	sendcancelSignature_7	b1
m2	sendqueryPackage_9	b2
m3	sendscheduleSignature_11	b3
n1	receivecancelSignature_7	b1
n2	receivequeryPackage_9	b2
n3	receivescheduleSignature_11	b3
o1	sendconfirmPackage_8	b1
o2	sendackSignature_10	b2
p1	receiveconfirmPackage_8	b1
p2	receiveackSignature_10	b2
q1	sendapproveRoute_12	
r1	receiveapproveRoute_12	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
    ├── m1 → n1 → o1 → p1 → q1 → r1
    ├── m2 → n2 → o2 → p2 → q1 → r1
    └── m3 → n3 → q1 → r1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m2 -> n2 -> o2 -> p2 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m3 -> n3 -> q1 -> r1
```

model_109 HospitalLabRequest — alt ใน loop และกิ่งแรกมี loop ซ้อน (เงื่อนไขซ้อนกัน)

Guard tree

a1	cycleNo < 8
b1	cycleNo mod 3 = 0
c1	attemptNo < 10
b2	cycleNo mod 3 ≠ 0

```
(root)
└── a1  cycleNo < 8
    ├── b1  cycleNo mod 3 = 0
    │   └── c1  attemptNo < 10
    └── b2  cycleNo mod 3 ≠ 0
```

เส้นทาง

```
a1 -> b1 -> c1
a1 -> b2
```

Event tree

รหัส	event	guard
a1	sendcheckOrder_1	
b1	receivecheckOrder_1	
c1	sendrequestResult_2	
d1	receiverequestResult_2	
e1	sendconfirmOrder_3	a1
f1	receiveconfirmOrder_3	a1
g1	sendscheduleOrder_4	a1 b1
g2	sendreportSample_9	a1 b2
h1	receivescheduleOrder_4	a1 b1
h2	receivereportSample_9	a1 b2
i1	sendackSample_5	a1 b1 c1
j1	receiveackSample_5	a1 b1 c1
k1	sendapproveSample_6	a1 b1 c1
l1	receiveapproveSample_6	a1 b1 c1
m1	sendissueResult_7	a1 b1 c1
n1	receiveissueResult_7	a1 b1 c1
o1	sendreportResult_8	a1 b1
o2	checkloop_ackSample_5_loop2	a1 b1 c1
p1	receivereportResult_8	a1 b1
q1	sendrequestSample_10	a1
r1	receiverequestSample_10	a1
s1	sendreserveSample_11	
s2	checkloop_confirmOrder_3_loop1	a1
t1	receivereserveSample_11	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1 → m1 → n1
    │   ├── o1 → p1 → q1 → r1
    │   │   ├── s1 → t1
    │   │   └── s2 ↺ e1
    │   └── o2 ↺ i1
    └── g2 → h2 → q1 → r1
        ├── s1 → t1
        └── s2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s2 ↺ e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o2 ↺ i1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> q1 -> r1 -> s1
-> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> q1 -> r1 -> s2 ↺
e1
```

5. ภาพรวมทั้งชุด

จำนวนเส้นทางใน guard tree	0	1	2	3	4	6
จำนวนแบบจำลอง	52	71	67	5	2	1

- 0 คือแบบจำลองที่ไม่มีเงื่อนไขจากแผนภาพ จึงมีเฉพาะ event tree
- tree ของแต่ละแบบจำลองอยู่ใน `models/<id>/tree.txt` และ `tree.json` ส่วนข้อมูลรวมทุกแบบจำลองอยู่ใน `data/trees.jsonl`, `trees/guard_codes.csv` และ `trees/event_codes.csv`

- **model_001** — model.txt มีเพียงชื่อ event จึงใช้ event และ guard จากไฟล์ .bpo ของ Rodin แทน มี 1 event ที่ไม่มีชื่อในไฟล์ (unnamed_event_N, guard `retry < 3`) และเนื่องจากแปลงด้วยสูตรรุ่นแรก event tree จึงแยกเป็นหลายกิ่งจาก root

6. tree ของแต่ละแบบจำลอง

แต่ละแบบจำลองแสดง 2 ส่วน ด้านซ้ายคือ guard (รหัส, tree, เส้นทาง) ด้านขวาคือ event (รหัส, tree แบบย่อ, เส้นทาง) คอลัมน์ guard ในตาราง event คือรหัส guard ของ event นั้น และ ↺ x หมายถึงวนกลับไปที่ event x

การเงินและการชำระเงิน · Finance & Payment (14)

model_001 Payment & Notification · pass

Guard

a1	Login = 1
b1	amount_1 ≥ 1000
b2	amount_1 < 1000
c1	retry < 3
c2	retry = 3
d1	par_1 = 2

```
(root)
├── a1  Login = 1
│   ├── b1  amount_1 ≥ 1000
│   │   └── c1  retry < 3
│   └── b2  amount_1 < 1000
│       └── c1  retry < 3
└── c2  retry = 3
    └── d1  par_1 = 2
```

เส้นทาง

```
a1 -> b1 -> c1
a1 -> b2 -> c1
c2 -> d1
```

Event

รหัส	event	guard
a1	sendsubmitPayment_1_opt1	a1
a2	sendvalidatePayment_7_loop3	c2
b1	receivesubmitPayment_1_opt1	a1
b2	receivevalidatePayment_7_loop3	c2
c1	sendpaymentDetails_2_opt1	a1
c2	sendvalidationResult_8_loop3	c2
d1	receivePaymentDetails_2_opt1	a1
d2	recivevalidationResult_8_loop3	c2
e1	sendprocessLargePayment_3_alt2	b1
e2	sendProcessStandarddPayment_5_alt2	b2
f1	receiveprocessLargePayment_3_alt2	b1
f2	receiveProcessStandarddPayment_5_alt2	b2
g1	sendapprovalRequired_4_alt2	b1
g2	sendpaymentConfirmation_6_alt2	b2
h1	reciveapprovalRequired_4_alt2	b1
h2	recivepaymentConfirmation_6_alt2	b2
i1	unnamed_event_N	c1
j1	sendsendSMSNotification_9_par4	
k1	sendsendEmailNotification_10_par4	
k2	receivedsendSMSNotification_9_par4	d1
l1	receivedsendEmailNotification_10_par4	d1
m1	sendpaymentStatus_11	
n1	receivepaymentStatus_11	

```
(root) INITIALISATION
├── a1 → b1 → c1 → d1
│   ├── e1 → f1 → g1 → h1 → i1
│   └── e2 → f2 → g2 → h2 → i1
└── a2 → b2 → c2 → d2 → j1
    ├── k1 → l1 → m1 → n1
    └── k2
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g2 -> h2 -> i1
a2 -> b2 -> c2 -> d2 -> j1 -> k1 -> l1 -> m1 -> n1
a2 -> b2 -> c2 -> d2 -> j1 -> k2
```

model_003 ATMWithdrawal · pass

Guard

a1	pinAttempt < 3
b1	pinValid = 1
b2	pinValid = 0

```
(root)
└── a1  pinAttempt < 3
    ├── b1  pinValid = 1
    └── b2  pinValid = 0
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

รหัส	event	guard
a1	sendinsertCard_1	
b1	receiveinsertCard_1	
c1	sendrequestPIN_2	
d1	receiverequestPIN_2	
e1	sendenterPIN_3	
f1	receiveenterPIN_3	
g1	sendinvalidPIN_4	a1
h1	receiveinvalidPIN_4	a1
i1	sendreenterPIN_5	a1
j1	receivereenterPIN_5	a1
k1	sendselectAmount_6	b1
k2	sendretainCard_8	b2
k3	checkloop_invalidPIN_4_loop1	a1
l1	receiveselectAmount_6	b1
l2	receiveretainCard_8	b2
m1	senddispenseCash_7	b1
n1	receivedispenseCash_7	b1

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1
    ├── k1 → l1 → m1 → n1
    ├── k2 → l2
    └── k3 ↺ g1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2
-> l2
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k3 ↺
g1
```

model_004 BankTransfer · fail

Guard

a1	riskFlag = 0
a2	riskFlag = 1

```
(root)
├── a1  riskFlag = 0
└── a2  riskFlag = 1
```

เส้นทาง

```
a1
a2
```

Event

รหัส	event	guard
a1	sendinitiateTransfer_1	
b1	receiveinitiateTransfer_1	
c1	sendcheckFraud_2	
d1	receivecheckFraud_2	
e1	sendriskScore_3	
f1	receiveriskScore_3	
g1	sendpostTransfer_4	a1
g2	sendholdTransfer_6	a2
h1	receivepostTransfer_4	a1
h2	receiveholdTransfer_6	a2
i1	sendpostingDone_5	a1
j1	receivepostingDone_5	a1
k1	sendtransferStatus_7	
l1	receivetransferStatus_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1
    └── g2 → h2 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1
```

model_025 InsuranceClaim · pass

Guard

a1	hasDocs = 1
b1	claimAmount ≤ 5000
b2	claimAmount > 5000

```
(root)
└── a1  hasDocs = 1
    ├── b1  claimAmount ≤ 5000
    └── b2  claimAmount > 5000
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

รหัส	event	guard
a1	sendsubmitClaim_1	
b1	receivesubmitClaim_1	
c1	senduploadDocs_2_opt1	a1
d1	receiveuploadDocs_2_opt1	a1
e1	sendapprovalNotice_3	b1
e2	sendassignAssessor_4	b2
f1	receiveapprovalNotice_3	b1
f2	receiveassignAssessor_4	b2
g1	sendpayout_8	
h1	receivepayout_8	
i1	sendtransferDone_9	
j1	receivetransferDone_9	
k1	sendnotifyClaimant_10	
l1	receivenotifyClaimant_10	
m1	sendsiteVisit_5	b2
n1	receivesiteVisit_5	b2
o1	sendvisitDone_6	b2
p1	receivevisitDone_6	b2
q1	sendassessmentReport_7	b2
r1	receiveassessmentReport_7	b2

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1 → k1 → l1
    └── e2 → f2 → m1 → n1 → o1 → p1 → q1 → r1 →
          g1 → h1 → i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> m1 -> n1 -> o1 -> p1 -> q1
-> r1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
```

model_056 BankBranchQueue · pass

Guard

a1	passNo < 11

```
(root)
└── a1  passNo < 11
```

เส้นทาง
a1

Event

รหัส	event	guard
a1	sendqueryCounter_1	
b1	receivequeryCounter_1	
c1	sendassignService_2	
d1	receiveassignService_2	
e1	sendscheduleService_3	a1
f1	receivescheduleService_3	a1
g1	sendsyncTicket_4	a1
h1	receivesyncTicket_4	a1
i1	sendfetchTicket_5	a1
j1	receivefetchTicket_5	a1
k1	sendackCounter_6	a1
l1	receiveackCounter_6	a1
m1	sendnotifyService_7	a1
n1	receivenotifyService_7	a1
o1	sendqueryTicket_8	
o2	checkloop_scheduleService_3_loop1	a1
p1	receivequeryTicket_8	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
    ├── o1 → p1
    └── o2 ↺ e1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o2 ↺ e1

model_066 CanteenPayment · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendupdateBalance_1	
b1	receiveupdateBalance_1	
c1	sendlogTopup_2	
d1	receivelogTopup_2	
e1	sendrequestTopup_3	
f1	receiverequestTopup_3	
g1	sendassignBalance_4	
h1	receiveassignBalance_4	
i1	sendcheckTopup_5	
j1	receivecheckTopup_5	
k1	sendreleaseMeal_6	
l1	receivereleaseMeal_6	
m1	sendreportMeal_7	
n1	receivereportMeal_7	
o1	sendupdateTopup_8	
p1	receiveupdateTopup_8	
q1	sendscheduleMeal_9	
r1	receivescheduleMeal_9	
s1	sendsendBalance_10	
t1	receivesendBalance_10	
u1	sendscheduleBalance_11	
v1	receivescheduleBalance_11	
w1	sendvalidateMeal_12	
x1	receivevalidateMeal_12	
y1	sendconfirmTopup_13	
z1	receiveconfirmTopup_13	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1 →
      q1 → r1 → s1 → t1 → u1 → v1 → w1 → x1 →
      y1 → z1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1 -> w1 -> x1 -> y1 -> z1

model_094 FleetFuelCard · fail

Guard

a1	temp < 50
a2	temp ≥ 50

```
(root)
├── a1  temp < 50
└── a2  temp ≥ 50
```

เส้นทาง

a1
a2

Event

รหัส	event	guard
a1	sendreleaseLimit_1	
b1	receivereleaseLimit_1	
c1	sendreportCard_2	
d1	receivereportCard_2	
e1	sendsendFuel_3	a1
e2	sendcancelFuel_8	a2
f1	receivesendFuel_3	a1
f2	receivecancelFuel_8	a2
g1	sendissueLimit_4	a1
h1	receiveissueLimit_4	a1
i1	sendissueCard_5	a1
j1	receiveissueCard_5	a1
k1	sendapproveCard_6	a1
l1	receiveapproveCard_6	a1
m1	sendissueFuel_7	a1
n1	receiveissueFuel_7	a1
o1	sendvalidateCard_9	
p1	receivevalidateCard_9	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1 → k1 → l1 →
    │     m1 → n1 → o1 → p1
    └── e2 → f2 → o1 → p1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> o1 -> p1

model_100 GasStationPrepay · pass

Guard

a1	express = 1
b1	attemptNo < 4

```
(root)
└── a1  express = 1
    └── b1  attemptNo < 4
```

เส้นทาง

a1 -> b1

Event

รหัส	event	guard
a1	sendnotifyPrepay_1	
b1	receivenotifyPrepay_1	
c1	sendvalidatePrepay_2	
d1	receivevalidatePrepay_2	
e1	sendlogFuel_3_opt1	a1
f1	receivelogFuel_3_opt1	a1
g1	sendreserveReceipt_4	
h1	receivereserveReceipt_4	
i1	sendassignFuel_5	
j1	receiveassignFuel_5	
k1	sendreleaseReceipt_6	b1
l1	receivereleaseReceipt_6	b1
m1	sendfetchFuel_7	b1
n1	receivefetchFuel_7	b1
o1	sendrequestFuel_8	
o2	checkloop_releaseReceipt_6_loop1	b1
p1	receiverequestFuel_8	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
    ├── o1 → p1
    └── o2 ↺ k1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o2 ↺ k1

model_113 InsuranceTelematics · pass

Guard

a1	express = 1
b1	roundNo < 6
c1	urgent = 1
c2	urgent = 0

```
(root)
└── a1  express = 1
    └── b1  roundNo < 6
        ├── c1  urgent = 1
        └── c2  urgent = 0
```

เส้นทาง

```
a1 -> b1 -> c1
a1 -> b1 -> c2
```

Event

รหัส	event	guard
a1	sendconfirmRating_1	
b1	receiveconfirmRating_1	
c1	sendfetchPremium_2	
d1	receivefetchPremium_2	
e1	sendsyncRating_3_opt1	a1
f1	receivesyncRating_3_opt1	a1
g1	sendnotifyRating_4_opt1	a1 b1
h1	receivenotifyRating_4_opt1	a1 b1
i1	sendconfirmTrip_5_opt1	a1
i2	checkloop_notifyRating_4_loop1	a1 b1
j1	receiveconfirmTrip_5_opt1	a1
k1	sendconfirmPremium_6	
l1	receiveconfirmPremium_6	
m1	sendackPremium_7	
n1	receiveackPremium_7	
o1	sendassignTrip_8	c1
o2	sendschedulePremium_9	c2
p1	receiveassignTrip_8	c1
p2	receiveschedulePremium_9	c2
q1	sendqueryRating_10	
r1	receivequeryRating_10	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1
    ├── i1 → j1 → k1 → l1 → m1 → n1
    │   ├── o1 → p1 → q1 → r1
    │   └── o2 → p2 → q1 → r1
    └── i2 ↺ g1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o2 -> p2 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i2 ↺ g1
```

model_124 MarketplaceEscrow · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendvalidateEscrow_1	
b1	receivevalidateEscrow_1	
c1	sendapproveEscrow_2	
d1	receiveapproveEscrow_2	
e1	sendlogEscrow_3	
f1	receivelogEscrow_3	
g1	sendrequestEscrow_4	
h1	receiverequestEscrow_4	
i1	sendfetchEscrow_5	
j1	receivefetchEscrow_5	
k1	sendnotifyEscrow_6	
l1	receivenotifyEscrow_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1
```

model_128 MobileTopUp · fail

Guard

a1	t_obs ≤ 13
a2	t_obs > 13

```
(root)
├── a1  t_obs ≤ 13
└── a2  t_obs > 13
```

เส้นทาง

a1
a2

Event

รหัส	event	guard
a1	sendnotifyCredit_1	
b1	receivenotifyCredit_1	
c1	sendconfirmBundle_2	
d1	receiveconfirmBundle_2	
e1	sendreportBundle_3	a1
e2	sendcheckBundle_5	a2
f1	receivereportBundle_3	a1
f2	receivecheckBundle_5	a2
g1	sendackCredit_4	a1
g2	sendassignCredit_6	a2
h1	receiveackCredit_4	a1
h2	receiveassignCredit_6	a2
i1	sendreleaseTopup_7	
j1	receivereleaseTopup_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1
    └── e2 → f2 → g2 → h2 → i1 → j1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g2 -> h2 -> i1 -> j1

model_131 OnlineBankingOTP · fail

Guard

a1	retryNo < 12
b1	retryNo mod 4 = 0
b2	retryNo mod 4 ≠ 0

```
(root)
└── a1  retryNo < 12
    ├── b1  retryNo mod 4 = 0
    └── b2  retryNo mod 4 ≠ 0
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

รหัส	event	guard
a1	sendconfirmTransfer_1	
b1	receiveconfirmTransfer_1	
c1	sendsendOtp_2	
d1	receivesendOtp_2	
e1	sendsendTransfer_3	a1
f1	receivesendTransfer_3	a1
g1	sendvalidateAccount_4	a1 b1
g2	sendsyncOtp_6	a1 b2
h1	receivevalidateAccount_4	a1 b1
h2	receivesyncOtp_6	a1 b2
i1	sendupdateOtp_5	a1 b1
i2	sendapproveOtp_7	a1 b2
j1	receiveupdateOtp_5	a1 b1
j2	receiveapproveOtp_7	a1 b2
k1	sendcancelTransfer_8	a1
l1	receivecancelTransfer_8	a1
m1	sendapproveAccount_9	a1
n1	receiveapproveAccount_9	a1
o1	sendsendAccount_10	a1
p1	receivesendAccount_10	a1
q1	sendissueOtp_11	
q2	checkloop_sendTransfer_3_loop1	a1
r1	receiveissueOtp_11	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1 → m1 → n1 →
    │     o1 → p1
    │   ├── q1 → r1
    │   └── q2 ↺ e1
    └── g2 → h2 → i2 → j2 → k1 → l1 → m1 → n1 →
          o1 → p1
        ├── q1 → r1
        └── q2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q2 ↺ e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> i2 -> j2 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> i2 -> j2 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q2 ↺ e1
```

model_132 OnlineDonation · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendsendReceipt_1	
b1	receivesendReceipt_1	
c1	sendqueryGift_2	
d1	receivequeryGift_2	
e1	sendapproveReceipt_3	
f1	receiveapproveReceipt_3	
g1	sendnotifyReceipt_4	
h1	receivenotifyReceipt_4	
i1	sendapproveGift_5	
j1	receiveapproveGift_5	
k1	sendrequestGift_6	
l1	receiverequestGift_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_189 TradingOrderGateway · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendcheckFill_1	
b1	receivecheckFill_1	
c1	sendassignOrder_2	
d1	receiveassignOrder_2	
e1	sendscheduleFill_3	
f1	receivescheduleFill_3	
g1	sendscheduleOrder_4	
h1	receivescheduleOrder_4	
i1	sendnotifyOrder_5	
j1	receivenotifyOrder_5	
k1	sendackMargin_6	
l1	receiveackMargin_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

สุขภาพและการแพทย์ · Healthcare & Medical (26)

model_006 BloodBankRequest · pass

Guard

a1	stockUnits ≥ 2
a2	stockUnits < 2

```
(root)
├── a1  stockUnits ≥ 2
└── a2  stockUnits < 2
```

เส้นทาง
a1
a2

Event

รหัส	event	guard
a1	sendrequestBlood_1	
b1	receiverequestBlood_1	
c1	sendcrossMatch_2	a1
c2	sendbackorder_5	a2
d1	receivecrossMatch_2	a1
d2	receivebackorder_5	a2
e1	sendprepareCourier_3	a1
f1	receiveprepareCourier_3	a1
g1	senddeliverBlood_4	a1
h1	receivedeliverBlood_4	a1
i1	sendconfirmReceipt_6	
j1	receiveconfirmReceipt_6	

```
(root) INITIALISATION
└── a1 → b1
    ├── c1 → d1 → e1 → f1 → g1 → h1 → i1 → j1
    └── c2 → d2 → i1 → j1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1
a1 -> b1 -> c2 -> d2 -> i1 -> j1

model_020 FitnessTracker · pass

Guard

a1	syncRound < 10

```
(root)
└── a1  syncRound < 10
```

เส้นทาง
a1

Event

รหัส	event	guard
a1	sendpairDevice_1	
b1	receivepairDevice_1	
c1	sendpairOK_2	
d1	receivepairOK_2	
e1	sendsyncSteps_3	a1
f1	receivesyncSteps_3	a1
g1	sendsyncAck_4	a1
h1	receivesyncAck_4	a1
i1	sendshowDashboard_5	
i2	checkloop_syncSteps_3_loop1	a1
j1	receiveshowDashboard_5	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1
    ├── i1 → j1
    └── i2 ↺ e1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i2 ↺ e1

model_023 HospitalTriage · pass

Guard

a1	severity = CRITICAL
a2	severity = NORMAL

```
(root)
├── a1  severity = CRITICAL
└── a2  severity = NORMAL
```

เส้นทาง

a1
a2

Event

รหัส	event	guard
a1	sendregisterPatient_1	
b1	receiveregisterPatient_1	
c1	sendmeasureVitals_2	
d1	receivemeasureVitals_2	
e1	sendcallDoctor_3	a1
e2	sendscheduleVisit_6	a2
f1	receivecallDoctor_3	a1
f2	receivescheduleVisit_6	a2
g1	sendorderLabTest_4	a1
h1	receiveorderLabTest_4	a1
i1	sendprescribeDrug_5	a1
j1	receiveprescribeDrug_5	a1
k1	senddischargeNote_7	
l1	receivedischargeNote_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1 → k1 → l1
    └── e2 → f2 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> k1 -> l1

model_026 LabSampleTesting · fail

Guard

a1	retestNo < 2

```
(root)
└── a1  retestNo < 2
```

เส้นทาง

a1

Event

รหัส	event	guard
a1	sendcollectSample_1	
b1	receivecollectSample_1	
c1	sendtransportSample_2	
d1	receivetransportSample_2	
e1	sendsampleReceived_3	
f1	receivesampleReceived_3	
g1	sendrequestRetest_4	a1
h1	receiverequestRetest_4	a1
i1	sendretestResult_5	a1
j1	receiveretestResult_5	a1
k1	sendfinalReport_6	
k2	checkloop_requestRetest_4_loop1	a1
l1	receivefinalReport_6	
m1	sendnotifyClinic_7	
n1	receivenotifyClinic_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1
    ├── k1 → l1 → m1 → n1
    └── k2 ↺ g1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 ↺ g1

model_029 MedicationDispenser · fail

Guard

a1	taken = 1
a2	taken = 0

```
(root)
├── a1  taken = 1
└── a2  taken = 0
```

เส้นทาง

a1
a2

Event

รหัส	event	guard
a1	sendscheduleDose_1	
b1	receivescheduleDose_1	
c1	senddoseAlert_2	
d1	receivedoseAlert_2	
e1	sendconfirmIntake_3	
f1	receiveconfirmIntake_3	
g1	sendlogDose_4	a1
g2	sendescalateMissed_6	a2
h1	receivelogDose_4	a1
h2	receiveescalateMissed_6	a2
i1	sendlogAck_5	a1
j1	receivelogAck_5	a1
k1	sendrefillCheck_7	
l1	receiverefillCheck_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1
    └── g2 → h2 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1

model_052 AmbulanceRouting · pass

Guard

a1	useCache = 1
b1	needAck = 1

```
(root)
└── a1  useCache = 1
    └── b1  needAck = 1
```

เส้นทาง

a1 -> b1

Event

รหัส	event	guard
a1	sendapproveEta_1	
b1	receiveapproveEta_1	
c1	sendackPatient_2	
d1	receiveackPatient_2	
e1	sendassignPatient_3_opt1	a1
f1	receiveassignPatient_3_opt1	a1
g1	sendnotifyRoute_4	
h1	receivenotifyRoute_4	
i1	sendreserveEta_5	
j1	receivereserveEta_5	
k1	sendrequestRoute_6_opt2	b1
l1	receiverequestRoute_6_opt2	b1
m1	sendcheckPatient_7	
n1	receivecheckPatient_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1

model_058 BloodDonationDrive · fail

Guard

a1	t_obs ≤ 8
b1	withExtra = 1
a2	t_obs > 8

```
(root)
├── a1  t_obs ≤ 8
│   └── b1  withExtra = 1
└── a2  t_obs > 8
```

เส้นทาง

```
a1 -> b1
a2
```

Event

รหัส	event	guard
a1	sendackScreening_1	
b1	receiveackScreening_1	
c1	sendlogSlot_2	
d1	receivelogSlot_2	
e1	sendsyncSlot_3	a1
e2	sendlogDonation_6	a2
f1	receivesyncSlot_3	a1
f2	receivelogDonation_6	a2
g1	sendnotifyDonation_4_opt1	a1 b1
h1	receivenotifyDonation_4_opt1	a1 b1
i1	sendvalidateScreening_5	a1
j1	receivevalidateScreening_5	a1
k1	sendfetchScreening_7	
l1	receivefetchScreening_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1 → k1 → l1
    └── e2 → f2 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> k1 -> l1
```

model_082 DentalAppointment · fail

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendqueryAppointment_1	
b1	receivequeryAppointment_1	
c1	sendassignTreatment_2	
d1	receiveassignTreatment_2	
e1	sendupdateTreatment_3	
f1	receiveupdateTreatment_3	
g1	sendqueryClaim_4	
h1	receivequeryClaim_4	
i1	sendconfirmAppointment_5	
j1	receiveconfirmAppointment_5	
k1	sendreserveClaim_6	
l1	receivereserveClaim_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1
```

model_101 GlucoseMeterSync · pass

Guard

a1	needAck = 1

```
(root)
└── a1  needAck = 1
```

เส้นทาง
a1

Event

รหัส	event	guard
a1	sendconfirmInsulin_1	
b1	receiveconfirmInsulin_1	
c1	sendsyncReading_2	
d1	receivesyncReading_2	
e1	sendupdateAlert_3	
f1	receiveupdateAlert_3	
g1	sendqueryReading_4	
h1	receivequeryReading_4	
i1	sendnotifyAlert_5	
j1	receivenotifyAlert_5	
k1	sendreleaseAlert_6	
l1	receivereleaseAlert_6	
m1	sendqueryAlert_7	
n1	receivequeryAlert_7	
o1	sendreleaseInsulin_8_opt1	a1
p1	receivereleaseInsulin_8_opt1	a1
q1	sendrequestAlert_9_opt1	a1
r1	receiverequestAlert_9_opt1	a1
s1	sendreportInsulin_10_opt1	a1
t1	receivereportInsulin_10_opt1	a1
u1	sendupdateInsulin_11_opt1	a1
v1	receiveupdateInsulin_11_opt1	a1
w1	sendsendAlert_12_opt1	a1
x1	receivesendAlert_12_opt1	a1
y1	sendapproveAlert_13_opt1	a1
z1	receiveapproveAlert_13_opt1	a1
aa1	sendreserveAlert_14	
ab1	receivereserveAlert_14	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1 →
      q1 → r1 → s1 → t1 → u1 → v1 → w1 → x1 →
      y1 → z1 → aa1 → ab1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1 -> w1 -> x1 -> y1 -> z1 -> aa1 -> ab1

model_106 HomeOxygenConcentrator · pass

Guard

a1	withExtra = 1
b1	needAck = 1

```
(root)
└── a1  withExtra = 1
    └── b1  needAck = 1
```

เส้นทาง

a1 -> b1

Event

รหัส	event	guard
a1	sendreserveFlow_1	
b1	receivereserveFlow_1	
c1	sendupdateAlert_2	
d1	receiveupdateAlert_2	
e1	sendreserveAlert_3_opt1	a1
f1	receivereserveAlert_3_opt1	a1
g1	sendnotifyAlert_4	
h1	receivenotifyAlert_4	
i1	sendsyncAlert_5	
j1	receivesyncAlert_5	
k1	sendrequestFlow_6_opt2	b1
l1	receiverequestFlow_6_opt2	b1
m1	sendlogFlow_7_opt2	b1
n1	receivelogFlow_7_opt2	b1
o1	sendschedulePurity_8	
p1	receiveschedulePurity_8	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1

model_108 HospitalBedAllocation · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendsendBed_1	
b1	receivesendBed_1	
c1	sendassignBed_2	
d1	receiveassignBed_2	
e1	sendsyncTransfer_3	
f1	receivesyncTransfer_3	
g1	sendscheduleBed_4	
h1	receivescheduleBed_4	
i1	sendlogTransfer_5	
j1	receivelogTransfer_5	
k1	sendreleasePatient_6	
l1	receivereleasePatient_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_109 HospitalLabRequest · fail

Guard

a1	cycleNo < 8
b1	cycleNo mod 3 = 0
c1	attemptNo < 10
b2	cycleNo mod 3 ≠ 0

```
(root)
└── a1  cycleNo < 8
    ├── b1  cycleNo mod 3 = 0
    │   └── c1  attemptNo < 10
    └── b2  cycleNo mod 3 ≠ 0
```

เส้นทาง

```
a1 -> b1 -> c1
a1 -> b2
```

Event

รหัส	event	guard
a1	sendcheckOrder_1	
b1	receivecheckOrder_1	
c1	sendrequestResult_2	
d1	receiverequestResult_2	
e1	sendconfirmOrder_3	a1
f1	receiveconfirmOrder_3	a1
g1	sendscheduleOrder_4	a1 b1
g2	sendreportSample_9	a1 b2
h1	receivescheduleOrder_4	a1 b1
h2	receivereportSample_9	a1 b2
i1	sendackSample_5	a1 b1 c1
j1	receiveackSample_5	a1 b1 c1
k1	sendapproveSample_6	a1 b1 c1
l1	receiveapproveSample_6	a1 b1 c1
m1	sendissueResult_7	a1 b1 c1
n1	receiveissueResult_7	a1 b1 c1
o1	sendreportResult_8	a1 b1
o2	checkloop_ackSample_5_loop2	a1 b1 c1
p1	receivereportResult_8	a1 b1
q1	sendrequestSample_10	a1
r1	receiverequestSample_10	a1
s1	sendreserveSample_11	
s2	checkloop_confirmOrder_3_loop1	a1
t1	receivereserveSample_11	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1 → m1 → n1
    │   ├── o1 → p1 → q1 → r1
    │   │   ├── s1 → t1
    │   │   └── s2 ↺ e1
    │   └── o2 ↺ i1
    └── g2 → h2 → q1 → r1
        ├── s1 → t1
        └── s2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s2 ↺ e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o2 ↺ i1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> q1 -> r1 -> s1
-> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> q1 -> r1 -> s2 ↺
e1
```

model_110 HospitalPharmacyRobot · pass

Guard

a1	t_obs ≤ 13
a2	t_obs > 13

```
(root)
├── a1  t_obs ≤ 13
└── a2  t_obs > 13
```

เส้นทาง

a1
a2

Event

รหัส	event	guard
a1	sendnotifyDrawer_1	
b1	receivenotifyDrawer_1	
c1	sendconfirmOrder_2	
d1	receiveconfirmOrder_2	
e1	sendreportOrder_3	a1
e2	sendcheckOrder_5	a2
f1	receivereportOrder_3	a1
f2	receivecheckOrder_5	a2
g1	sendackDrawer_4	a1
g2	sendassignDrawer_6	a2
h1	receiveackDrawer_4	a1
h2	receiveassignDrawer_6	a2
i1	sendreleaseDose_7	
j1	receivereleaseDose_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1
    └── e2 → f2 → g2 → h2 → i1 → j1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g2 -> h2 -> i1 -> j1

model_125 MedicalImagingPACS · fail

Guard

a1	roundNo < 6
b1	roundNo mod 3 = 0
b2	roundNo mod 3 ≠ 0

```
(root)
└── a1  roundNo < 6
    ├── b1  roundNo mod 3 = 0
    └── b2  roundNo mod 3 ≠ 0
```

เส้นทาง

a1 -> b1
a1 -> b2

Event

รหัส	event	guard
a1	sendcheckImage_1	
b1	receivecheckImage_1	
c1	sendassignImage_2	
d1	receiveassignImage_2	
e1	sendassignReport_3	a1
f1	receiveassignReport_3	a1
g1	sendapproveReport_4	a1 b1
g2	sendrequestImage_5	a1 b2
h1	receiveapproveReport_4	a1 b1
h2	receiverequestImage_5	a1 b2
i1	sendvalidateImage_6	a1
j1	receivevalidateImage_6	a1
k1	sendsyncReport_7	
k2	checkloop_assignReport_3_loop1	a1
l1	receivesyncReport_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1
    │   ├── k1 → l1
    │   └── k2 ↺ e1
    └── g2 → h2 → i1 → j1
        ├── k1 → l1
        └── k2 ↺ e1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 ↺ e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> i1 -> j1 -> k2 ↺ e1

model_126 MedicalTranscription · pass

Guard

a1	optIn = 1

```
(root)
└── a1  optIn = 1
```

เส้นทาง
a1

Event

รหัส	event	guard
a1	sendcheckTranscript_1	
b1	receivecheckTranscript_1	
c1	sendlogTranscript_2	
d1	receivelogTranscript_2	
e1	sendsendNote_3_opt1	a1
f1	receivesendNote_3_opt1	a1
g1	sendupdateDictation_4_opt1	a1
h1	receiveupdateDictation_4_opt1	a1
i1	sendnotifyDictation_5	
j1	receivenotifyDictation_5	
k1	sendconfirmDictation_6	
l1	receiveconfirmDictation_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_139 PatientIntakeForm · fail

Guard

a1	attemptNo < 5
b1	attemptNo mod 2 = 0

```
(root)
└── a1  attemptNo < 5
    └── b1  attemptNo mod 2 = 0
```

เส้นทาง
a1 -> b1

Event

รหัส	event	guard
a1	sendreleaseForm_1	
b1	receivereleaseForm_1	
c1	sendcancelForm_2	
d1	receivecancelForm_2	
e1	sendconfirmForm_3	a1
f1	receiveconfirmForm_3	a1
g1	sendreportForm_4_opt1	a1 b1
h1	receivereportForm_4_opt1	a1 b1
i1	sendscheduleConsent_5	a1
j1	receivescheduleConsent_5	a1
k1	sendconfirmConsent_6	a1
l1	receiveconfirmConsent_6	a1
m1	sendissueForm_7	a1
n1	receiveissueForm_7	a1
o1	sendassignRecord_8	
o2	checkloop_confirmForm_3_loop1	a1
p1	receiveassignRecord_8	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
    ├── o1 → p1
    └── o2 ↺ e1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o2 ↺ e1

model_141 PharmacyDelivery · fail

Guard

a1	needAck = 1
b1	mode = GOLD
b2	mode = SILVER
b3	mode = BRONZE

```
(root)
└── a1  needAck = 1
    ├── b1  mode = GOLD
    ├── b2  mode = SILVER
    └── b3  mode = BRONZE
```

เส้นทาง

```
a1 -> b1
a1 -> b2
a1 -> b3
```

Event

รหัส	event	guard
a1	sendsyncSignature_1	
b1	receivesyncSignature_1	
c1	sendvalidateSignature_2	
d1	receivevalidateSignature_2	
e1	sendfetchRoute_3_opt1	a1
f1	receivefetchRoute_3_opt1	a1
g1	sendqueryRoute_4_opt1	a1
h1	receivequeryRoute_4_opt1	a1
i1	sendconfirmRoute_5	
j1	receiveconfirmRoute_5	
k1	sendissuePackage_6	
l1	receiveissuePackage_6	
m1	sendcancelSignature_7	b1
m2	sendqueryPackage_9	b2
m3	sendscheduleSignature_11	b3
n1	receivecancelSignature_7	b1
n2	receivequeryPackage_9	b2
n3	receivescheduleSignature_11	b3
o1	sendconfirmPackage_8	b1
o2	sendackSignature_10	b2
p1	receiveconfirmPackage_8	b1
p2	receiveackSignature_10	b2
q1	sendapproveRoute_12	
r1	receiveapproveRoute_12	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
    ├── m1 → n1 → o1 → p1 → q1 → r1
    ├── m2 → n2 → o2 → p2 → q1 → r1
    └── m3 → n3 → q1 → r1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m2 -> n2 -> o2 -> p2 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m3 -> n3 -> q1 -> r1
```

model_142 PharmacyRefill · fail

Guard

a1	express = 1

```
(root)
└── a1  express = 1
```

เส้นทาง
a1

Event

รหัส	event	guard
a1	sendissuePrescription_1	
b1	receiveissuePrescription_1	
c1	sendcheckClaim_2	
d1	receivecheckClaim_2	
e1	sendschedulePrescription_3_opt1	a1
f1	receiveschedulePrescription_3_opt1	a1
g1	sendqueryRefill_4_opt1	a1
h1	receivequeryRefill_4_opt1	a1
i1	sendackRefill_5	
j1	receiveackRefill_5	
k1	sendreservePrescription_6	
l1	receivereservePrescription_6	
m1	sendcancelRefill_7	
n1	receivecancelRefill_7	
o1	sendvalidateClaim_8	
p1	receivevalidateClaim_8	
q1	sendconfirmPrescription_9	
r1	receiveconfirmPrescription_9	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1 →
      q1 → r1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1

model_153 RemotePatientMonitor · fail

Guard

a1	passNo < 12

```
(root)
└── a1  passNo < 12
```

เส้นทาง
a1

Event

รหัส	event	guard
a1	sendsyncAlert_1	
b1	receivesyncAlert_1	
c1	sendcancelReport_2	
d1	receivecancelReport_2	
e1	sendapproveVital_3	a1
f1	receiveapproveVital_3	a1
g1	sendscheduleReport_4	a1
h1	receivescheduleReport_4	a1
i1	sendreleaseReport_5	
i2	checkloop_approveVital_3_loop1	a1
j1	receivereleaseReport_5	
k1	sendsendAlert_6	
l1	receivesendAlert_6	
m1	sendreportVital_7	
n1	receivereportVital_7	
o1	sendnotifyAlert_8	
p1	receivenotifyAlert_8	
q1	sendfetchVital_9	
r1	receivefetchVital_9	
s1	sendreportAlert_10	
t1	receivereportAlert_10	
u1	sendreportReport_11	
v1	receivereportReport_11	
w1	sendconfirmVital_12	
x1	receiveconfirmVital_12	
y1	sendassignAlert_13	
z1	receiveassignAlert_13	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1
    ├── i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1 →
    │     q1 → r1 → s1 → t1 → u1 → v1 → w1 → x1 →
    │     y1 → z1
    └── i2 ↺ e1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1 -> w1 -> x1 -> y1 -> z1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i2 ↺ e1

model_158 RuralTelehealthKiosk · fail

Guard

a1	passNo < 6
b1	passNo mod 4 = 0
b2	passNo mod 4 ≠ 0

```
(root)
└── a1  passNo < 6
    ├── b1  passNo mod 4 = 0
    └── b2  passNo mod 4 ≠ 0
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

รหัส	event	guard
a1	sendreportVital_1	
b1	receivereportVital_1	
c1	sendvalidateVital_2	
d1	receivevalidateVital_2	
e1	sendfetchPrescription_3	a1
f1	receivefetchPrescription_3	a1
g1	sendschedulePrescription_4	a1 b1
g2	sendreportPrescription_11	a1 b2
h1	receiveschedulePrescription_4	a1 b1
h2	receivereportPrescription_11	a1 b2
i1	sendcancelPrescription_5	a1 b1
j1	receivecancelPrescription_5	a1 b1
k1	sendnotifyConsult_6	a1 b1
l1	receivenotifyConsult_6	a1 b1
m1	sendsendVital_7	a1 b1
n1	receivesendVital_7	a1 b1
o1	sendfetchVital_8	a1 b1
p1	receivefetchVital_8	a1 b1
q1	sendackConsult_9	a1 b1
r1	receiveackConsult_9	a1 b1
s1	sendupdateConsult_10	a1 b1
t1	receiveupdateConsult_10	a1 b1
u1	sendconfirmVital_12	a1
v1	receiveconfirmVital_12	a1
w1	sendreleaseVital_13	
w2	checkloop_fetchPrescription_3_loop1	a1
x1	receivereleaseVital_13	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1 → m1 → n1 →
    │     o1 → p1 → q1 → r1 → s1 → t1 → u1 → v1
    │   ├── w1 → x1
    │   └── w2 ↺ e1
    └── g2 → h2 → u1 → v1
        ├── w1 → x1
        └── w2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 ->
v1 -> w1 -> x1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 ->
v1 -> w2 ↺ e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> u1 -> v1 -> w1
-> x1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> u1 -> v1 -> w2 ↺
e1
```

model_172 SmartMirrorFitness · fail

Guard

a1	attemptNo < 6
b1	attemptNo mod 2 = 0

```
(root)
└── a1  attemptNo < 6
    └── b1  attemptNo mod 2 = 0
```

เส้นทาง

```
a1 -> b1
```

Event

รหัส	event	guard
a1	sendreleaseWorkout_1	
b1	receivereleaseWorkout_1	
c1	sendreportRep_2	
d1	receivereportRep_2	
e1	sendapprovePlan_3	a1
f1	receiveapprovePlan_3	a1
g1	sendconfirmWorkout_4_opt1	a1 b1
h1	receiveconfirmWorkout_4_opt1	a1 b1
i1	sendlogWorkout_5_opt1	a1 b1
j1	receivelogWorkout_5_opt1	a1 b1
k1	sendissueWorkout_6_opt1	a1 b1
l1	receiveissueWorkout_6_opt1	a1 b1
m1	sendcancelRep_7_opt1	a1 b1
n1	receivecancelRep_7_opt1	a1 b1
o1	sendackRep_8_opt1	a1 b1
p1	receiveackRep_8_opt1	a1 b1
q1	sendreservePlan_9_opt1	a1 b1
r1	receivereservePlan_9_opt1	a1 b1
s1	sendsendWorkout_10_opt1	a1 b1
t1	receivesendWorkout_10_opt1	a1 b1
u1	sendscheduleWorkout_11	a1
v1	receivescheduleWorkout_11	a1
w1	sendsyncPlan_12	
w2	checkloop_approvePlan_3_loop1	a1
x1	receivesyncPlan_12	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1 →
      q1 → r1 → s1 → t1 → u1 → v1
    ├── w1 → x1
    └── w2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 ->
v1 -> w1 -> x1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 ->
v1 -> w2 ↺ e1
```

model_173 SmartPillBox · pass

Guard

a1	needAck = 1
b1	mode = GOLD
b2	mode = SILVER
b3	mode = BRONZE

```
(root)
└── a1  needAck = 1
    ├── b1  mode = GOLD
    ├── b2  mode = SILVER
    └── b3  mode = BRONZE
```

เส้นทาง

```
a1 -> b1
a1 -> b2
a1 -> b3
```

Event

รหัส	event	guard
a1	sendsyncRefill_1	
b1	receivesyncRefill_1	
c1	sendvalidateRefill_2	
d1	receivevalidateRefill_2	
e1	sendfetchReminder_3_opt1	a1
f1	receivefetchReminder_3_opt1	a1
g1	sendqueryReminder_4_opt1	a1
h1	receivequeryReminder_4_opt1	a1
i1	sendconfirmReminder_5	
j1	receiveconfirmReminder_5	
k1	sendissueDose_6	
l1	receiveissueDose_6	
m1	sendcancelRefill_7	b1
m2	sendqueryDose_9	b2
m3	sendscheduleRefill_11	b3
n1	receivecancelRefill_7	b1
n2	receivequeryDose_9	b2
n3	receivescheduleRefill_11	b3
o1	sendconfirmDose_8	b1
o2	sendackRefill_10	b2
p1	receiveconfirmDose_8	b1
p2	receiveackRefill_10	b2
q1	sendapproveReminder_12	
r1	receiveapproveReminder_12	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
    ├── m1 → n1 → o1 → p1 → q1 → r1
    ├── m2 → n2 → o2 → p2 → q1 → r1
    └── m3 → n3 → q1 → r1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m2 -> n2 -> o2 -> p2 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m3 -> n3 -> q1 -> r1
```

model_186 TelemedicineVisit · fail

Guard

a1	stepNo < 2
b1	stepNo mod 2 = 0
b2	stepNo mod 2 ≠ 0

```
(root)
└── a1  stepNo < 2
    ├── b1  stepNo mod 2 = 0
    └── b2  stepNo mod 2 ≠ 0
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

รหัส	event	guard
a1	sendissuePrescription_1	
b1	receiveissuePrescription_1	
c1	sendconfirmPrescription_2	
d1	receiveconfirmPrescription_2	
e1	sendupdateVisit_3	a1
f1	receiveupdateVisit_3	a1
g1	sendschedulePrescription_4	a1 b1
g2	sendrequestVisit_5	a1 b2
h1	receiveschedulePrescription_4	a1 b1
h2	receiverequestVisit_5	a1 b2
i1	sendsyncPrescription_6	a1
j1	receivesyncPrescription_6	a1
k1	sendsendNote_7	
k2	checkloop_updateVisit_3_loop1	a1
l1	receivesendNote_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1
    │   ├── k1 → l1
    │   └── k2 ↺ e1
    └── g2 → h2 → i1 → j1
        ├── k1 → l1
        └── k2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 ↺
e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> i1 -> j1 -> k1
-> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> i1 -> j1 -> k2 ↺
e1
```

model_187 ThermalCameraScreening · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendcancelScan_1	
b1	receivecancelScan_1	
c1	sendackScan_2	
d1	receiveackScan_2	
e1	sendassignAlert_3	
f1	receiveassignAlert_3	
g1	sendconfirmAlert_4	
h1	receiveconfirmAlert_4	
i1	sendackAlert_5	
j1	receiveackAlert_5	
k1	sendqueryFever_6	
l1	receivequeryFever_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1
```

model_192 VaccineColdBox · fail

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendreleaseAlarm_1	
b1	receivereleaseAlarm_1	
c1	sendackVaccine_2	
d1	receiveackVaccine_2	
e1	sendsyncReading_3	
f1	receivesyncReading_3	
g1	sendsyncAlarm_4	
h1	receivesyncAlarm_4	
i1	sendapproveAlarm_5	
j1	receiveapproveAlarm_5	
k1	sendsendVaccine_6	
l1	receivesendVaccine_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_193 VeterinaryClinic · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendreleaseAppointment_1	
b1	receivereleaseAppointment_1	
c1	sendcancelAppointment_2	
d1	receivecancelAppointment_2	
e1	sendreportInvoice_3	
f1	receivereportInvoice_3	
g1	sendsendAppointment_4	
h1	receivesendAppointment_4	
i1	sendapproveAppointment_5	
j1	receiveapproveAppointment_5	
k1	sendvalidateInvoice_6	
l1	receivevalidateInvoice_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

การเดินทางและยานพาหนะ · Transportation & Mobility (26)

model_002 AirportBaggage · fail

Guard

a1	bagCleared = 1
a2	bagCleared = 0

```
(root)
├── a1  bagCleared = 1
└── a2  bagCleared = 0
```

เส้นทาง

a1
a2

Event

รหัส	event	guard
a1	sendtagBag_1	
b1	receivetagBag_1	
c1	sendrouteBag_2	
d1	receiverouteBag_2	
e1	sendsorted_3	
f1	receivesorted_3	
g1	sendloadBag_4	a1
g2	senddivertBag_6	a2
h1	receiveloadBag_4	a1
h2	receivedivertBag_6	a2
i1	sendloadedAck_5	a1
j1	receiveloadedAck_5	a1
k1	sendmanifestUpdate_7	
l1	receivemanifestUpdate_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1
    └── g2 → h2 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1

model_007 CarRentalBooking · pass

Guard

a1	searchCount < 3
b1	insurance = 1
c1	t_pay ≤ 10
c2	t_pay > 10

```
(root)
└── a1  searchCount < 3
    └── b1  insurance = 1
        ├── c1  t_pay ≤ 10
        └── c2  t_pay > 10
```

เส้นทาง

a1 -> b1 -> c1
a1 -> b1 -> c2

Event

รหัส	event	guard
a1	sendopenApp_1	
b1	receiveopenApp_1	
c1	sendsearchCar_2	a1
d1	receivesearchCar_2	a1
e1	sendcarList_3	a1
f1	receivecarList_3	a1
g1	sendaddInsurance_4_opt1	b1
g2	checkloop_searchCar_2_loop1	a1
h1	receiveaddInsurance_4_opt1	b1
i1	sendrequestPayment_5	
j1	receiverequestPayment_5	
k1	sendpaymentResult_6	
l1	receivepaymentResult_6	
m1	sendconfirmBooking_7	c1
m2	sendcancelBooking_8	c2
n1	receiveconfirmBooking_7	c1
n2	receivecancelBooking_8	c2

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1
    │   ├── m1 → n1
    │   └── m2 → n2
    └── g2 ↺ c1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m2 -> n2
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 ↺ c1

model_019 FerryTicketGate · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendtapTicket_1	
b1	receivetapTicket_1	
c1	sendvalidate_2	
d1	receivevalidate_2	
e1	sendopenBarrier_3	
f1	receiveopenBarrier_3	
g1	sendpassThrough_4	
h1	receivepassThrough_4	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1

model_040 SmartParking · pass

Guard

a1	slotFree = 1

```
(root)
└── a1  slotFree = 1
```

เส้นทาง

a1

Event

รหัส	event	guard
a1	sendrequestEntry_1	
b1	receiverequestEntry_1	
c1	sendcheckSlot_2	
d1	receivecheckSlot_2	
e1	sendslotStatus_3	
f1	receiveslotStatus_3	
g1	sendopenGate_4_opt1	a1
h1	receiveopenGate_4_opt1	a1
i1	sendissueTicket_5_opt1	a1
j1	receiveissueTicket_5_opt1	a1
k1	sendlogEntry_6	
l1	receivelogEntry_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_042 TaxiDispatch · fail

Guard

a1	shareRide = 1

```
(root)
└── a1  shareRide = 1
```

เส้นทาง
a1

Event

รหัส	event	guard
a1	sendrequestRide_1	
b1	receiverequestRide_1	
c1	sendfindDriver_2	
d1	receivefindDriver_2	
e1	senddriverAccept_3	
f1	receivedriverAccept_3	
g1	sendrideConfirmed_4	
h1	receiverideConfirmed_4	
i1	sendaddPassenger_5_opt1	a1
j1	receiveaddPassenger_5_opt1	a1
k1	sendtripComplete_6	
l1	receivetripComplete_6	
m1	sendchargeRider_7	
n1	receivechargeRider_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1

model_045 TrafficLightController · fail

Guard

a1	cycleCount < 10

```
(root)
└── a1  cycleCount < 10
```

เส้นทาง
a1

Event

รหัส	event	guard
a1	sendsetGreen_1	
b1	receivesetGreen_1	
c1	sendgreenAck_2	
d1	receivegreenAck_2	
e1	sendsetYellow_3	a1
f1	receivesetYellow_3	a1
g1	sendsetRed_4	a1
h1	receivesetRed_4	a1
i1	sendphaseDone_5	a1
j1	receivephaseDone_5	a1
k1	sendshutdown_6	
k2	checkloop_setYellow_3_loop1	a1
l1	receiveshutdown_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1
    ├── k1 → l1
    └── k2 ↺ e1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 ↺ e1

model_049 AirportSecurityLane · fail

Guard

a1	usage < 10
a2	usage ≥ 10

```
(root)
├── a1  usage < 10
└── a2  usage ≥ 10
```

เส้นทาง

a1
a2

Event

รหัส	event	guard
a1	sendcancelTray_1	
b1	receivecancelTray_1	
c1	sendvalidateTray_2	
d1	receivevalidateTray_2	
e1	sendrequestScan_3	a1
e2	sendreleaseScan_4	a2
f1	receiverequestScan_3	a1
f2	receivereleaseScan_4	a2
g1	sendreserveTray_6	
h1	receivereserveTray_6	
i1	sendupdateTray_5	a2
j1	receiveupdateTray_5	a2

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1
    └── e2 → f2 → i1 → j1 → g1 → h1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> i1 -> j1 -> g1 -> h1

model_050 AirportShuttle · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendsendRoute_1	
b1	receivesendRoute_1	
c1	sendnotifyPickup_2	
d1	receivenotifyPickup_2	
e1	sendupdateSeat_3	
f1	receiveupdateSeat_3	
g1	sendconfirmPickup_4	
h1	receiveconfirmPickup_4	
i1	sendscheduleRoute_5	
j1	receivescheduleRoute_5	
k1	sendfetchSeat_6	
l1	receivefetchSeat_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1

model_057 BikeSharing · pass

Guard

a1	t_obs ≤ 6
b1	useCache = 1
a2	t_obs > 6

```
(root)
├── a1  t_obs ≤ 6
│   └── b1  useCache = 1
└── a2  t_obs > 6
```

เส้นทาง

```
a1 -> b1
a2
```

Event

รหัส	event	guard
a1	sendfetchRide_1	
b1	receivefetchRide_1	
c1	sendconfirmBike_2	
d1	receiveconfirmBike_2	
e1	sendsyncFee_3	a1
e2	sendvalidateFee_7	a2
f1	receivesyncFee_3	a1
f2	receivevalidateFee_7	a2
g1	sendfetchBike_4_opt1	a1 b1
h1	receivefetchBike_4_opt1	a1 b1
i1	sendreserveBike_5_opt1	a1 b1
j1	receivereserveBike_5_opt1	a1 b1
k1	sendapproveBike_6	a1
l1	receiveapproveBike_6	a1
m1	sendnotifyFee_8	
n1	receivenotifyFee_8	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1 → k1 → l1 →
    │     m1 → n1
    └── e2 → f2 → m1 → n1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> m1 -> n1
```

model_062 BusTicketBooking · fail

Guard

a1	retryNo < 3
b1	retryNo mod 3 = 0

```
(root)
└── a1  retryNo < 3
    └── b1  retryNo mod 3 = 0
```

เส้นทาง

```
a1 -> b1
```

Event

รหัส	event	guard
a1	sendvalidateSeat_1	
b1	receivevalidateSeat_1	
c1	sendlogTicket_2	
d1	receivelogTicket_2	
e1	sendsendSeat_3	a1
f1	receivesendSeat_3	a1
g1	sendreserveSeat_4_opt1	a1 b1
h1	receivereserveSeat_4_opt1	a1 b1
i1	sendrequestRoute_5_opt1	a1 b1
j1	receiverequestRoute_5_opt1	a1 b1
k1	sendlogSeat_6	a1
l1	receivelogSeat_6	a1
m1	sendscheduleTicket_7	a1
n1	receivescheduleTicket_7	a1
o1	sendapproveSeat_8	
o2	checkloop_sendSeat_3_loop1	a1
p1	receiveapproveSeat_8	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
    ├── o1 → p1
    └── o2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o2 ↺ e1
```

model_064 CampusShuttleBus · pass

Guard

a1	optIn = 1
b1	load < 10
b2	load ≥ 10

```
(root)
└── a1  optIn = 1
    ├── b1  load < 10
    └── b2  load ≥ 10
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

รหัส	event	guard
a1	sendscheduleArrival_1	
b1	receivescheduleArrival_1	
c1	sendscheduleSeat_2	
d1	receivescheduleSeat_2	
e1	sendvalidateArrival_3	
f1	receivevalidateArrival_3	
g1	sendupdateStop_4	
h1	receiveupdateStop_4	
i1	sendvalidateStop_5	
j1	receivevalidateStop_5	
k1	sendreserveSeat_6	
l1	receivereserveSeat_6	
m1	sendupdateArrival_7_opt1	a1
n1	receiveupdateArrival_7_opt1	a1
o1	sendapproveArrival_8_opt1	a1 b1
o2	sendsendStop_10_opt1	a1 b2
p1	receiveapproveArrival_8_opt1	a1 b1
p2	receivesendStop_10_opt1	a1 b2
q1	sendquerySeat_9_opt1	a1 b1
r1	receivequerySeat_9_opt1	a1 b1
s1	sendissueStop_11_opt1	a1
t1	receiveissueStop_11_opt1	a1
u1	sendissueArrival_12	
v1	receiveissueArrival_12	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
    ├── o1 → p1 → q1 → r1 → s1 → t1 → u1 → v1
    └── o2 → p2 → s1 → t1 → u1 → v1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 ->
v1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o2 -> p2 -> s1 -> t1 -> u1 -> v1
```

model_068 CarRentalReturn · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendapproveDamage_1	
b1	receiveapproveDamage_1	
c1	sendfetchCar_2	
d1	receivefetchCar_2	
e1	sendvalidateDamage_3	
f1	receivevalidateDamage_3	
g1	sendcancelCar_4	
h1	receivecancelCar_4	
i1	sendfetchDamage_5	
j1	receivefetchDamage_5	
k1	sendqueryInvoice_6	
l1	receivequeryInvoice_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1
```

model_087 DroneTrafficControl · pass

Guard

a1	approved = 1
b1	express = 1
a2	approved = 0

```
(root)
├── a1  approved = 1
│   └── b1  express = 1
└── a2  approved = 0
```

เส้นทาง

```
a1 -> b1
a2
```

Event

รหัส	event	guard
a1	sendreportCorridor_1	
b1	receivereportCorridor_1	
c1	sendsyncFlight_2	
d1	receivesyncFlight_2	
e1	sendackFlight_3	a1
e2	sendcancelCorridor_7	a2
f1	receiveackFlight_3	a1
f2	receivecancelCorridor_7	a2
g1	sendupdateCorridor_4_opt1	a1 b1
h1	receiveupdateCorridor_4_opt1	a1 b1
i1	sendqueryCorridor_5_opt1	a1 b1
j1	receivequeryCorridor_5_opt1	a1 b1
k1	sendconfirmFlight_6	a1
l1	receiveconfirmFlight_6	a1
m1	sendconfirmCorridor_8	
n1	receiveconfirmCorridor_8	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1 → k1 → l1 →
    │     m1 → n1
    └── e2 → f2 → m1 → n1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> m1 -> n1
```

model_095 FlightCheckIn · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendapproveSeat_1	
b1	receiveapproveSeat_1	
c1	sendcancelSeat_2	
d1	receivecancelSeat_2	
e1	sendfetchBoarding_3	
f1	receivefetchBoarding_3	
g1	sendcancelBag_4	
h1	receivecancelBag_4	
i1	sendassignSeat_5	
j1	receiveassignSeat_5	
k1	sendrequestBoarding_6	
l1	receiverequestBoarding_6	
m1	sendackBag_7	
n1	receiveackBag_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1
```

model_096 FlightCrewScheduling · fail

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendcancelDuty_1	
b1	receivecancelDuty_1	
c1	sendackDuty_2	
d1	receiveackDuty_2	
e1	sendassignSwap_3	
f1	receiveassignSwap_3	
g1	sendconfirmSwap_4	
h1	receiveconfirmSwap_4	
i1	sendackSwap_5	
j1	receiveackSwap_5	
k1	sendqueryRoster_6	
l1	receivequeryRoster_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_099 GarageGateControl · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendsendTicket_1	
b1	receivesendTicket_1	
c1	sendassignTicket_2	
d1	receiveassignTicket_2	
e1	sendreserveTicket_3	
f1	receivereserveTicket_3	
g1	sendreserveFee_4	
h1	receivereserveFee_4	
i1	sendrequestSlot_5	
j1	receiverequestSlot_5	
k1	sendassignFee_6	
l1	receiveassignFee_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_137 ParkingPermitPortal · pass

Guard

a1	retryNo < 9

```
(root)
└── a1  retryNo < 9
```

เส้นทาง
a1

Event

รหัส	event	guard
a1	sendfetchPermit_1	
b1	receivefetchPermit_1	
c1	sendapprovePlate_2	
d1	receiveapprovePlate_2	
e1	sendqueryFine_3	a1
f1	receivequeryFine_3	a1
g1	sendconfirmPlate_4	a1
h1	receiveconfirmPlate_4	a1
i1	sendfetchPlate_5	a1
j1	receivefetchPlate_5	a1
k1	sendreservePermit_6	
k2	checkloop_queryFine_3_loop1	a1
l1	receivereservePermit_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1
    ├── k1 → l1
    └── k2 ↺ e1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 ↺ e1

model_150 RailFareGate · pass

Guard

a1	needAck = 1
b1	withExtra = 1

```
(root)
└── a1  needAck = 1
    └── b1  withExtra = 1
```

เส้นทาง

```
a1 -> b1
```

Event

รหัส	event	guard
a1	sendqueryBalance_1	
b1	receivequeryBalance_1	
c1	sendassignBalance_2	
d1	receiveassignBalance_2	
e1	sendcheckFare_3_opt1	a1
f1	receivecheckFare_3_opt1	a1
g1	sendvalidateCard_4_opt1	a1
h1	receivevalidateCard_4_opt1	a1
i1	sendcheckCard_5	
j1	receivecheckCard_5	
k1	sendqueryFare_6	
l1	receivequeryFare_6	
m1	sendvalidateFare_7_opt2	b1
n1	receivevalidateFare_7_opt2	b1
o1	sendrequestCard_8_opt2	b1
p1	receiverequestCard_8_opt2	b1
q1	sendreportBalance_9	
r1	receivereportBalance_9	
s1	sendreleaseFare_10	
t1	receivereleaseFare_10	
u1	sendissueFare_11	
v1	receiveissueFare_11	
w1	sendlogBalance_12	
x1	receivelogBalance_12	
y1	sendvalidateBalance_13	
z1	receivevalidateBalance_13	
aa1	sendassignFare_14	
ab1	receiveassignFare_14	
ac1	sendcancelFare_15	
ad1	receivecancelFare_15	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1 →
      q1 → r1 → s1 → t1 → u1 → v1 → w1 → x1 →
      y1 → z1 → aa1 → ab1 → ac1 → ad1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 ->
v1 -> w1 -> x1 -> y1 -> z1 -> aa1 -> ab1 -> ac1 -> ad1
```

model_154 RentalScooter · pass

Guard

a1	express = 1

```
(root)
└── a1  express = 1
```

เส้นทาง
a1

Event

รหัส	event	guard
a1	sendfetchBattery_1	
b1	receivefetchBattery_1	
c1	sendrequestBattery_2	
d1	receiverequestBattery_2	
e1	sendupdateUnlock_3_opt1	a1
f1	receiveupdateUnlock_3_opt1	a1
g1	sendissueTrip_4	
h1	receiveissueTrip_4	
i1	sendscheduleTrip_5	
j1	receivescheduleTrip_5	
k1	sendfetchTrip_6	
l1	receivefetchTrip_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_156 RideHailingSurge · fail

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendnotifyFare_1	
b1	receivenotifyFare_1	
c1	sendcancelSurge_2	
d1	receivecancelSurge_2	
e1	sendreleaseTrip_3	
f1	receivereleaseTrip_3	
g1	sendrequestTrip_4	
h1	receiverequestTrip_4	
i1	sendlogSurge_5	
j1	receivelogSurge_5	
k1	sendfetchFare_6	
l1	receivefetchFare_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_160 SchoolBusTracking · pass

Guard

a1	t_obs ≤ 2
a2	t_obs > 2

```
(root)
├── a1  t_obs ≤ 2
└── a2  t_obs > 2
```

เส้นทาง

a1
a2

Event

รหัส	event	guard
a1	sendupdateArrival_1	
b1	receiveupdateArrival_1	
c1	sendconfirmStop_2	
d1	receiveconfirmStop_2	
e1	sendreportBus_3	a1
e2	sendsyncBus_5	a2
f1	receivereportBus_3	a1
f2	receivesyncBus_5	a2
g1	sendcancelStop_4	a1
h1	receivecancelStop_4	a1
i1	sendcheckBus_6	
j1	receivecheckBus_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1
    └── e2 → f2 → i1 → j1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> i1 -> j1

model_163 ShipLockOperation · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendqueryLock_1	
b1	receivequeryLock_1	
c1	sendscheduleLock_2	
d1	receivescheduleLock_2	
e1	sendreleaseLevel_3	
f1	receivereleaseLevel_3	
g1	sendapproveLevel_4	
h1	receiveapproveLevel_4	
i1	sendconfirmLock_5	
j1	receiveconfirmLock_5	
k1	sendassignLevel_6	
l1	receiveassignLevel_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_183 StreetParkingMeter · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendreserveFee_1	
b1	receivereserveFee_1	
c1	sendlogFee_2	
d1	receivelogFee_2	
e1	sendreportFee_3	
f1	receivereportFee_3	
g1	sendreserveSession_4	
h1	receivereserveSession_4	
i1	sendqueryFine_5	
j1	receivequeryFine_5	
k1	sendissueSession_6	
l1	receiveissueSession_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_188 TollPlazaETC · pass

Guard

a1	withExtra = 1

```
(root)
└── a1  withExtra = 1
```

เส้นทาง

a1

Event

รหัส	event	guard
a1	sendrequestPlate_1	
b1	receiverequestPlate_1	
c1	sendschedulePlate_2	
d1	receiveschedulePlate_2	
e1	sendvalidatePlate_3_opt1	a1
f1	receivevalidatePlate_3_opt1	a1
g1	sendvalidateToll_4_opt1	a1
h1	receivevalidateToll_4_opt1	a1
i1	sendqueryAccount_5	
j1	receivequeryAccount_5	
k1	sendreservePlate_6	
l1	receivereservePlate_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_190 TrafficSignalPriority · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendqueryPriority_1	
b1	receivequeryPriority_1	
c1	sendassignGreen_2	
d1	receiveassignGreen_2	
e1	sendupdateGreen_3	
f1	receiveupdateGreen_3	
g1	sendqueryLane_4	
h1	receivequeryLane_4	
i1	sendconfirmPriority_5	
j1	receiveconfirmPriority_5	
k1	sendreserveLane_6	
l1	receivereserveLane_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_191 TrainSeatReservation · fail

Guard

a1	passNo < 11
b1	passNo mod 2 = 0
b2	passNo mod 2 ≠ 0

```
(root)
└── a1  passNo < 11
    ├── b1  passNo mod 2 = 0
    └── b2  passNo mod 2 ≠ 0
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

รหัส	event	guard
a1	sendqueryTicket_1	
b1	receivequeryTicket_1	
c1	sendassignCoach_2	
d1	receiveassignCoach_2	
e1	sendscheduleCoach_3	a1
f1	receivescheduleCoach_3	a1
g1	sendconfirmSeat_4	a1 b1
g2	sendapproveTicket_8	a1 b2
h1	receiveconfirmSeat_4	a1 b1
h2	receiveapproveTicket_8	a1 b2
i1	sendreserveTicket_5	a1 b1
j1	receivereserveTicket_5	a1 b1
k1	sendissueTicket_6	a1 b1
l1	receiveissueTicket_6	a1 b1
m1	sendrequestCoach_7	a1 b1
n1	receiverequestCoach_7	a1 b1
o1	sendcheckTicket_9	a1
p1	receivecheckTicket_9	a1
q1	sendcancelTicket_10	a1
r1	receivecancelTicket_10	a1
s1	sendreportCoach_11	a1
t1	receivereportCoach_11	a1
u1	sendissueCoach_12	
u2	checkloop_scheduleCoach_3_loop1	a1
v1	receiveissueCoach_12	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1 → m1 → n1 →
    │     o1 → p1 → q1 → r1 → s1 → t1
    │   ├── u1 → v1
    │   └── u2 ↺ e1
    └── g2 → h2 → o1 → p1 → q1 → r1 → s1 → t1
        ├── u1 → v1
        └── u2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 ->
v1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u2 ↺ e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> o1 -> p1 -> q1
-> r1 -> s1 -> t1 -> u1 -> v1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> o1 -> p1 -> q1
-> r1 -> s1 -> t1 -> u2 ↺ e1
```

โลจิสติกส์และคลังสินค้า · Logistics & Supply Chain (14)

model_011 ColdChainMonitor · fail

Guard

a1	tempBreach = 1
a2	tempBreach = 0

```
(root)
├── a1  tempBreach = 1
└── a2  tempBreach = 0
```

เส้นทาง

a1
a2

Event

รหัส	event	guard
a1	sendlogTemperature_1	
b1	receivelogTemperature_1	
c1	senduploadLog_2	
d1	receiveuploadLog_2	
e1	senduploadAck_3	
f1	receiveuploadAck_3	
g1	sendraiseAlert_4	a1
g2	senddailySummary_6	a2
h1	receiveraiseAlert_4	a1
h2	receivedailySummary_6	a2
i1	sendalertAck_5	a1
j1	receivealertAck_5	a1
k1	sendcloseTrip_7	
l1	receivecloseTrip_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1
    └── g2 → h2 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1

model_014 DroneDelivery · pass

Guard

a1	t_deliver ≤ 20
a2	t_deliver > 20

```
(root)
├── a1  t_deliver ≤ 20
└── a2  t_deliver > 20
```

เส้นทาง

a1
a2

Event

รหัส	event	guard
a1	sendplaceOrder_1	
b1	receiveplaceOrder_1	
c1	sendprepareParcel_2	
d1	receiveprepareParcel_2	
e1	sendassignDrone_3	
f1	receiveassignDrone_3	
g1	sendpickupParcel_4	
h1	receivepickupParcel_4	
i1	senddeliverParcel_5	
j1	receivedeliverParcel_5	
k1	sendconfirmDelivery_6	a1
k2	sendlateCompensation_7	a2
l1	receiveconfirmDelivery_6	a1
l2	receivelateCompensation_7	a2
m1	sendreturnToBase_8	
n1	receivereturnToBase_8	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1
    ├── k1 → l1 → m1 → n1
    └── k2 → l2 → m1 → n1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 -> l2 -> m1 -> n1

model_032 PackageTracking · pass

Guard

a1	customsCheck = 1
b1	delivered = YES
b2	delivered = NO

```
(root)
└── a1  customsCheck = 1
    ├── b1  delivered = YES
    └── b2  delivered = NO
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

รหัส	event	guard
a1	sendcreateShipment_1	
b1	receivecreateShipment_1	
c1	sendpickupConfirm_2	
d1	receivepickupConfirm_2	
e1	sendarriveHub_3	
f1	receivearriveHub_3	
g1	sendscanIn_4	
h1	receivescanIn_4	
i1	sendcustomsHold_5_opt1	a1
j1	receivecustomsHold_5_opt1	a1
k1	senddispatchOut_6	
l1	receivedispatchOut_6	
m1	senddeliver_7	b1
m2	sendleaveNotice_9	b2
n1	receivedeliver_7	b1
n2	receiveleaveNotice_9	b2
o1	sendsignReceipt_8	b1
p1	receivesignReceipt_8	b1
q1	sendcloseShipment_10	
r1	receivecloseShipment_10	
s1	sendrateService_11	
t1	receiverateService_11	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
    ├── m1 → n1 → o1 → p1 → q1 → r1 → s1 → t1
    └── m2 → n2 → q1 → r1 → s1 → t1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m2 -> n2 -> q1 -> r1 -> s1 -> t1
```

model_033 ParcelLocker · fail

Guard

a1	codeValid = 1

```
(root)
└── a1  codeValid = 1
```

เส้นทาง

```
a1
```

Event

รหัส	event	guard
a1	senddepositParcel_1	
b1	receivedepositParcel_1	
c1	sendlockerReceipt_2	
d1	receivelockerReceipt_2	
e1	sendsendPickupCode_3	
f1	receivesendPickupCode_3	
g1	sendenterCode_4	
h1	receiveenterCode_4	
i1	sendopenSlot_5_opt1	a1
j1	receiveopenSlot_5_opt1	a1
k1	sendslotClosed_6_opt1	a1
l1	receiveslotClosed_6_opt1	a1
m1	sendnotifyCourier_7	
n1	receivenotifyCourier_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1
```

model_048 WarehouseRobotFleet · pass

Guard

a1	shift < 8
b1	lowBattery = 1

```
(root)
└── a1  shift < 8
    └── b1  lowBattery = 1
```

เส้นทาง

```
a1 -> b1
```

Event

รหัส	event	guard
a1	sendstartShift_1	
b1	receivestartShift_1	
c1	sendassignTaskA_2	a1
d1	receiveassignTaskA_2	a1
e1	sendassignTaskB_3	a1
f1	receiveassignTaskB_3	a1
g1	sendtaskDone_4	a1
h1	receivetaskDone_4	a1
i1	sendreserveDock_5_opt1	a1 b1
j1	receivereserveDock_5_opt1	a1 b1
k1	senddockReady_6_opt1	a1 b1
l1	receivedockReady_6_opt1	a1 b1
m1	sendendShift_7	
m2	checkloop_assignTaskA_2_loop1	a1
n1	receiveendShift_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
    ├── m1 → n1
    └── m2 ↺ c1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m2 ↺ c1
```

model_067 CargoShipTracking · fail

Guard

a1	cycleNo < 7
b1	useCache = 1

```
(root)
└── a1  cycleNo < 7
    └── b1  useCache = 1
```

เส้นทาง

```
a1 -> b1
```

Event

รหัส	event	guard
a1	sendsyncManifest_1	
b1	receivesyncManifest_1	
c1	sendassignContainer_2	
d1	receiveassignContainer_2	
e1	sendackContainer_3	a1
f1	receiveackContainer_3	a1
g1	sendupdateManifest_4_opt1	a1 b1
h1	receiveupdateManifest_4_opt1	a1 b1
i1	sendupdateBerth_5	a1
j1	receiveupdateBerth_5	a1
k1	sendreleaseContainer_6	
k2	checkloop_ackContainer_3_loop1	a1
l1	receivereleaseContainer_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1
    ├── k1 → l1
    └── k2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 ↺
e1
```

model_081 CustomsClearance · pass

Guard

a1	useCache = 1

```
(root)
└── a1  useCache = 1
```

เส้นทาง
a1

Event

รหัส	event	guard
a1	sendsendRelease_1	
b1	receivesendRelease_1	
c1	sendrequestDuty_2	
d1	receiverequestDuty_2	
e1	sendcancelDuty_3_opt1	a1
f1	receivecancelDuty_3_opt1	a1
g1	sendapproveDeclaration_4	
h1	receiveapproveDeclaration_4	
i1	sendsyncDeclaration_5	
j1	receivesyncDeclaration_5	
k1	sendfetchRelease_6	
l1	receivefetchRelease_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_114 InventoryReorder · fail

Guard

a1	retryNo < 12
b1	retryNo mod 3 = 0
c1	express = 1

```
(root)
└── a1  retryNo < 12
    └── b1  retryNo mod 3 = 0
        └── c1  express = 1
```

เส้นทาง
a1 -> b1 -> c1

Event

รหัส	event	guard
a1	sendcheckInvoice_1	
b1	receivecheckInvoice_1	
c1	sendreportInvoice_2	
d1	receivereportInvoice_2	
e1	sendsendInvoice_3	a1
f1	receivesendInvoice_3	a1
g1	sendackInvoice_4_opt1	a1 b1
h1	receiveackInvoice_4_opt1	a1 b1
i1	sendlogStock_5_opt1	a1 b1
j1	receivelogStock_5_opt1	a1 b1
k1	sendrequestInvoice_6	a1
l1	receiverequestInvoice_6	a1
m1	sendscheduleStock_7_opt2	a1 c1
n1	receivescheduleStock_7_opt2	a1 c1
o1	sendscheduleOrder_8_opt2	a1 c1
p1	receivescheduleOrder_8_opt2	a1 c1
q1	sendupdateInvoice_9	a1
r1	receiveupdateInvoice_9	a1
s1	sendissueOrder_10	a1
t1	receiveissueOrder_10	a1
u1	sendfetchOrder_11	
u2	checkloop_sendInvoice_3_loop1	a1
v1	receivefetchOrder_11	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1 →
      q1 → r1 → s1 → t1
    ├── u1 → v1
    └── u2 ↺ e1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u2 ↺ e1

model_122 MailroomSorting · pass

Guard

a1	withExtra = 1
b1	pollNo < 5

```
(root)
└── a1  withExtra = 1
    └── b1  pollNo < 5
```

เส้นทาง

```
a1 -> b1
```

Event

รหัส	event	guard
a1	sendassignSlot_1	
b1	receiveassignSlot_1	
c1	sendackNotice_2	
d1	receiveackNotice_2	
e1	sendcancelNotice_3	
f1	receivecancelNotice_3	
g1	sendissueNotice_4	
h1	receiveissueNotice_4	
i1	sendfetchSlot_5	
j1	receivefetchSlot_5	
k1	sendcheckParcel_6	
l1	receivecheckParcel_6	
m1	sendapproveNotice_7_opt1	a1
n1	receiveapproveNotice_7_opt1	a1
o1	sendlogParcel_8_opt1	a1 b1
p1	receivelogParcel_8_opt1	a1 b1
q1	sendcancelParcel_9_opt1	a1 b1
r1	receivecancelParcel_9_opt1	a1 b1
s1	sendreserveParcel_10_opt1	a1 b1
t1	receivereserveParcel_10_opt1	a1 b1
u1	sendreportParcel_11_opt1	a1
u2	checkloop_logParcel_8_loop1	a1 b1
v1	receivereportParcel_11_opt1	a1
w1	sendvalidateSlot_12	
x1	receivevalidateSlot_12	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1 →
      q1 → r1 → s1 → t1
    ├── u1 → v1 → w1 → x1
    └── u2 ↺ o1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 ->
v1 -> w1 -> x1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u2 ↺ o1
```

model_135 ParcelReturnDropoff · fail

Guard

a1	useCache = 1

```
(root)
└── a1  useCache = 1
```

เส้นทาง
a1

Event

รหัส	event	guard
a1	sendsyncReturn_1	
b1	receivesyncReturn_1	
c1	sendsendRefund_2	
d1	receivesendRefund_2	
e1	sendnotifyReturn_3_opt1	a1
f1	receivenotifyReturn_3_opt1	a1
g1	sendscheduleReturn_4	
h1	receivescheduleReturn_4	
i1	sendackReturn_5	
j1	receiveackReturn_5	
k1	sendreportReturn_6	
l1	receivereportReturn_6	
m1	sendlogRefund_7	
n1	receivelogRefund_7	
o1	sendapproveRefund_8	
p1	receiveapproveRefund_8	
q1	sendassignLabel_9	
r1	receiveassignLabel_9	
s1	sendvalidateRefund_10	
t1	receivevalidateRefund_10	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1 →
      q1 → r1 → s1 → t1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1

model_146 PortCraneControl · pass

Guard

a1	express = 1

```
(root)
└── a1  express = 1
```

เส้นทาง
a1

Event

รหัส	event	guard
a1	sendreportContainer_1	
b1	receivereportContainer_1	
c1	sendcheckSlot_2	
d1	receivecheckSlot_2	
e1	sendrequestLift_3_opt1	a1
f1	receiverequestLift_3_opt1	a1
g1	sendackContainer_4_opt1	a1
h1	receiveackContainer_4_opt1	a1
i1	sendassignLift_5	
j1	receiveassignLift_5	
k1	sendupdateSlot_6	
l1	receiveupdateSlot_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_161 SeaportPilotBooking · pass

Guard

a1	needAck = 1

```
(root)
└── a1  needAck = 1
```

เส้นทาง
a1

Event

รหัส	event	guard
a1	sendnotifyArrival_1	
b1	receivenotifyArrival_1	
c1	sendreserveArrival_2	
d1	receivereserveArrival_2	
e1	sendsendBerth_3_opt1	a1
f1	receivesendBerth_3_opt1	a1
g1	sendlogPilot_4	
h1	receivelogPilot_4	
i1	sendrequestBerth_5	
j1	receiverequestBerth_5	
k1	sendcheckArrival_6	
l1	receivecheckArrival_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_195 WarehouseForklift · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendscheduleAisle_1	
b1	receivescheduleAisle_1	
c1	sendcancelAisle_2	
d1	receivecancelAisle_2	
e1	sendvalidatePallet_3	
f1	receivevalidatePallet_3	
g1	sendapprovePallet_4	
h1	receiveapprovePallet_4	
i1	sendfetchAisle_5	
j1	receivefetchAisle_5	
k1	sendapproveAisle_6	
l1	receiveapproveAisle_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_196 WarehousePicking · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendreportTote_1	
b1	receivereportTote_1	
c1	sendsendOrder_2	
d1	receivesendOrder_2	
e1	sendlogTote_3	
f1	receivelogTote_3	
g1	sendlogBin_4	
h1	receivelogBin_4	
i1	sendassignBin_5	
j1	receiveassignBin_5	
k1	sendreserveTote_6	
l1	receivereserveTote_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

บ้านและอาคารอัจฉริยะ · Smart Home & Building (21)

model_015 ElevatorControl · pass

Guard

a1	doorState = CLEAR
a2	doorState = BLOCKED
b1	retryDoor < 3

```
(root)
├── a1  doorState = CLEAR
└── a2  doorState = BLOCKED
    └── b1  retryDoor < 3
```

เส้นทาง

a1
a2 -> b1

Event

รหัส	event	guard
a1	sendcallElevator_1	
b1	receivecallElevator_1	
c1	sendmoveCar_2	
d1	receivemoveCar_2	
e1	sendarrived_3	
f1	receivearrived_3	
g1	sendopenDoor_4	a1
g2	sendcloseDoor_6	a2 b1
h1	receiveopenDoor_4	a1
h2	receivecloseDoor_6	a2 b1
i1	senddoorOpened_5	a1
i2	senddoorSensor_7	a2 b1
j1	receivedoorOpened_5	a1
j2	receivedoorSensor_7	a2 b1
k1	sendnotifyPassenger_8	
k2	checkloop_closeDoor_6_loop1	a2 b1
l1	receivenotifyPassenger_8	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1
    └── g2 → h2 → i2 → j2
        ├── k1 → l1
        └── k2 ↺ g2
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> i2 -> j2 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> i2 -> j2 -> k2 ↺ g2

model_037 SmartDoorLock · pass

Guard

a1	authResult = PASS
b1	autoLock = 1
a2	authResult = FAIL

```
(root)
├── a1  authResult = PASS
│   └── b1  autoLock = 1
└── a2  authResult = FAIL
```

เส้นทาง

a1 -> b1
a2

Event

รหัส	event	guard
a1	sendunlockRequest_1	
b1	receiveunlockRequest_1	
c1	sendchallenge_2	
d1	receivechallenge_2	
e1	sendrespond_3	
f1	receiverespond_3	
g1	senddoorUnlocked_4	a1
g2	sendaccessDenied_6	a2
h1	receivedoorUnlocked_4	a1
h2	receiveaccessDenied_6	a2
i1	sendscheduleRelock_5_opt1	a1 b1
j1	receivescheduleRelock_5_opt1	a1 b1
k1	sendstatusUpdate_7	
l1	receivestatusUpdate_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1
    └── g2 → h2 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1

model_041 SmartPlugControl · fail

Guard

a1	relayOn = 1
a2	relayOn = 0

```
(root)
├── a1  relayOn = 1
└── a2  relayOn = 0
```

เส้นทาง

a1
a2

Event

รหัส	event	guard
a1	sendqueryState_1	
b1	receivequeryState_1	
c1	sendplugState_2	
d1	receiveplugState_2	
e1	sendturnOff_3	a1
e2	sendturnOn_5	a2
f1	receiveturnOff_3	a1
f2	receiveturnOn_5	a2
g1	sendoffAck_4	a1
h1	receiveoffAck_4	a1
i1	sendenergyReport_6	
j1	receiveenergyReport_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1
    └── e2 → f2 → i1 → j1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> i1 -> j1

model_043 ThermostatControl · pass

Guard

a1	cycle < 5
b1	overheat = 1
b2	overheat = 0

```
(root)
└── a1  cycle < 5
    ├── b1  overheat = 1
    └── b2  overheat = 0
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

รหัส	event	guard
a1	sendstartHeating_1	
b1	receivestartHeating_1	
c1	sendheatingAck_2	
d1	receiveheatingAck_2	
e1	sendqueryTemp_3	a1
f1	receivequeryTemp_3	a1
g1	sendtempReport_4	a1
h1	receivetempReport_4	a1
i1	sendreducePower_5	a1 b1
i2	sendkeepPower_6	a1 b2
j1	receivereducePower_5	a1 b1
j2	receivekeepPower_6	a1 b2
k1	sendstopHeating_7	
k2	checkloop_queryTemp_3_loop1	a1
l1	receivestopHeating_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1
    ├── i1 → j1
    │   ├── k1 → l1
    │   └── k2 ↺ e1
    └── i2 → j2
        ├── k1 → l1
        └── k2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 ↺
e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i2 -> j2 -> k1
-> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i2 -> j2 -> k2 ↺
e1
```

model_047 VoiceAssistant · pass

Guard

a1	intent = MUSIC
a2	intent = WEATHER
a3	intent = UNKNOWN

```
(root)
├── a1  intent = MUSIC
├── a2  intent = WEATHER
└── a3  intent = UNKNOWN
```

เส้นทาง

a1
a2
a3

Event

รหัส	event	guard
a1	sendwakeWord_1	
b1	receivewakeWord_1	
c1	sendstreamAudio_2	
d1	receivestreamAudio_2	
e1	sendintentResult_3	
f1	receiveintentResult_3	
g1	sendplaySong_4	a1
g2	sendfetchWeather_5	a2
g3	sendaskRepeat_8	a3
h1	receiveplaySong_4	a1
h2	receivefetchWeather_5	a2
h3	receiveaskRepeat_8	a3
i1	sendsleepMode_9	
j1	receivesleepMode_9	
k1	sendweatherInfo_6	a2
l1	receiveweatherInfo_6	a2
m1	sendspeakWeather_7	a2
n1	receivespeakWeather_7	a2

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1
    ├── g2 → h2 → k1 → l1 → m1 → n1 → i1 → j1
    └── g3 → h3 → i1 → j1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1 -> m1 -> n1 -> i1 -> j1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g3 -> h3 -> i1 -> j1

model_053 ApartmentIntercom · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendsyncCall_1	
b1	receivesyncCall_1	
c1	sendreserveCall_2	
d1	receivereserveCall_2	
e1	sendreserveDoor_3	
f1	receivereserveDoor_3	
g1	sendcheckDoor_4	
h1	receivecheckDoor_4	
i1	sendqueryDoor_5	
j1	receivequeryDoor_5	
k1	sendcancelCall_6	
l1	receivecancelCall_6	
m1	sendlogCode_7	
n1	receivelogCode_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1

model_060 BuildingVisitorPass · pass

Guard

a1	t_obs ≤ 4
a2	t_obs > 4

```
(root)
├── a1  t_obs ≤ 4
└── a2  t_obs > 4
```

เส้นทาง

a1
a2

Event

รหัส	event	guard
a1	sendsendBadge_1	
b1	receivesendBadge_1	
c1	sendcheckBadge_2	
d1	receivecheckBadge_2	
e1	sendqueryVisit_3	a1
e2	sendnotifyPass_5	a2
f1	receivequeryVisit_3	a1
f2	receivenotifyPass_5	a2
g1	sendreleasePass_4	a1
g2	sendvalidateVisit_6	a2
h1	receivereleasePass_4	a1
h2	receivevalidateVisit_6	a2
i1	sendsendPass_7	
j1	receivesendPass_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1
    └── e2 → f2 → g2 → h2 → i1 → j1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g2 -> h2 -> i1 -> j1

model_086 DormLaundry · pass

Guard

a1	cycleNo < 3

```
(root)
└── a1  cycleNo < 3
```

เส้นทาง

a1

Event

รหัส	event	guard
a1	sendackCredit_1	
b1	receiveackCredit_1	
c1	sendconfirmCredit_2	
d1	receiveconfirmCredit_2	
e1	sendreserveMachine_3	a1
f1	receivereserveMachine_3	a1
g1	sendnotifyCredit_4	a1
h1	receivenotifyCredit_4	a1
i1	sendreleaseCycle_5	
i2	checkloop_reserveMachine_3_loop1	a1
j1	receivereleaseCycle_5	
k1	sendfetchCredit_6	
l1	receivefetchCredit_6	
m1	sendsendCredit_7	
n1	receivesendCredit_7	
o1	sendfetchCycle_8	
p1	receivefetchCycle_8	
q1	sendcancelCycle_9	
r1	receivecancelCycle_9	
s1	sendqueryCredit_10	
t1	receivequeryCredit_10	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1
    ├── i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1 →
    │     q1 → r1 → s1 → t1
    └── i2 ↺ e1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i2 ↺ e1

model_093 FireSprinklerSystem · pass

Guard

a1	optIn = 1

```
(root)
└── a1  optIn = 1
```

เส้นทาง
a1

Event

รหัส	event	guard
a1	sendcheckValve_1	
b1	receivecheckValve_1	
c1	sendlogValve_2	
d1	receivelogValve_2	
e1	sendsendAlarm_3_opt1	a1
f1	receivesendAlarm_3_opt1	a1
g1	sendupdateFire_4_opt1	a1
h1	receiveupdateFire_4_opt1	a1
i1	sendnotifyFire_5	
j1	receivenotifyFire_5	
k1	sendconfirmFire_6	
l1	receiveconfirmFire_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_107 HomeSecurityAlarm · fail

Guard

a1	needAck = 1

```
(root)
└── a1  needAck = 1
```

เส้นทาง
a1

Event

รหัส	event	guard
a1	sendlogCode_1	
b1	receivelogCode_1	
c1	sendreportAlert_2	
d1	receivereportAlert_2	
e1	sendcheckAlert_3_opt1	a1
f1	receivecheckAlert_3_opt1	a1
g1	sendnotifyZone_4_opt1	a1
h1	receivenotifyZone_4_opt1	a1
i1	sendsyncZone_5	
j1	receivesyncZone_5	
k1	sendqueryCode_6	
l1	receivequeryCode_6	
m1	sendassignAlert_7	
n1	receiveassignAlert_7	
o1	sendqueryZone_8	
p1	receivequeryZone_8	
q1	sendscheduleZone_9	
r1	receivescheduleZone_9	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1 →
      q1 → r1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1

model_115 IoTDoorbell · pass

Guard

a1	useCache = 1
b1	needAck = 1

```
(root)
└── a1  useCache = 1
    └── b1  needAck = 1
```

เส้นทาง

```
a1 -> b1
```

Event

รหัส	event	guard
a1	sendapproveMessage_1	
b1	receiveapproveMessage_1	
c1	sendackVideo_2	
d1	receiveackVideo_2	
e1	sendassignVideo_3_opt1	a1
f1	receiveassignVideo_3_opt1	a1
g1	sendnotifyRing_4	
h1	receivenotifyRing_4	
i1	sendreserveMessage_5	
j1	receivereserveMessage_5	
k1	sendrequestRing_6_opt2	b1
l1	receiverequestRing_6_opt2	b1
m1	sendcheckVideo_7	
n1	receivecheckVideo_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1
```

model_119 LiftEmergencyCall · fail

Guard

a1	needAck = 1
b1	t_obs ≤ 23
b2	t_obs > 23

```
(root)
└── a1  needAck = 1
    ├── b1  t_obs ≤ 23
    └── b2  t_obs > 23
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

รหัส	event	guard
a1	sendreleaseCall_1	
b1	receivereleaseCall_1	
c1	sendqueryCall_2	
d1	receivequeryCall_2	
e1	sendsyncFault_3_opt1	a1
f1	receivesyncFault_3_opt1	a1
g1	sendlogCall_4	
h1	receivelogCall_4	
i1	sendnotifyFault_5	
j1	receivenotifyFault_5	
k1	sendreportFault_6	b1
k2	sendcancelFault_7	b2
l1	receivereportFault_6	b1
l2	receivecancelFault_7	b2
m1	sendsendFault_8	
n1	receivesendFault_8	
o1	sendsyncCall_9	
p1	receivesyncCall_9	
q1	sendackFault_10	
r1	receiveackFault_10	
s1	sendnotifyDispatch_11	
t1	receivenotifyDispatch_11	
u1	sendconfirmDispatch_12	
v1	receiveconfirmDispatch_12	
w1	sendvalidateFault_13	
x1	receivevalidateFault_13	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1
    ├── k1 → l1 → m1 → n1 → o1 → p1 → q1 → r1 →
    │     s1 → t1 → u1 → v1 → w1 → x1
    └── k2 → l2 → m1 → n1 → o1 → p1 → q1 → r1 →
          s1 → t1 → u1 → v1 → w1 → x1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 ->
v1 -> w1 -> x1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2
-> l2 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 ->
v1 -> w1 -> x1
```

model_120 LiftMaintenance · fail

Guard

a1	attemptNo < 5

```
(root)
└── a1  attemptNo < 5
```

เส้นทาง
a1

Event

รหัส	event	guard
a1	sendlogJob_1	
b1	receivelogJob_1	
c1	sendvalidateJob_2	
d1	receivevalidateJob_2	
e1	sendrequestJob_3	a1
f1	receiverequestJob_3	a1
g1	sendcheckJob_4	a1
h1	receivecheckJob_4	a1
i1	sendsyncLock_5	a1
j1	receivesyncLock_5	a1
k1	sendcancelFault_6	
k2	checkloop_requestJob_3_loop1	a1
l1	receivecancelFault_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1
    ├── k1 → l1
    └── k2 ↺ e1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 ↺ e1

model_130 OfficeLightingControl · fail

Guard

a1	needAck = 1
b1	express = 1

```
(root)
└── a1  needAck = 1
    └── b1  express = 1
```

เส้นทาง
a1 -> b1

Event

รหัส	event	guard
a1	sendlogDim_1	
b1	receivelogDim_1	
c1	sendsyncSchedule_2	
d1	receivesyncSchedule_2	
e1	sendreleaseOccupancy_3_opt1	a1
f1	receivereleaseOccupancy_3_opt1	a1
g1	sendapproveOccupancy_4	
h1	receiveapproveOccupancy_4	
i1	sendlogSchedule_5	
j1	receivelogSchedule_5	
k1	sendqueryDim_6_opt2	b1
l1	receivequeryDim_6_opt2	b1
m1	sendfetchSchedule_7	
n1	receivefetchSchedule_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1

model_140 PetDoorController · pass

Guard

a1	t_obs ≤ 58
a2	t_obs > 58

```
(root)
├── a1  t_obs ≤ 58
└── a2  t_obs > 58
```

เส้นทาง

a1
a2

Event

รหัส	event	guard
a1	sendrequestLog_1	
b1	receiverequestLog_1	
c1	sendscheduleEntry_2	
d1	receivescheduleEntry_2	
e1	sendcheckLog_3	
f1	receivecheckLog_3	
g1	sendcancelCurfew_4	
h1	receivecancelCurfew_4	
i1	sendreportCurfew_5	
j1	receivereportCurfew_5	
k1	sendqueryEntry_6	
l1	receivequeryEntry_6	
m1	sendrequestCurfew_7	
n1	receiverequestCurfew_7	
o1	sendapproveLog_8	a1
o2	sendupdateEntry_10	a2
p1	receiveapproveLog_8	a1
p2	receiveupdateEntry_10	a2
q1	sendsendCurfew_9	a1
q2	sendsyncCurfew_11	a2
r1	receivesendCurfew_9	a1
r2	receivesyncCurfew_11	a2
s1	sendsendEntry_12	
t1	receivesendEntry_12	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
    ├── o1 → p1 → q1 → r1 → s1 → t1
    └── o2 → p2 → q2 → r2 → s1 → t1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o2 -> p2 -> q2 -> r2 -> s1 -> t1
```

model_157 RoboticLawnMower · fail

Guard

a1	useCache = 1

```
(root)
└── a1  useCache = 1
```

เส้นทาง

a1

Event

รหัส	event	guard
a1	sendsendDock_1	
b1	receivesendDock_1	
c1	sendrequestBlade_2	
d1	receiverequestBlade_2	
e1	sendcancelBlade_3_opt1	a1
f1	receivecancelBlade_3_opt1	a1
g1	sendapproveZone_4	
h1	receiveapproveZone_4	
i1	sendsyncZone_5	
j1	receivesyncZone_5	
k1	sendfetchDock_6	
l1	receivefetchDock_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1
```

model_164 SmartAquarium · fail

Guard

a1	cycleNo < 4
b1	cycleNo mod 2 = 0
b2	cycleNo mod 2 ≠ 0

```
(root)
└── a1  cycleNo < 4
    ├── b1  cycleNo mod 2 = 0
    └── b2  cycleNo mod 2 ≠ 0
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

รหัส	event	guard
a1	sendvalidatePh_1	
b1	receivevalidatePh_1	
c1	sendupdateFeed_2	
d1	receiveupdateFeed_2	
e1	sendreportFeed_3	a1
f1	receivereportFeed_3	a1
g1	sendapproveLight_4	a1 b1
g2	sendnotifyFeed_6	a1 b2
h1	receiveapproveLight_4	a1 b1
h2	receivenotifyFeed_6	a1 b2
i1	sendqueryFeed_5	a1 b1
j1	receivequeryFeed_5	a1 b1
k1	sendsendLight_7	a1
l1	receivesendLight_7	a1
m1	sendsyncLight_8	
m2	checkloop_reportFeed_3_loop1	a1
n1	receivesyncLight_8	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1
    │   ├── m1 → n1
    │   └── m2 ↺ e1
    └── g2 → h2 → k1 → l1
        ├── m1 → n1
        └── m2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m2 ↺ e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1 -> m1
-> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1 -> m2 ↺
e1
```

model_165 SmartBabyMonitor · pass

Guard

a1	useCache = 1
b1	score < 50
b2	score ≥ 50

```
(root)
└── a1  useCache = 1
    ├── b1  score < 50
    └── b2  score ≥ 50
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

รหัส	event	guard
a1	sendqueryClimate_1	
b1	receivequeryClimate_1	
c1	sendconfirmCry_2	
d1	receiveconfirmCry_2	
e1	sendreserveCry_3_opt1	a1
f1	receivereserveCry_3_opt1	a1
g1	sendcheckAlert_4_opt1	a1 b1
g2	sendscheduleAlert_9_opt1	a1 b2
h1	receivecheckAlert_4_opt1	a1 b1
h2	receivescheduleAlert_9_opt1	a1 b2
i1	sendapproveCry_5_opt1	a1 b1
j1	receiveapproveCry_5_opt1	a1 b1
k1	sendreserveAlert_6_opt1	a1 b1
l1	receivereserveAlert_6_opt1	a1 b1
m1	sendvalidateAlert_7_opt1	a1 b1
n1	receivevalidateAlert_7_opt1	a1 b1
o1	sendupdateClimate_8_opt1	a1 b1
p1	receiveupdateClimate_8_opt1	a1 b1
q1	sendsyncCry_10_opt1	a1
r1	receivesyncCry_10_opt1	a1
s1	sendcancelCry_11	
t1	receivecancelCry_11	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1 → m1 → n1 →
    │     o1 → p1 → q1 → r1 → s1 → t1
    └── g2 → h2 → q1 → r1 → s1 → t1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> q1 -> r1 -> s1
-> t1
```

model_167 SmartCurtainControl · fail

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendapprovePosition_1	
b1	receiveapprovePosition_1	
c1	sendupdateScene_2	
d1	receiveupdateScene_2	
e1	sendconfirmLux_3	
f1	receiveconfirmLux_3	
g1	sendfetchScene_4	
h1	receivefetchScene_4	
i1	sendreleaseLux_5	
j1	receivereleaseLux_5	
k1	sendcheckScene_6	
l1	receivecheckScene_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1
```

model_168 SmartFridgeInventory · fail

Guard

a1	attemptNo < 2
b1	tier = GREEN
b2	tier = RED
c1	stepNo < 11

```
(root)
└── a1  attemptNo < 2
    ├── b1  tier = GREEN
    │   └── c1  stepNo < 11
    └── b2  tier = RED
        └── c1  stepNo < 11
```

เส้นทาง

```
a1 -> b1 -> c1
a1 -> b2 -> c1
```

Event

รหัส	event	guard
a1	sendapproveOrder_1	
b1	receiveapproveOrder_1	
c1	sendassignExpiry_2	
d1	receiveassignExpiry_2	
e1	sendsendOrder_3	a1
f1	receivesendOrder_3	a1
g1	sendassignItem_4	a1 b1
g2	sendlogOrder_6	a1 b2
h1	receiveassignItem_4	a1 b1
h2	receivelogOrder_6	a1 b2
i1	sendscheduleOrder_5	a1 b1
j1	receivescheduleOrder_5	a1 b1
k1	sendapproveExpiry_7	a1
l1	receiveapproveExpiry_7	a1
m1	sendqueryItem_8	
m2	checkloop_sendOrder_3_loop1	a1
n1	receivequeryItem_8	
o1	sendcheckOrder_9	
p1	receivecheckOrder_9	
q1	sendissueExpiry_10	c1
r1	receiveissueExpiry_10	c1
s1	sendsyncOrder_11	
s2	checkloop_issueExpiry_10_loop2	c1
t1	receivesyncOrder_11	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1
    │   ├── m1 → n1 → o1 → p1 → q1 → r1
    │   │   ├── s1 → t1
    │   │   └── s2 ↺ q1
    │   └── m2 ↺ e1
    └── g2 → h2 → k1 → l1
        ├── m1 → n1 → o1 → p1 → q1 → r1
        │   ├── s1 → t1
        │   └── s2 ↺ q1
        └── m2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s2 ↺ q1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m2 ↺ e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1 -> m1
-> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1 -> m1
-> n1 -> o1 -> p1 -> q1 -> r1 -> s2 ↺ q1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1 -> m2 ↺
e1
```

model_197 WaterLeakDetection · fail

Guard

a1	cycleNo < 10
b1	cycleNo mod 2 = 0

```
(root)
└── a1  cycleNo < 10
    └── b1  cycleNo mod 2 = 0
```

เส้นทาง

```
a1 -> b1
```

Event

รหัส	event	guard
a1	sendfetchTicket_1	
b1	receivefetchTicket_1	
c1	sendupdateLeak_2	
d1	receiveupdateLeak_2	
e1	sendreserveValve_3	a1
f1	receivereserveValve_3	a1
g1	sendreleaseLeak_4_opt1	a1 b1
h1	receivereleaseLeak_4_opt1	a1 b1
i1	sendreleaseTicket_5	a1
j1	receivereleaseTicket_5	a1
k1	sendreportValve_6	a1
l1	receivereportValve_6	a1
m1	sendsyncTicket_7	a1
n1	receivesyncTicket_7	a1
o1	sendissueTicket_8	a1
p1	receiveissueTicket_8	a1
q1	sendcancelLeak_9	a1
r1	receivecancelLeak_9	a1
s1	sendackLeak_10	a1
t1	receiveackLeak_10	a1
u1	sendreportLeak_11	
u2	checkloop_reserveValve_3_loop1	a1
v1	receivereportLeak_11	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1 →
      q1 → r1 → s1 → t1
    ├── u1 → v1
    └── u2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 ->
v1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u2 ↺ e1
```

อุตสาหกรรม พลังงาน และโครงสร้างพื้นฐาน · Industrial, Energy & Infrastructure (13)

model_005 BatteryChargeCycle · fail

Guard

a1	pulseNo < 8

```
(root)
└── a1  pulseNo < 8
```

เส้นทาง
a1

Event

รหัส	event	guard
a1	sendstartCharge_1	
b1	receivestartCharge_1	
c1	sendchargeAck_2	
d1	receivechargeAck_2	
e1	sendchargePulse_3	a1
f1	receivechargePulse_3	a1
g1	sendcellVoltage_4	a1
h1	receivecellVoltage_4	a1
i1	sendstopCharge_5	
i2	checkloop_chargePulse_3_loop1	a1
j1	receivestopCharge_5	
k1	sendfullReport_6	
l1	receivefullReport_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1
    ├── i1 → j1 → k1 → l1
    └── i2 ↺ e1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i2 ↺ e1

model_018 FactoryRobotArm · pass

Guard

a1	pickCount < 10
b1	defect = 1

```
(root)
└── a1  pickCount < 10
    └── b1  defect = 1
```

เส้นทาง
a1 -> b1

Event

รหัส	event	guard
a1	sendstartBatch_1	
b1	receivestartBatch_1	
c1	sendinitArm_2	
d1	receiveinitArm_2	
e1	sendcalibrateCamera_3	
f1	receivecalibrateCamera_3	
g1	sendlogStart_4	
h1	receivelogStart_4	
i1	sendpickItem_5	a1
j1	receivepickItem_5	a1
k1	sendinspectItem_6	a1
l1	receiveinspectItem_6	a1
m1	sendinspectionResult_7	a1
n1	receiveinspectionResult_7	a1
o1	senddiscardItem_8_opt1	a1 b1
p1	receivediscardItem_8_opt1	a1 b1
q1	sendreportSummary_9	
q2	checkloop_pickItem_5_loop1	a1
r1	receivereportSummary_9	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1
    ├── q1 → r1
    └── q2 ↺ i1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q2 ↺ i1

model_039 SmartMeterBilling · fail

Guard

a1	autoPay = 1

```
(root)
└── a1  autoPay = 1
```

เส้นทาง
a1

Event

รหัส	event	guard
a1	sendsendUsage_1	
b1	receivesendUsage_1	
c1	sendusageAck_2	
d1	receiveusageAck_2	
e1	sendgenerateBill_3	
f1	receivegenerateBill_3	
g1	sendpayBill_4	
h1	receivepayBill_4	
i1	sendsetAutoPay_5_opt1	a1
j1	receivesetAutoPay_5_opt1	a1
k1	sendautoPayOK_6_opt1	a1
l1	receiveautoPayOK_6_opt1	a1
m1	sendreceiptIssued_7	
n1	receivereceiptIssued_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1

model_079 CraneLoadMonitor · fail

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendcancelLimit_1	
b1	receivecancelLimit_1	
c1	sendvalidateAlarm_2	
d1	receivevalidateAlarm_2	
e1	sendackLimit_3	
f1	receiveackLimit_3	
g1	sendissueLimit_4	
h1	receiveissueLimit_4	
i1	sendcancelLoad_5	
j1	receivecancelLoad_5	
k1	sendscheduleLoad_6	
l1	receivescheduleLoad_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_088 EVBatterySwap · pass

Guard

a1	useCache = 1
b1	optIn = 1
c1	retryNo < 11

```
(root)
└── a1  useCache = 1
    └── b1  optIn = 1
        └── c1  retryNo < 11
```

เส้นทาง

```
a1 -> b1 -> c1
```

Event

รหัส	event	guard
a1	sendreportCredit_1	
b1	receivereportCredit_1	
c1	sendissueCredit_2	
d1	receiveissueCredit_2	
e1	sendreserveSwap_3_opt1	a1
f1	receivereserveSwap_3_opt1	a1
g1	sendnotifyCredit_4	
h1	receivenotifyCredit_4	
i1	sendreportBattery_5	
j1	receivereportBattery_5	
k1	sendissueSwap_6_opt2	b1
l1	receiveissueSwap_6_opt2	b1
m1	sendcheckCredit_7	
n1	receivecheckCredit_7	
o1	sendscheduleSwap_8	
p1	receivescheduleSwap_8	
q1	sendlogBattery_9	c1
r1	receivelogBattery_9	c1
s1	sendfetchSwap_10	c1
t1	receivefetchSwap_10	c1
u1	sendconfirmBattery_11	
u2	checkloop_logBattery_9_loop1	c1
v1	receiveconfirmBattery_11	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1 →
      q1 → r1 → s1 → t1
    ├── u1 → v1
    └── u2 ↺ q1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 ->
v1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u2 ↺ q1
```

model_089 EVChargingSession · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendcancelTariff_1	
b1	receivecancelTariff_1	
c1	sendnotifySession_2	
d1	receivenotifySession_2	
e1	sendupdateTariff_3	
f1	receiveupdateTariff_3	
g1	sendrequestEnergy_4	
h1	receiverequestEnergy_4	
i1	sendvalidateSession_5	
j1	receivevalidateSession_5	
k1	sendackSession_6	
l1	receiveackSession_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1
```

model_091 FactoryQualityGate · pass

Guard

a1	roundNo < 7

```
(root)
└── a1  roundNo < 7
```

เส้นทาง
a1

Event

รหัส	event	guard
a1	sendcancelPart_1	
b1	receivecancelPart_1	
c1	sendfetchDefect_2	
d1	receivefetchDefect_2	
e1	sendlogDefect_3	a1
f1	receivelogDefect_3	a1
g1	sendnotifyPart_4	a1
h1	receivenotifyPart_4	a1
i1	sendreportPart_5	
i2	checkloop_logDefect_3_loop1	a1
j1	receivereportPart_5	
k1	sendreleaseBatch_6	
l1	receivereleaseBatch_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1
    ├── i1 → j1 → k1 → l1
    └── i2 ↺ e1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i2 ↺ e1

model_105 HomeEnergyBattery · pass

Guard

a1	express = 1
b1	attemptNo < 4

```
(root)
└── a1  express = 1
    └── b1  attemptNo < 4
```

เส้นทาง
a1 -> b1

Event

รหัส	event	guard
a1	sendnotifyExport_1	
b1	receivenotifyExport_1	
c1	sendvalidateExport_2	
d1	receivevalidateExport_2	
e1	sendlogCharge_3_opt1	a1
f1	receivelogCharge_3_opt1	a1
g1	sendreserveTariff_4	
h1	receivereserveTariff_4	
i1	sendassignCharge_5	
j1	receiveassignCharge_5	
k1	sendreleaseTariff_6	b1
l1	receivereleaseTariff_6	b1
m1	sendfetchCharge_7	b1
n1	receivefetchCharge_7	b1
o1	sendrequestCharge_8	
o2	checkloop_releaseTariff_6_loop1	b1
p1	receiverequestCharge_8	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
    ├── o1 → p1
    └── o2 ↺ k1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o2 ↺ k1

model_144 PipelineLeakSCADA · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendscheduleWork_1	
b1	receivescheduleWork_1	
c1	sendcancelWork_2	
d1	receivecancelWork_2	
e1	sendvalidatePressure_3	
f1	receivevalidatePressure_3	
g1	sendapprovePressure_4	
h1	receiveapprovePressure_4	
i1	sendfetchWork_5	
j1	receivefetchWork_5	
k1	sendapproveWork_6	
l1	receiveapproveWork_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_170 SmartHelmetMining · pass

Guard

a1	temp < 50
a2	temp ≥ 50

```
(root)
├── a1  temp < 50
└── a2  temp ≥ 50
```

เส้นทาง

a1
a2

Event

รหัส	event	guard
a1	sendsyncAlarm_1	
b1	receivesyncAlarm_1	
c1	sendackGas_2	
d1	receiveackGas_2	
e1	sendscheduleGas_3	a1
e2	sendcheckAlarm_4	a2
f1	receivescheduleGas_3	a1
f2	receivecheckAlarm_4	a2
g1	sendupdateAlarm_6	
h1	receiveupdateAlarm_6	
i1	sendreserveLocation_5	a2
j1	receivereserveLocation_5	a2

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1
    └── e2 → f2 → i1 → j1 → g1 → h1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> i1 -> j1 -> g1 -> h1

model_176 SmartStreetLight · fail

Guard

a1	urgent = 1
a2	urgent = 0

```
(root)
├── a1  urgent = 1
└── a2  urgent = 0
```

เส้นทาง

a1
a2

Event

รหัส	event	guard
a1	sendcancelDim_1	
b1	receivecancelDim_1	
c1	sendqueryDim_2	
d1	receivequeryDim_2	
e1	sendrequestFault_3	
f1	receiverequestFault_3	
g1	sendreleaseLamp_4	
h1	receivereleaseLamp_4	
i1	sendsendDim_5	
j1	receivesendDim_5	
k1	sendlogLamp_6	
l1	receivelogLamp_6	
m1	sendconfirmDim_7	
n1	receiveconfirmDim_7	
o1	sendconfirmFault_8	a1
o2	sendapproveDim_10	a2
p1	receiveconfirmFault_8	a1
p2	receiveapproveDim_10	a2
q1	sendnotifyDim_9	a1
r1	receivenotifyDim_9	a1
s1	sendackFault_11	
t1	receiveackFault_11	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
    ├── o1 → p1 → q1 → r1 → s1 → t1
    └── o2 → p2 → s1 → t1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o2 -> p2 -> s1 -> t1
```

model_181 SolarPanelMonitor · fail

Guard

a1	attemptNo < 8
b1	attemptNo mod 2 = 0
b2	attemptNo mod 2 ≠ 0

```
(root)
└── a1  attemptNo < 8
    ├── b1  attemptNo mod 2 = 0
    └── b2  attemptNo mod 2 ≠ 0
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

รหัส	event	guard
a1	sendackPower_1	
b1	receiveackPower_1	
c1	sendapproveExport_2	
d1	receiveapproveExport_2	
e1	sendissuePower_3	a1
f1	receiveissuePower_3	a1
g1	sendconfirmFault_4	a1 b1
g2	sendcheckPower_6	a1 b2
h1	receiveconfirmFault_4	a1 b1
h2	receivecheckPower_6	a1 b2
i1	sendreportFault_5	a1 b1
j1	receivereportFault_5	a1 b1
k1	sendreportExport_7	a1
l1	receivereportExport_7	a1
m1	sendreserveExport_8	
m2	checkloop_issuePower_3_loop1	a1
n1	receivereserveExport_8	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1
    │   ├── m1 → n1
    │   └── m2 ↺ e1
    └── g2 → h2 → k1 → l1
        ├── m1 → n1
        └── m2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m2 ↺ e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1 -> m1
-> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1 -> m2 ↺
e1
```

model_198 WindTurbineMonitor · pass

Guard

a1	withExtra = 1
b1	needAck = 1

```
(root)
└── a1  withExtra = 1
    └── b1  needAck = 1
```

เส้นทาง

```
a1 -> b1
```

Event

รหัส	event	guard
a1	sendreserveRpm_1	
b1	receivereserveRpm_1	
c1	sendupdatePower_2	
d1	receiveupdatePower_2	
e1	sendreservePower_3_opt1	a1
f1	receivereservePower_3_opt1	a1
g1	sendnotifyPower_4	
h1	receivenotifyPower_4	
i1	sendsyncPower_5	
j1	receivesyncPower_5	
k1	sendrequestRpm_6_opt2	b1
l1	receiverequestRpm_6_opt2	b1
m1	sendlogRpm_7_opt2	b1
n1	receivelogRpm_7_opt2	b1
o1	sendscheduleFault_8	
p1	receivescheduleFault_8	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1
```

ค้าปลีก อาหาร และ e-commerce · Retail, Food & E-commerce (14)

model_010 CoffeeOrderKiosk · pass

Guard

a1	addShot = 1

```
(root)
└── a1  addShot = 1
```

เส้นทาง
a1

Event

รหัส	event	guard
a1	sendchooseDrink_1	
b1	receivechooseDrink_1	
c1	sendshowPrice_2	
d1	receiveshowPrice_2	
e1	sendaddExtraShot_3_opt1	a1
f1	receiveaddExtraShot_3_opt1	a1
g1	sendpayCash_4	
h1	receivepayCash_4	
i1	sendserveDrink_5	
j1	receiveserveDrink_5	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1

model_021 FoodOrderApp · pass

Guard

a1	hasCoupon = 1

```
(root)
└── a1  hasCoupon = 1
```

เส้นทาง
a1

Event

รหัส	event	guard
a1	sendbrowseMenu_1	
b1	receivebrowseMenu_1	
c1	sendmenuList_2	
d1	receivemenuList_2	
e1	sendapplyCoupon_3_opt1	a1
f1	receiveapplyCoupon_3_opt1	a1
g1	senddiscountApplied_4_opt1	a1
h1	receivediscountApplied_4_opt1	a1
i1	sendsubmitOrder_5	
j1	receivesubmitOrder_5	
k1	sendforwardOrder_6	
l1	receiveforwardOrder_6	
m1	sendorderReady_7	
n1	receiveorderReady_7	
o1	sendnotifyCustomer_8	
p1	receivenotifyCustomer_8	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1

model_030 OnlineAuction · fail

Guard

a1	roundNo < 5

```
(root)
└── a1  roundNo < 5
```

เส้นทาง
a1

Event

รหัส	event	guard
a1	sendplaceBid_1	
b1	receiveplaceBid_1	
c1	sendbidAccepted_2	
d1	receivebidAccepted_2	
e1	sendoutbidNotice_3	a1
f1	receiveoutbidNotice_3	a1
g1	sendraiseBid_4	a1
h1	receiveraiseBid_4	a1
i1	sendcloseAuction_5	
i2	checkloop_outbidNotice_3_loop1	a1
j1	receivecloseAuction_5	
k1	sendsaleConfirm_6	
l1	receivesaleConfirm_6	
m1	sendwinnerNotice_7	
n1	receivewinnerNotice_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1
    ├── i1 → j1 → k1 → l1 → m1 → n1
    └── i2 ↺ e1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i2 ↺ e1

model_044 ToolRentalLocker · fail

Guard

a1	lateReturn = 1

```
(root)
└── a1  lateReturn = 1
```

เส้นทาง
a1

Event

รหัส	event	guard
a1	sendreserveTool_1	
b1	receivereserveTool_1	
c1	sendreservationCode_2	
d1	receivereservationCode_2	
e1	sendunlockLocker_3	
f1	receiveunlockLocker_3	
g1	sendverifyCode_4	
h1	receiveverifyCode_4	
i1	sendcodeValid_5	
j1	receivecodeValid_5	
k1	sendreleaseTool_6	
l1	receivereleaseTool_6	
m1	sendchargeLateFee_7_opt1	a1
n1	receivechargeLateFee_7_opt1	a1

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1

model_046 VendingMachine · pass

Guard

a1	coinStatus = ENOUGH
a2	coinStatus = SHORT
a3	coinStatus = INVALID

```
(root)
├── a1  coinStatus = ENOUGH
├── a2  coinStatus = SHORT
└── a3  coinStatus = INVALID
```

เส้นทาง

a1
a2
a3

Event

รหัส	event	guard
a1	sendinsertCoin_1	
b1	receiveinsertCoin_1	
c1	sendselectItem_2	
d1	receiveselectItem_2	
e1	senddispenseItem_3	a1
e2	sendrequestMoreCoins_5	a2
e3	sendrejectCoin_6	a3
f1	receivedispenseItem_3	a1
f2	receiverequestMoreCoins_5	a2
f3	receiverejectCoin_6	a3
g1	sendreturnChange_4	a1
h1	receivereturnChange_4	a1
i1	sendshowIdle_7	
j1	receiveshowIdle_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1
    ├── e2 → f2 → i1 → j1
    └── e3 → f3 → i1 → j1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> i1 -> j1
a1 -> b1 -> c1 -> d1 -> e3 -> f3 -> i1 -> j1

model_063 CafeLoyaltyPoints · fail

Guard

a1	roundNo < 12
b1	roundNo mod 2 = 0
c1	needAck = 1

```
(root)
└── a1  roundNo < 12
    └── b1  roundNo mod 2 = 0
        └── c1  needAck = 1
```

เส้นทาง

a1 -> b1 -> c1

Event

รหัส	event	guard
a1	sendapproveVisit_1	
b1	receiveapproveVisit_1	
c1	sendfetchPoint_2	
d1	receivefetchPoint_2	
e1	sendcheckPoint_3	a1
f1	receivecheckPoint_3	a1
g1	sendissueVisit_4_opt1	a1 b1
h1	receiveissueVisit_4_opt1	a1 b1
i1	sendupdateVisit_5	a1
j1	receiveupdateVisit_5	a1
k1	sendreleaseVisit_6_opt2	a1 c1
l1	receivereleaseVisit_6_opt2	a1 c1
m1	sendsendPoint_7	a1
n1	receivesendPoint_7	a1
o1	sendsyncReward_8	a1
p1	receivesyncReward_8	a1
q1	sendfetchReward_9	
q2	checkloop_checkPoint_3_loop1	a1
r1	receivefetchReward_9	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1
    ├── q1 → r1
    └── q2 ↺ e1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q2 ↺ e1

model_075 ConcessionStandOrder · pass

Guard

a1	retryNo < 8
b1	roundNo < 6
c1	pollNo < 11

```
(root)
└── a1  retryNo < 8
    └── b1  roundNo < 6
        └── c1  pollNo < 11
```

เส้นทาง

```
a1 -> b1 -> c1
```

Event

รหัส	event	guard
a1	sendsyncPickup_1	
b1	receivesyncPickup_1	
c1	sendreportPickup_2	
d1	receivereportPickup_2	
e1	sendconfirmOrder_3	a1
f1	receiveconfirmOrder_3	a1
g1	sendapproveSnack_4	
g2	checkloop_confirmOrder_3_loop1	a1
h1	receiveapproveSnack_4	
i1	sendassignOrder_5	
j1	receiveassignOrder_5	
k1	sendscheduleSnack_6	b1
l1	receivescheduleSnack_6	b1
m1	sendnotifySnack_7	b1 c1
n1	receivenotifySnack_7	b1 c1
o1	sendnotifyPickup_8	b1 c1
p1	receivenotifyPickup_8	b1 c1
q1	sendschedulePickup_9	b1
q2	checkloop_notifySnack_7_loop3	b1 c1
r1	receiveschedulePickup_9	b1
s1	sendupdatePickup_10	
s2	checkloop_scheduleSnack_6_loop2	b1
t1	receiveupdatePickup_10	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1 → m1 → n1 →
    │     o1 → p1
    │   ├── q1 → r1
    │   │   ├── s1 → t1
    │   │   └── s2 ↺ k1
    │   └── q2 ↺ m1
    └── g2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s2 ↺ k1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q2 ↺ m1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 ↺ e1
```

model_097 FoodTruckOrdering · pass

Guard

a1	pollNo < 5
b1	needAck = 1
c1	express = 1
d1	tier = GREEN
d2	tier = RED

```
(root)
└── a1  pollNo < 5
    └── b1  needAck = 1
        └── c1  express = 1
            ├── d1  tier = GREEN
            └── d2  tier = RED
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1
a1 -> b1 -> c1 -> d2
```

Event

รหัส	event	guard
a1	sendvalidateOrder_1	
b1	receivevalidateOrder_1	
c1	sendcheckPickup_2	
d1	receivecheckPickup_2	
e1	sendreleasePickup_3	a1
f1	receivereleasePickup_3	a1
g1	sendissueDish_4_opt1	a1 b1
h1	receiveissueDish_4_opt1	a1 b1
i1	sendcheckDish_5	a1
j1	receivecheckDish_5	a1
k1	sendcancelDish_6_opt2	a1 c1
l1	receivecancelDish_6_opt2	a1 c1
m1	sendcancelPickup_7_opt2	a1 c1 d1
m2	sendassignPickup_8_opt2	a1 c1 d2
n1	receivecancelPickup_7_opt2	a1 c1 d1
n2	receiveassignPickup_8_opt2	a1 c1 d2
o1	sendapprovePickup_9_opt2	a1 c1
p1	receiveapprovePickup_9_opt2	a1 c1
q1	sendackOrder_10	a1
r1	receiveackOrder_10	a1
s1	sendcheckOrder_11	
s2	checkloop_releasePickup_3_loop1	a1
t1	receivecheckOrder_11	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
    ├── m1 → n1 → o1 → p1 → q1 → r1
    │   ├── s1 → t1
    │   └── s2 ↺ e1
    └── m2 → n2 → o1 → p1 → q1 → r1
        ├── s1 → t1
        └── s2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s2 ↺ e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m2 -> n2 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m2 -> n2 -> o1 -> p1 -> q1 -> r1 -> s2 ↺ e1
```

model_104 GroceryDeliverySlot · fail

Guard

a1	retryNo < 8
b1	retryNo mod 3 = 0
b2	retryNo mod 3 ≠ 0

```
(root)
└── a1  retryNo < 8
    ├── b1  retryNo mod 3 = 0
    └── b2  retryNo mod 3 ≠ 0
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

รหัส	event	guard
a1	sendapproveOrder_1	
b1	receiveapproveOrder_1	
c1	sendfetchSlot_2	
d1	receivefetchSlot_2	
e1	sendreleaseOrder_3	a1
f1	receivereleaseOrder_3	a1
g1	sendqueryOrder_4	a1 b1
g2	sendreportSlot_5	a1 b2
h1	receivequeryOrder_4	a1 b1
h2	receivereportSlot_5	a1 b2
i1	sendnotifySlot_12	a1
j1	receivenotifySlot_12	a1
k1	sendcancelSlot_13	
k2	checkloop_releaseOrder_3_loop1	a1
l1	receivecancelSlot_13	
m1	sendsyncSlot_6	a1 b2
n1	receivesyncSlot_6	a1 b2
o1	sendnotifyOrder_7	a1 b2
p1	receivenotifyOrder_7	a1 b2
q1	sendreportDelivery_8	a1 b2
r1	receivereportDelivery_8	a1 b2
s1	sendreserveOrder_9	a1 b2
t1	receivereserveOrder_9	a1 b2
u1	sendassignDelivery_10	a1 b2
v1	receiveassignDelivery_10	a1 b2
w1	sendfetchDelivery_11	a1 b2
x1	receivefetchDelivery_11	a1 b2

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1
    │   ├── k1 → l1
    │   └── k2 ↺ e1
    └── g2 → h2 → m1 → n1 → o1 → p1 → q1 → r1 →
          s1 → t1 → u1 → v1 → w1 → x1 → i1 → j1
        ├── k1 → l1
        └── k2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 ↺
e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> m1 -> n1 -> o1
-> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1 -> w1 -> x1 -> i1 ->
j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> m1 -> n1 -> o1
-> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1 -> w1 -> x1 -> i1 ->
j1 -> k2 ↺ e1
```

model_133 OnlineGroceryPicking · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendlogOrder_1	
b1	receivelogOrder_1	
c1	sendassignOrder_2	
d1	receiveassignOrder_2	
e1	sendcancelOrder_3	
f1	receivecancelOrder_3	
g1	sendfetchOrder_4	
h1	receivefetchOrder_4	
i1	sendfetchItem_5	
j1	receivefetchItem_5	
k1	sendcancelItem_6	
l1	receivecancelItem_6	
m1	sendconfirmBag_7	
n1	receiveconfirmBag_7	
o1	sendapproveOrder_8	
p1	receiveapproveOrder_8	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1

model_143 PhotoPrintKiosk · pass

Guard

a1	t_obs ≤ 13
a2	t_obs > 13

```
(root)
├── a1  t_obs ≤ 13
└── a2  t_obs > 13
```

เส้นทาง

a1
a2

Event

รหัส	event	guard
a1	sendcheckPrint_1	
b1	receivecheckPrint_1	
c1	sendassignPrint_2	
d1	receiveassignPrint_2	
e1	sendassignOrder_3	a1
e2	sendupdatePhoto_5	a2
f1	receiveassignOrder_3	a1
f2	receiveupdatePhoto_5	a2
g1	sendreservePhoto_4	a1
h1	receivereservePhoto_4	a1
i1	sendschedulePrint_6	
j1	receiveschedulePrint_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1
    └── e2 → f2 → i1 → j1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> i1 -> j1

model_162 SelfCheckoutLane · pass

Guard

a1	t_obs ≤ 10
a2	t_obs > 10
b1	stepNo < 3

```
(root)
├── a1  t_obs ≤ 10
│   └── b1  stepNo < 3
└── a2  t_obs > 10
    └── b1  stepNo < 3
```

เส้นทาง

```
a1 -> b1
a2 -> b1
```

Event

รหัส	event	guard
a1	sendreportItem_1	
b1	receivereportItem_1	
c1	sendassignItem_2	
d1	receiveassignItem_2	
e1	sendcancelItem_3	a1
e2	sendscheduleItem_4	a2
f1	receivecancelItem_3	a1
f2	receivescheduleItem_4	a2
g1	sendnotifyReceipt_5	
h1	receivenotifyReceipt_5	
i1	sendissueItem_6	
j1	receiveissueItem_6	
k1	sendissueReceipt_7	b1
l1	receiveissueReceipt_7	b1
m1	sendassignReceipt_8	b1
n1	receiveassignReceipt_8	b1
o1	sendreportReceipt_9	
o2	checkloop_issueReceipt_7_loop1	b1
p1	receivereportReceipt_9	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1 → k1 → l1 →
    │     m1 → n1
    │   ├── o1 → p1
    │   └── o2 ↺ k1
    └── e2 → f2 → g1 → h1 → i1 → j1 → k1 → l1 →
          m1 → n1
        ├── o1 → p1
        └── o2 ↺ k1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o2 ↺ k1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o2 ↺ k1
```

model_175 SmartShoppingCart · pass

Guard

a1	useCache = 1

```
(root)
└── a1  useCache = 1
```

เส้นทาง

```
a1
```

Event

รหัส	event	guard
a1	sendsyncTotal_1	
b1	receivesyncTotal_1	
c1	sendcancelCoupon_2	
d1	receivecancelCoupon_2	
e1	sendapproveCoupon_3_opt1	a1
f1	receiveapproveCoupon_3_opt1	a1
g1	sendqueryItem_4	
h1	receivequeryItem_4	
i1	sendupdateCoupon_5	
j1	receiveupdateCoupon_5	
k1	sendissueCoupon_6	
l1	receiveissueCoupon_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1
```

model_178 SmartVendingFridge · fail

Guard

a1	attemptNo < 9
b1	attemptNo mod 4 = 0

```
(root)
└── a1  attemptNo < 9
    └── b1  attemptNo mod 4 = 0
```

เส้นทาง

```
a1 -> b1
```

Event

รหัส	event	guard
a1	sendackItem_1	
b1	receiveackItem_1	
c1	sendreportCharge_2	
d1	receivereportCharge_2	
e1	sendassignItem_3	a1
f1	receiveassignItem_3	a1
g1	sendfetchItem_4_opt1	a1 b1
h1	receivefetchItem_4_opt1	a1 b1
i1	sendsendRestock_5_opt1	a1 b1
j1	receivesendRestock_5_opt1	a1 b1
k1	sendreserveCharge_6	a1
l1	receivereserveCharge_6	a1
m1	sendlogRestock_7	
m2	checkloop_assignItem_3_loop1	a1
n1	receivelogRestock_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
    ├── m1 → n1
    └── m2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m2 ↺ e1
```

การจอง ที่พัก และกิจกรรม · Booking, Hospitality & Events (19)

model_009 CinemaTicketing · pass

Guard

a1	seatTry < 3
b1	wantSnack = 1

```
(root)
└── a1  seatTry < 3
    └── b1  wantSnack = 1
```

เส้นทาง

```
a1 -> b1
```

Event

รหัส	event	guard
a1	sendselectMovie_1	
b1	receiveselectMovie_1	
c1	sendshowtimes_2	
d1	receiveshowtimes_2	
e1	sendpickSeat_3	a1
f1	receivepickSeat_3	a1
g1	sendseatStatus_4	a1
h1	receiveseatStatus_4	a1
i1	sendreserveSeat_5	
i2	checkloop_pickSeat_3_loop1	a1
j1	receivereserveSeat_5	
k1	sendaddSnack_6_opt1	b1
l1	receiveaddSnack_6_opt1	b1
m1	sendissueTicket_7	
n1	receiveissueTicket_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1
    ├── i1 → j1 → k1 → l1 → m1 → n1
    └── i2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i2 ↺ e1
```

model_012 ConferenceRoomBooking · fail

Guard

a1	needDisplay = 1

```
(root)
└── a1  needDisplay = 1
```

เส้นทาง
a1

Event

รหัส	event	guard
a1	sendsearchRoom_1	
b1	receivesearchRoom_1	
c1	sendavailability_2	
d1	receiveavailability_2	
e1	sendbookRoom_3	
f1	receivebookRoom_3	
g1	sendcreateEvent_4	
h1	receivecreateEvent_4	
i1	sendeventCreated_5	
j1	receiveeventCreated_5	
k1	sendupdatePanel_6_opt1	a1
l1	receiveupdatePanel_6_opt1	a1
m1	sendbookingDone_7	
n1	receivebookingDone_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1

model_022 GymCheckIn · fail

Guard

a1	membership = ACTIVE
a2	membership = EXPIRED

```
(root)
├── a1  membership = ACTIVE
└── a2  membership = EXPIRED
```

เส้นทาง
a1
a2

Event

รหัส	event	guard
a1	sendscanQR_1	
b1	receivescanQR_1	
c1	sendaccessResult_2	
d1	receiveaccessResult_2	
e1	sendopenGate_3	a1
e2	sendrenewPrompt_4	a2
f1	receiveopenGate_3	a1
f2	receiverenewPrompt_4	a2
g1	sendexitScan_5	
h1	receiveexitScan_5	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1
    └── e2 → f2 → g1 → h1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g1 -> h1

model_024 HotelCheckIn · pass

Guard

a1	roomReady = 1
a2	roomReady = 0

```
(root)
├── a1  roomReady = 1
└── a2  roomReady = 0
```

เส้นทาง

a1
a2

Event

รหัส	event	guard
a1	sendarrive_1	
b1	receivearrive_1	
c1	sendlookupBooking_2	
d1	receivelookupBooking_2	
e1	sendbookingInfo_3	
f1	receivebookingInfo_3	
g1	sendassignRoom_4	a1
g2	sendofferLounge_6	a2
h1	receiveassignRoom_4	a1
h2	receiveofferLounge_6	a2
i1	sendissueKeycard_5	a1
j1	receiveissueKeycard_5	a1
k1	sendupdateOccupancy_7	
l1	receiveupdateOccupancy_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1
    └── g2 → h2 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1

model_055 ArtGalleryTicketing · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendvalidateEntry_1	
b1	receivevalidateEntry_1	
c1	sendapproveEntry_2	
d1	receiveapproveEntry_2	
e1	sendlogEntry_3	
f1	receivelogEntry_3	
g1	sendrequestEntry_4	
h1	receiverequestEntry_4	
i1	sendfetchEntry_5	
j1	receivefetchEntry_5	
k1	sendnotifyEntry_6	
l1	receivenotifyEntry_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_069 CarWashBooking · pass

Guard

a1	roundNo < 12
b1	withExtra = 1

```
(root)
└── a1  roundNo < 12
    └── b1  withExtra = 1
```

เส้นทาง

```
a1 -> b1
```

Event

รหัส	event	guard
a1	sendcancelSlot_1	
b1	receivecancelSlot_1	
c1	sendfetchWash_2	
d1	receivefetchWash_2	
e1	sendcancelPayment_3	a1
f1	receivecancelPayment_3	a1
g1	sendapproveWash_4	a1
h1	receiveapproveWash_4	a1
i1	sendissueWash_5_opt1	a1 b1
j1	receiveissueWash_5_opt1	a1 b1
k1	sendconfirmPayment_6	a1
l1	receiveconfirmPayment_6	a1
m1	sendconfirmWash_7	
m2	checkloop_cancelPayment_3_loop1	a1
n1	receiveconfirmWash_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
    ├── m1 → n1
    └── m2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m2 ↺ e1
```

model_074 ConcertTicketQueue · fail

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendsyncSeat_1	
b1	receivesyncSeat_1	
c1	sendassignSeat_2	
d1	receiveassignSeat_2	
e1	sendfetchTicket_3	
f1	receivefetchTicket_3	
g1	sendvalidateQueue_4	
h1	receivevalidateQueue_4	
i1	sendquerySeat_5	
j1	receivequerySeat_5	
k1	sendreserveQueue_6	
l1	receivereserveQueue_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1
```

model_076 ConferenceBadgeScan · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendreleaseBadge_1	
b1	receivereleaseBadge_1	
c1	sendscheduleSession_2	
d1	receivescheduleSession_2	
e1	sendissueBadge_3	
f1	receiveissueBadge_3	
g1	sendquerySession_4	
h1	receivequerySession_4	
i1	sendcancelSession_5	
j1	receivecancelSession_5	
k1	sendvalidateSession_6	
l1	receivevalidateSession_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_078 CoworkingDeskBooking · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendsendDesk_1	
b1	receivesendDesk_1	
c1	sendreserveBooking_2	
d1	receivereserveBooking_2	
e1	sendvalidateBooking_3	
f1	receivevalidateBooking_3	
g1	sendissuePass_4	
h1	receiveissuePass_4	
i1	sendconfirmPass_5	
j1	receiveconfirmPass_5	
k1	sendqueryPass_6	
l1	receivequeryPass_6	
m1	sendsyncBooking_7	
n1	receivesyncBooking_7	
o1	sendqueryDesk_8	
p1	receivequeryDesk_8	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1

model_090 EventCheckInQR · fail

Guard

a1	passNo < 8
b1	passNo mod 4 = 0
b2	passNo mod 4 ≠ 0
c1	withExtra = 1

```
(root)
└── a1  passNo < 8
    ├── b1  passNo mod 4 = 0
    │   └── c1  withExtra = 1
    └── b2  passNo mod 4 ≠ 0
        └── c1  withExtra = 1
```

เส้นทาง

```
a1 -> b1 -> c1
a1 -> b2 -> c1
```

Event

รหัส	event	guard
a1	sendreleaseBadge_1	
b1	receivereleaseBadge_1	
c1	sendsyncBadge_2	
d1	receivesyncBadge_2	
e1	sendnotifyEntry_3	a1
f1	receivenotifyEntry_3	a1
g1	sendsendBadge_4	a1 b1
g2	sendupdateBadge_6	a1 b2
h1	receivesendBadge_4	a1 b1
h2	receiveupdateBadge_6	a1 b2
i1	sendreportEntry_5	a1 b1
j1	receivereportEntry_5	a1 b1
k1	sendsyncPass_7	a1
l1	receivesyncPass_7	a1
m1	sendfetchBadge_8	
m2	checkloop_notifyEntry_3_loop1	a1
n1	receivefetchBadge_8	
o1	sendreportBadge_9	
p1	receivereportBadge_9	
q1	sendassignEntry_10_opt1	c1
r1	receiveassignEntry_10_opt1	c1
s1	sendlogEntry_11_opt1	c1
t1	receivelogEntry_11_opt1	c1
u1	sendissuePass_12	
v1	receiveissuePass_12	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1
    │   ├── m1 → n1 → o1 → p1 → q1 → r1 → s1 → t1 →
    │   │     u1 → v1
    │   └── m2 ↺ e1
    └── g2 → h2 → k1 → l1
        ├── m1 → n1 → o1 → p1 → q1 → r1 → s1 → t1 →
        │     u1 → v1
        └── m2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 ->
v1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m2 ↺ e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1 -> m1
-> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1 -> m2 ↺
e1
```

model_102 GolfCartFleet · fail

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendfetchFee_1	
b1	receivefetchFee_1	
c1	sendscheduleRound_2	
d1	receivescheduleRound_2	
e1	sendfetchCart_3	
f1	receivefetchCart_3	
g1	sendconfirmCart_4	
h1	receiveconfirmCart_4	
i1	sendapproveCart_5	
j1	receiveapproveCart_5	
k1	sendissueCart_6	
l1	receiveissueCart_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_111 HotelKeylessEntry · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendfetchAccess_1	
b1	receivefetchAccess_1	
c1	sendscheduleRoom_2	
d1	receivescheduleRoom_2	
e1	sendfetchKey_3	
f1	receivefetchKey_3	
g1	sendconfirmKey_4	
h1	receiveconfirmKey_4	
i1	sendapproveKey_5	
j1	receiveapproveKey_5	
k1	sendissueKey_6	
l1	receiveissueKey_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_112 HotelRoomService · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendconfirmMeal_1	
b1	receiveconfirmMeal_1	
c1	sendackBill_2	
d1	receiveackBill_2	
e1	sendvalidateMeal_3	
f1	receivevalidateMeal_3	
g1	sendapproveBill_4	
h1	receiveapproveBill_4	
i1	sendrequestBill_5	
j1	receiverequestBill_5	
k1	sendupdateBill_6	
l1	receiveupdateBill_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_123 MarathonTiming · fail

Guard

a1	roundNo < 5
b1	roundNo mod 2 = 0
c1	load < 10
d1	retryNo < 11
c2	load ≥ 10

```
(root)
└── a1  roundNo < 5
    └── b1  roundNo mod 2 = 0
        ├── c1  load < 10
        │   └── d1  retryNo < 11
        └── c2  load ≥ 10
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1
a1 -> b1 -> c2
```

Event

รหัส	event	guard
a1	sendreleaseSplit_1	
b1	receivereleaseSplit_1	
c1	sendquerySplit_2	
d1	receivequerySplit_2	
e1	sendcancelBib_3	a1
f1	receivecancelBib_3	a1
g1	sendrequestBib_4_opt1	a1 b1
h1	receiverequestBib_4_opt1	a1 b1
i1	sendrequestFinish_5	a1
j1	receiverequestFinish_5	a1
k1	sendassignSplit_6	a1
l1	receiveassignSplit_6	a1
m1	sendcancelFinish_7	
m2	checkloop_cancelBib_3_loop1	a1
n1	receivecancelFinish_7	
o1	sendsendFinish_8	
p1	receivesendFinish_8	
q1	sendfetchFinish_9	c1
q2	sendreleaseFinish_13	c2
r1	receivefetchFinish_9	c1
r2	receivereleaseFinish_13	c2
s1	sendreserveSplit_10	c1 d1
t1	receivereserveSplit_10	c1 d1
u1	sendreserveFinish_11	c1 d1
v1	receivereserveFinish_11	c1 d1
w1	sendvalidateFinish_12	c1
w2	checkloop_reserveSplit_10_loop2	c1 d1
x1	receivevalidateFinish_12	c1
y1	sendsendBib_14	
z1	receivesendBib_14	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
    ├── m1 → n1 → o1 → p1
    │   ├── q1 → r1 → s1 → t1 → u1 → v1
    │   │   ├── w1 → x1 → y1 → z1
    │   │   └── w2 ↺ s1
    │   └── q2 → r2 → y1 → z1
    └── m2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 ->
v1 -> w1 -> x1 -> y1 -> z1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 ->
v1 -> w2 ↺ s1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q2 -> r2 -> y1 -> z1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m2 ↺ e1
```

model_127 MeetingRoomDisplay · pass

Guard

a1	isVip = 1
a2	isVip = 0

```
(root)
├── a1  isVip = 1
└── a2  isVip = 0
```

เส้นทาง

a1
a2

Event

รหัส	event	guard
a1	sendcancelMeeting_1	
b1	receivecancelMeeting_1	
c1	sendconfirmCheckin_2	
d1	receiveconfirmCheckin_2	
e1	sendconfirmMeeting_3	a1
e2	sendrequestMeeting_5	a2
f1	receiveconfirmMeeting_3	a1
f2	receiverequestMeeting_5	a2
g1	sendupdateRoom_4	a1
h1	receiveupdateRoom_4	a1
i1	sendconfirmRoom_6	
j1	receiveconfirmRoom_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1
    └── e2 → f2 → i1 → j1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> i1 -> j1

model_129 MuseumAudioGuide · pass

Guard

a1	useCache = 1
b1	stepNo < 11

```
(root)
└── a1  useCache = 1
    └── b1  stepNo < 11
```

เส้นทาง

a1 -> b1

Event

รหัส	event	guard
a1	sendconfirmTour_1	
b1	receiveconfirmTour_1	
c1	sendnotifyTour_2	
d1	receivenotifyTour_2	
e1	sendreserveTour_3_opt1	a1
f1	receivereserveTour_3_opt1	a1
g1	sendcancelTour_4	
h1	receivecancelTour_4	
i1	sendvalidateTour_5	
j1	receivevalidateTour_5	
k1	sendcheckTour_6	b1
l1	receivecheckTour_6	b1
m1	sendvalidateAudio_7	
m2	checkloop_checkTour_6_loop1	b1
n1	receivevalidateAudio_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
    ├── m1 → n1
    └── m2 ↺ k1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m2 ↺ k1

model_155 RestaurantReservation · pass

Guard

a1	express = 1
b1	roundNo < 8
c1	needAck = 1

```
(root)
└── a1  express = 1
    └── b1  roundNo < 8
        └── c1  needAck = 1
```

เส้นทาง

```
a1 -> b1 -> c1
```

Event

รหัส	event	guard
a1	sendfetchMenu_1	
b1	receivefetchMenu_1	
c1	sendreserveTable_2	
d1	receivereserveTable_2	
e1	sendqueryBooking_3_opt1	a1
f1	receivequeryBooking_3_opt1	a1
g1	sendsyncMenu_4	
h1	receivesyncMenu_4	
i1	sendlogMenu_5	
j1	receivelogMenu_5	
k1	sendapproveMenu_6	b1
l1	receiveapproveMenu_6	b1
m1	sendackBooking_7_opt2	b1 c1
n1	receiveackBooking_7_opt2	b1 c1
o1	sendreportMenu_8	b1
p1	receivereportMenu_8	b1
q1	sendassignBooking_9	
q2	checkloop_approveMenu_6_loop1	b1
r1	receiveassignBooking_9	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1
    ├── q1 → r1
    └── q2 ↺ k1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q2 ↺ k1
```

model_169 SmartGymLocker · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendsendLocker_1	
b1	receivesendLocker_1	
c1	sendreservePin_2	
d1	receivereservePin_2	
e1	sendvalidatePin_3	
f1	receivevalidatePin_3	
g1	sendissueSession_4	
h1	receiveissueSession_4	
i1	sendconfirmSession_5	
j1	receiveconfirmSession_5	
k1	sendquerySession_6	
l1	receivequerySession_6	
m1	sendsyncPin_7	
n1	receivesyncPin_7	
o1	sendqueryLocker_8	
p1	receivequeryLocker_8	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1
```

model_182 StadiumEntryGate · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendreserveSeat_1	
b1	receivereserveSeat_1	
c1	sendlogSeat_2	
d1	receivelogSeat_2	
e1	sendconfirmEntry_3	
f1	receiveconfirmEntry_3	
g1	sendlogEntry_4	
h1	receivelogEntry_4	
i1	sendsyncTicket_5	
j1	receivesyncTicket_5	
k1	sendscheduleEntry_6	
l1	receivescheduleEntry_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

การศึกษาและบริการภาครัฐ · Education & Public Services (17)

model_013 CourseEnrollment · fail

Guard

a1	needsApproval = 1

```
(root)
└── a1  needsApproval = 1
```

เส้นทาง

a1

Event

รหัส	event	guard
a1	sendbrowseCourses_1	
b1	receivebrowseCourses_1	
c1	sendcourseList_2	
d1	receivecourseList_2	
e1	sendenrollRequest_3	
f1	receiveenrollRequest_3	
g1	sendcheckPrereq_4	
h1	receivecheckPrereq_4	
i1	sendprereqOK_5	
j1	receiveprereqOK_5	
k1	sendrequestApproval_6_opt1	a1
l1	receiverequestApproval_6_opt1	a1
m1	sendapproved_7_opt1	a1
n1	receiveapproved_7_opt1	a1
o1	sendenrollConfirm_8	
p1	receiveenrollConfirm_8	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1

model_017 EmergencyDispatch · pass

Guard

a1	priority = HIGH
b1	updateCount < 4
c1	trafficJam = 1
a2	priority = LOW

```
(root)
├── a1  priority = HIGH
│   └── b1  updateCount < 4
│       └── c1  trafficJam = 1
└── a2  priority = LOW
```

เส้นทาง

```
a1 -> b1 -> c1
a2
```

Event

รหัส	event	guard
a1	sendreportEmergency_1	
b1	receivereportEmergency_1	
c1	sendaskDetails_2	
d1	receiveaskDetails_2	
e1	sendprovideDetails_3	
f1	receiveprovideDetails_3	
g1	senddispatchAmbulance_4	a1
g2	sendscheduleCallback_8	a2
h1	receivedispatchAmbulance_4	a1
h2	receivescheduleCallback_8	a2
i1	sendsendLocation_5	a1 b1
j1	receivesendLocation_5	a1 b1
k1	sendrerouteRequest_6_opt1	a1 b1 c1
l1	receivererouteRequest_6_opt1	a1 b1 c1
m1	sendnotifyHospital_7	a1
m2	checkloop_sendLocation_5_loop1	a1 b1
n1	receivenotifyHospital_7	a1
o1	sendcloseCase_9	
p1	receivecloseCase_9	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1
    │   ├── m1 → n1 → o1 → p1
    │   └── m2 ↺ i1
    └── g2 → h2 → o1 → p1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m2 ↺ i1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> o1 -> p1
```

model_027 LibraryLoan · pass

Guard

a1	fine ≤ 100
b1	wantReminder = 1
a2	fine > 100

```
(root)
├── a1  fine ≤ 100
│   └── b1  wantReminder = 1
└── a2  fine > 100
```

เส้นทาง

```
a1 -> b1
a2
```

Event

รหัส	event	guard
a1	sendscanCard_1	
b1	receivescanCard_1	
c1	sendcheckMember_2	
d1	receivecheckMember_2	
e1	sendmemberInfo_3	
f1	receivememberInfo_3	
g1	sendborrowBook_4	a1
g2	sendrejectLoan_7	a2
h1	receiveborrowBook_4	a1
h2	receiverejectLoan_7	a2
i1	sendrecordLoan_5	a1
j1	receiverecordLoan_5	a1
k1	sendsetReminder_6_opt1	a1 b1
l1	receivesetReminder_6_opt1	a1 b1
m1	sendlogout_8	
n1	receivelogout_8	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1 → m1 → n1
    └── g2 → h2 → m1 → n1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> m1 -> n1
```

model_031 OnlineExam · pass

Guard

a1	questionNo < 20
b1	autosaveCount < 3

```
(root)
└── a1  questionNo < 20
    └── b1  autosaveCount < 3
```

เส้นทาง

```
a1 -> b1
```

Event

รหัส	event	guard
a1	sendlogin_1	
b1	receivelogin_1	
c1	sendfetchQuestion_2	a1
d1	receivefetchQuestion_2	a1
e1	sendquestionData_3	a1
f1	receivequestionData_3	a1
g1	sendautosave_4	a1 b1
h1	receiveautosave_4	a1 b1
i1	sendsaveAck_5	a1 b1
j1	receivesaveAck_5	a1 b1
k1	sendsubmitAnswer_6	a1
k2	checkloop_autosave_4_loop2	a1 b1
l1	receivesubmitAnswer_6	a1
m1	sendflagBehavior_7	
m2	checkloop_fetchQuestion_2_loop1	a1
n1	receiveflagBehavior_7	
o1	sendexamResult_8	
p1	receiveexamResult_8	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1
    ├── k1 → l1
    │   ├── m1 → n1 → o1 → p1
    │   └── m2 ↺ c1
    └── k2 ↺ g1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m2 ↺ c1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 ↺
g1
```

model_059 BoardingSchoolGate · fail

Guard

a1	passNo < 5
b1	urgent = 1
b2	urgent = 0

```
(root)
└── a1  passNo < 5
    ├── b1  urgent = 1
    └── b2  urgent = 0
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

รหัส	event	guard
a1	sendsendPass_1	
b1	receivesendPass_1	
c1	sendscheduleNotice_2	
d1	receivescheduleNotice_2	
e1	sendrequestExit_3	a1
f1	receiverequestExit_3	a1
g1	sendapproveExit_4	
g2	checkloop_requestExit_3_loop1	a1
h1	receiveapproveExit_4	
i1	sendvalidateExit_5	
j1	receivevalidateExit_5	
k1	sendapproveNotice_6	b1
k2	sendcancelExit_8	b2
l1	receiveapproveNotice_6	b1
l2	receivecancelExit_8	b2
m1	sendnotifyNotice_7	b1
m2	sendsyncExit_9	b2
n1	receivenotifyNotice_7	b1
n2	receivesyncExit_9	b2
o1	sendupdateExit_10	
p1	receiveupdateExit_10	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1
    │   ├── k1 → l1 → m1 → n1 → o1 → p1
    │   └── k2 → l2 → m2 → n2 → o1 → p1
    └── g2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2
-> l2 -> m2 -> n2 -> o1 -> p1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 ↺ e1
```

model_071 ChildcareCheckin · fail

Guard

a1	approved = 1
a2	approved = 0
b1	retryNo < 4

```
(root)
├── a1  approved = 1
│   └── b1  retryNo < 4
└── a2  approved = 0
    └── b1  retryNo < 4
```

เส้นทาง

```
a1 -> b1
a2 -> b1
```

Event

รหัส	event	guard
a1	sendissuePickup_1	
b1	receiveissuePickup_1	
c1	sendschedulePickup_2	
d1	receiveschedulePickup_2	
e1	sendqueryNote_3	a1
e2	sendreserveNote_5	a2
f1	receivequeryNote_3	a1
f2	receivereserveNote_5	a2
g1	sendackPickup_4	a1
h1	receiveackPickup_4	a1
i1	sendscheduleChild_6	
j1	receivescheduleChild_6	
k1	sendvalidateNote_7	
l1	receivevalidateNote_7	
m1	sendcheckChild_8	b1
n1	receivecheckChild_8	b1
o1	sendcancelChild_9	
o2	checkloop_checkChild_8_loop1	b1
p1	receivecancelChild_9	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1 → k1 → l1 →
    │     m1 → n1
    │   ├── o1 → p1
    │   └── o2 ↺ m1
    └── e2 → f2 → i1 → j1 → k1 → l1 → m1 → n1
        ├── o1 → p1
        └── o2 ↺ m1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o2 ↺ m1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> i1 -> j1 -> k1 -> l1 -> m1
-> n1 -> o1 -> p1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> i1 -> j1 -> k1 -> l1 -> m1
-> n1 -> o2 ↺ m1
```

model_077 CourseAutoGrading · pass

Guard

a1	mode = MANUAL
a2	mode = AUTO
b1	t_obs ≤ 27
b2	t_obs > 27

```
(root)
├── a1  mode = MANUAL
│   ├── b1  t_obs ≤ 27
│   └── b2  t_obs > 27
└── a2  mode = AUTO
    ├── b1  t_obs ≤ 27
    └── b2  t_obs > 27
```

เส้นทาง

```
a1 -> b1
a1 -> b2
a2 -> b1
a2 -> b2
```

Event

รหัส	event	guard
a1	sendissueFeedback_1	
b1	receiveissueFeedback_1	
c1	sendassignFeedback_2	
d1	receiveassignFeedback_2	
e1	sendreserveGrade_3	a1
e2	sendlogGrade_4	a2
f1	receivereserveGrade_3	a1
f2	receivelogGrade_4	a2
g1	sendackGrade_6	
h1	receiveackGrade_6	
i1	sendnotifyGrade_7	
j1	receivenotifyGrade_7	
k1	sendupdateGrade_8	b1
k2	sendvalidateSubmission_9	b2
l1	receiveupdateGrade_8	b1
l2	receivevalidateSubmission_9	b2
m1	sendreleaseSubmission_11	
n1	receivereleaseSubmission_11	
o1	sendackSubmission_10	b2
p1	receiveackSubmission_10	b2
q1	sendcheckSubmission_5	a2
r1	receivecheckSubmission_5	a2

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1
    │   ├── k1 → l1 → m1 → n1
    │   └── k2 → l2 → o1 → p1 → m1 → n1
    └── e2 → f2 → q1 → r1 → g1 → h1 → i1 → j1
        ├── k1 → l1 → m1 → n1
        └── k2 → l2 → o1 → p1 → m1 → n1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2
-> l2 -> o1 -> p1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> q1 -> r1 -> g1 -> h1 -> i1
-> j1 -> k1 -> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> q1 -> r1 -> g1 -> h1 -> i1
-> j1 -> k2 -> l2 -> o1 -> p1 -> m1 -> n1
```

model_083 DigitalLibraryAccess · fail

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendsyncLoan_1	
b1	receivesyncLoan_1	
c1	sendfetchLoan_2	
d1	receivefetchLoan_2	
e1	sendsendLicense_3	
f1	receivesendLicense_3	
g1	sendreportLoan_4	
h1	receivereportLoan_4	
i1	sendreleaseLoan_5	
j1	receivereleaseLoan_5	
k1	sendcheckEbook_6	
l1	receivecheckEbook_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_084 DigitalVotingBooth · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendcancelBallot_1	
b1	receivecancelBallot_1	
c1	sendscheduleVote_2	
d1	receivescheduleVote_2	
e1	sendlogVote_3	
f1	receivelogVote_3	
g1	sendupdateReceipt_4	
h1	receiveupdateReceipt_4	
i1	sendvalidateReceipt_5	
j1	receivevalidateReceipt_5	
k1	sendsendVote_6	
l1	receivesendVote_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_085 DisasterAlertBroadcast · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendassignQuake_1	
b1	receiveassignQuake_1	
c1	sendfetchQuake_2	
d1	receivefetchQuake_2	
e1	sendreportShelter_3	
f1	receivereportShelter_3	
g1	sendfetchShelter_4	
h1	receivefetchShelter_4	
i1	sendqueryAlert_5	
j1	receivequeryAlert_5	
k1	sendissueAlert_6	
l1	receiveissueAlert_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_118 LabEquipmentBooking · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendreserveSample_1	
b1	receivereserveSample_1	
c1	sendreleaseSample_2	
d1	receivereleaseSample_2	
e1	sendsendSample_3	
f1	receivesendSample_3	
g1	sendcheckSample_4	
h1	receivecheckSample_4	
i1	sendsendSlot_5	
j1	receivesendSlot_5	
k1	sendissueSlot_6	
l1	receiveissueSlot_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_138 PassportRenewalKiosk · pass

Guard

a1	t_obs ≤ 4
a2	t_obs > 4

```
(root)
├── a1  t_obs ≤ 4
└── a2  t_obs > 4
```

เส้นทาง

a1
a2

Event

รหัส	event	guard
a1	sendackPhoto_1	
b1	receiveackPhoto_1	
c1	sendreportForm_2	
d1	receivereportForm_2	
e1	sendsendPassport_3	a1
e2	sendcancelPassport_4	a2
f1	receivesendPassport_3	a1
f2	receivecancelPassport_4	a2
g1	sendreserveForm_5	
h1	receivereserveForm_5	
i1	sendlogPassport_6	
j1	receivelogPassport_6	
k1	sendfetchPassport_7	
l1	receivefetchPassport_7	
m1	sendvalidatePhoto_8	
n1	receivevalidatePhoto_8	
o1	sendvalidateForm_9	
p1	receivevalidateForm_9	
q1	sendcheckPhoto_10	
r1	receivecheckPhoto_10	
s1	sendsendForm_11	
t1	receivesendForm_11	
u1	sendsyncPhoto_12	
v1	receivesyncPhoto_12	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1 → k1 → l1 →
    │     m1 → n1 → o1 → p1 → q1 → r1 → s1 → t1 →
    │     u1 → v1
    └── e2 → f2 → g1 → h1 → i1 → j1 → k1 → l1 →
          m1 → n1 → o1 → p1 → q1 → r1 → s1 → t1 →
          u1 → v1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1

model_148 PrisonVisitBooking · fail

Guard

a1	passNo < 9
b1	passNo mod 2 = 0
c1	optIn = 1

```
(root)
└── a1  passNo < 9
    └── b1  passNo mod 2 = 0
        └── c1  optIn = 1
```

เส้นทาง

```
a1 -> b1 -> c1
```

Event

รหัส	event	guard
a1	sendackVisit_1	
b1	receiveackVisit_1	
c1	sendsendSlot_2	
d1	receivesendSlot_2	
e1	sendqueryClearance_3	a1
f1	receivequeryClearance_3	a1
g1	sendcancelSlot_4_opt1	a1 b1
h1	receivecancelSlot_4_opt1	a1 b1
i1	sendissueVisit_5	a1
j1	receiveissueVisit_5	a1
k1	sendissueClearance_6_opt2	a1 c1
l1	receiveissueClearance_6_opt2	a1 c1
m1	sendrequestClearance_7	a1
n1	receiverequestClearance_7	a1
o1	sendsyncVisit_8	a1
p1	receivesyncVisit_8	a1
q1	sendapproveClearance_9	
q2	checkloop_queryClearance_3_loop1	a1
r1	receiveapproveClearance_9	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1
    ├── q1 → r1
    └── q2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q2 ↺ e1
```

model_152 RemoteExamProctor · fail

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendackExam_1	
b1	receiveackExam_1	
c1	sendapproveExam_2	
d1	receiveapproveExam_2	
e1	sendscheduleSession_3	
f1	receivescheduleSession_3	
g1	sendrequestExam_4	
h1	receiverequestExam_4	
i1	sendconfirmFlag_5	
j1	receiveconfirmFlag_5	
k1	sendnotifyFlag_6	
l1	receivenotifyFlag_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1
```

model_171 SmartLibraryShelf · fail

Guard

a1	roundNo < 12
b1	roundNo mod 4 = 0
c1	grade = ACTIVE
c2	grade = IDLE

```
(root)
└── a1  roundNo < 12
    └── b1  roundNo mod 4 = 0
        ├── c1  grade = ACTIVE
        └── c2  grade = IDLE
```

เส้นทาง

```
a1 -> b1 -> c1
a1 -> b1 -> c2
```

Event

รหัส	event	guard
a1	sendsyncShelf_1	
b1	receivesyncShelf_1	
c1	sendreserveShelf_2	
d1	receivereserveShelf_2	
e1	sendnotifyHold_3	a1
f1	receivenotifyHold_3	a1
g1	sendrequestBook_4_opt1	a1 b1
h1	receiverequestBook_4_opt1	a1 b1
i1	sendupdateHold_5_opt1	a1 b1
j1	receiveupdateHold_5_opt1	a1 b1
k1	sendapproveBook_6	a1
l1	receiveapproveBook_6	a1
m1	sendreportBook_7	a1
n1	receivereportBook_7	a1
o1	sendsyncBook_8	
o2	checkloop_notifyHold_3_loop1	a1
p1	receivesyncBook_8	
q1	sendrequestHold_9	
r1	receiverequestHold_9	
s1	sendcancelShelf_10	c1
s2	sendcancelHold_12	c2
t1	receivecancelShelf_10	c1
t2	receivecancelHold_12	c2
u1	sendlogBook_11	c1
v1	receivelogBook_11	c1
w1	sendreserveBook_13	
x1	receivereserveBook_13	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
    ├── o1 → p1 → q1 → r1
    │   ├── s1 → t1 → u1 → v1 → w1 → x1
    │   └── s2 → t2 → w1 → x1
    └── o2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 ->
v1 -> w1 -> x1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s2 -> t2 -> w1 ->
x1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o2 ↺ e1
```

model_184 StudentAttendance · pass

Guard

a1	useCache = 1

```
(root)
└── a1  useCache = 1
```

เส้นทาง
a1

Event

รหัส	event	guard
a1	sendsyncReport_1	
b1	receivesyncReport_1	
c1	sendcancelAbsence_2	
d1	receivecancelAbsence_2	
e1	sendapproveAbsence_3_opt1	a1
f1	receiveapproveAbsence_3_opt1	a1
g1	sendqueryCheckin_4	
h1	receivequeryCheckin_4	
i1	sendupdateAbsence_5	
j1	receiveupdateAbsence_5	
k1	sendissueAbsence_6	
l1	receiveissueAbsence_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_185 TaxFilingPortal · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendreportReturn_1	
b1	receivereportReturn_1	
c1	sendreleaseRefund_2	
d1	receivereleaseRefund_2	
e1	sendassignRefund_3	
f1	receiveassignRefund_3	
g1	sendschedulePayment_4	
h1	receiveschedulePayment_4	
i1	sendupdatePayment_5	
j1	receiveupdatePayment_5	
k1	sendissueReturn_6	
l1	receiveissueReturn_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

ไอทีและบริการดิจิทัล · IT & Digital Services (19)

model_008 ChatMessaging · fail

Guard

a1	typing = 1

```
(root)
└── a1  typing = 1
```

เส้นทาง
a1

Event

รหัส	event	guard
a1	sendsendText_1	
b1	receivesendText_1	
c1	senddeliverText_2	
d1	receivedeliverText_2	
e1	sendreadReceipt_3	
f1	receivereadReceipt_3	
g1	sendforwardReceipt_4	
h1	receiveforwardReceipt_4	
i1	sendtypingIndicator_5_opt1	a1
j1	receivetypingIndicator_5_opt1	a1
k1	sendrelayTyping_6_opt1	a1
l1	receiverelayTyping_6_opt1	a1
m1	sendreplyText_7	
n1	receivereplyText_7	
o1	senddeliverReply_8	
p1	receivedeliverReply_8	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1

model_016 EmailVerification · pass

Guard

a1	resendCount < 3
b1	verifyAttempt < 5

```
(root)
└── a1  resendCount < 3
    └── b1  verifyAttempt < 5
```

เส้นทาง
a1 -> b1

Event

รหัส	event	guard
a1	sendregister_1	
b1	receiveregister_1	
c1	sendsendCode_2	
d1	receivesendCode_2	
e1	sendresendRequest_3	a1
f1	receiveresendRequest_3	a1
g1	sendresendCode_4	a1
h1	receiveresendCode_4	a1
i1	sendsubmitCode_5	b1
i2	checkloop_resendRequest_3_loop1	a1
j1	receivesubmitCode_5	b1
k1	sendverifyResult_6	b1
l1	receiveverifyResult_6	b1
m1	sendaccountActivated_7	
m2	checkloop_submitCode_5_loop2	b1
n1	receiveaccountActivated_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1
    ├── i1 → j1 → k1 → l1
    │   ├── m1 → n1
    │   └── m2 ↺ i1
    └── i2 ↺ e1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m2 ↺ i1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i2 ↺ e1

model_028 LoginRateLimiter · fail

Guard

a1	failCount < 5
b1	failCount mod 2 = 0

```
(root)
└── a1  failCount < 5
    └── b1  failCount mod 2 = 0
```

เส้นทาง

```
a1 -> b1
```

Event

รหัส	event	guard
a1	sendloginForm_1	
b1	receiveloginForm_1	
c1	sendsubmitLogin_2	a1
d1	receivesubmitLogin_2	a1
e1	sendverify_3	a1
f1	receiveverify_3	a1
g1	sendverifyResult_4	a1
h1	receiveverifyResult_4	a1
i1	sendloginPage_5	a1
j1	receiveloginPage_5	a1
k1	sendlockAccount_6_opt1	b1
k2	checkloop_submitLogin_2_loop1	a1
l1	receivelockAccount_6_opt1	b1
m1	sendsessionToken_7	
n1	receivesessionToken_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1
    ├── k1 → l1 → m1 → n1
    └── k2 ↺ c1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 ↺
c1
```

model_034 PrinterQueue · fail

Guard

a1	jobCount < 5
b1	retryPrint < 3

```
(root)
└── a1  jobCount < 5
    └── b1  retryPrint < 3
```

เส้นทาง

```
a1 -> b1
```

Event

รหัส	event	guard
a1	sendsubmitJob_1	
b1	receivesubmitJob_1	
c1	sendjobQueued_2	
d1	receivejobQueued_2	
e1	sendpagePrinted_3	a1
f1	receivepagePrinted_3	a1
g1	sendnextPage_4	a1
h1	receivenextPage_4	a1
i1	sendreprint_5	b1
i2	checkloop_pagePrinted_3_loop1	a1
j1	receivereprint_5	b1
k1	sendreprintDone_6	b1
l1	receivereprintDone_6	b1
m1	sendjobComplete_7	
m2	checkloop_reprint_5_loop2	b1
n1	receivejobComplete_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1
    ├── i1 → j1 → k1 → l1
    │   ├── m1 → n1
    │   └── m2 ↺ i1
    └── i2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m2 ↺ i1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i2 ↺ e1
```

model_035 SensorDataPipeline · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendpublishReading_1	
b1	receivepublishReading_1	
c1	sendackReading_2	
d1	receiveackReading_2	
e1	sendaggregateBatch_3	
f1	receiveaggregateBatch_3	
g1	sendstoreConfirm_4	
h1	receivestoreConfirm_4	
i1	sendsyncClock_5	
j1	receivesyncClock_5	
k1	sendclockSynced_6	
l1	receiveclockSynced_6	
m1	sendheartbeat_7	
n1	receiveheartbeat_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1
```

model_036 ServerAutoScaling · pass

Guard

a1	checkNo < 12
b1	cpu > 80
b2	cpu ≤ 80

```
(root)
└── a1  checkNo < 12
    ├── b1  cpu > 80
    └── b2  cpu ≤ 80
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

รหัส	event	guard
a1	sendstartMonitoring_1	
b1	receivestartMonitoring_1	
c1	sendreportLoad_2	a1
d1	receivereportLoad_2	a1
e1	sendaddVM_3	a1 b1
e2	sendkeepCapacity_6	a1 b2
f1	receiveaddVM_3	a1 b1
f2	receivekeepCapacity_6	a1 b2
g1	sendvmReady_4	a1 b1
h1	receivevmReady_4	a1 b1
i1	sendregisterVM_5	a1 b1
j1	receiveregisterVM_5	a1 b1
k1	sendstopMonitoring_7	
k2	checkloop_reportLoad_2_loop1	a1
l1	receivestopMonitoring_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1
    │   ├── k1 → l1
    │   └── k2 ↺ c1
    └── e2 → f2
        ├── k1 → l1
        └── k2 ↺ c1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 ↺
c1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> k2 ↺ c1
```

model_054 ArchiveDocumentScan · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendreserveBatch_1	
b1	receivereserveBatch_1	
c1	sendreleaseBatch_2	
d1	receivereleaseBatch_2	
e1	sendsendBatch_3	
f1	receivesendBatch_3	
g1	sendcheckBatch_4	
h1	receivecheckBatch_4	
i1	sendsendDocument_5	
j1	receivesendDocument_5	
k1	sendissueDocument_6	
l1	receiveissueDocument_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1
```

model_061 BulkSMSGateway · fail

Guard

a1	retryNo < 11
b1	retryNo mod 3 = 0

```
(root)
└── a1  retryNo < 11
    └── b1  retryNo mod 3 = 0
```

เส้นทาง

```
a1 -> b1
```

Event

รหัส	event	guard
a1	sendconfirmDelivery_1	
b1	receiveconfirmDelivery_1	
c1	sendapproveCampaign_2	
d1	receiveapproveCampaign_2	
e1	sendlogCampaign_3	a1
f1	receivelogCampaign_3	a1
g1	sendlogDelivery_4_opt1	a1 b1
h1	receivelogDelivery_4_opt1	a1 b1
i1	sendnotifyCampaign_5	a1
j1	receivenotifyCampaign_5	a1
k1	sendlogMessage_6	a1
l1	receivelogMessage_6	a1
m1	sendconfirmCampaign_7	
m2	checkloop_logCampaign_3_loop1	a1
n1	receiveconfirmCampaign_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
    ├── m1 → n1
    └── m2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m2 ↺ e1
```

model_065 CampusWifiAccess · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendcancelSession_1	
b1	receivecancelSession_1	
c1	sendvalidateToken_2	
d1	receivevalidateToken_2	
e1	sendsendToken_3	
f1	receivesendToken_3	
g1	sendissueSession_4	
h1	receiveissueSession_4	
i1	sendreserveToken_5	
j1	receivereserveToken_5	
k1	sendsendSession_6	
l1	receivesendSession_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_072 CloudBackupJob · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendupdateSnapshot_1	
b1	receiveupdateSnapshot_1	
c1	sendconfirmSnapshot_2	
d1	receiveconfirmSnapshot_2	
e1	sendcheckJob_3	
f1	receivecheckJob_3	
g1	sendrequestChunk_4	
h1	receiverequestChunk_4	
i1	sendnotifyChunk_5	
j1	receivenotifyChunk_5	
k1	sendsyncJob_6	
l1	receivesyncJob_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_073 CloudCIPipeline · pass

Guard

a1	useCache = 1
b1	optIn = 1
c1	retryNo < 11

```
(root)
└── a1  useCache = 1
    └── b1  optIn = 1
        └── c1  retryNo < 11
```

เส้นทาง

```
a1 -> b1 -> c1
```

Event

รหัส	event	guard
a1	sendreportImage_1	
b1	receivereportImage_1	
c1	sendissueImage_2	
d1	receiveissueImage_2	
e1	sendreserveBuild_3_opt1	a1
f1	receivereserveBuild_3_opt1	a1
g1	sendnotifyImage_4	
h1	receivenotifyImage_4	
i1	sendreportCommit_5	
j1	receivereportCommit_5	
k1	sendissueBuild_6_opt2	b1
l1	receiveissueBuild_6_opt2	b1
m1	sendcheckImage_7	
n1	receivecheckImage_7	
o1	sendscheduleBuild_8	
p1	receivescheduleBuild_8	
q1	sendlogCommit_9	c1
r1	receivelogCommit_9	c1
s1	sendfetchBuild_10	c1
t1	receivefetchBuild_10	c1
u1	sendconfirmCommit_11	
u2	checkloop_logCommit_9_loop1	c1
v1	receiveconfirmCommit_11	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1 →
      q1 → r1 → s1 → t1
    ├── u1 → v1
    └── u2 ↺ q1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 ->
v1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u2 ↺ q1
```

model_117 JobApplicationPortal · pass

Guard

a1	stage = ACTIVE
a2	stage = IDLE

```
(root)
├── a1  stage = ACTIVE
└── a2  stage = IDLE
```

เส้นทาง

a1
a2

Event

รหัส	event	guard
a1	sendassignInterview_1	
b1	receiveassignInterview_1	
c1	sendcheckResume_2	
d1	receivecheckResume_2	
e1	sendvalidateInterview_3	
f1	receivevalidateInterview_3	
g1	sendreleaseInterview_4	
h1	receivereleaseInterview_4	
i1	sendackOffer_5	
j1	receiveackOffer_5	
k1	sendrequestOffer_6	
l1	receiverequestOffer_6	
m1	sendrequestInterview_7	
n1	receiverequestInterview_7	
o1	sendupdateOffer_8	a1
o2	sendcancelResume_9	a2
p1	receiveupdateOffer_8	a1
p2	receivecancelResume_9	a2
q1	sendassignResume_10	
r1	receiveassignResume_10	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
    ├── o1 → p1 → q1 → r1
    └── o2 → p2 → q1 → r1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o2 -> p2 -> q1 -> r1

model_134 OnlineQuizGame · pass

Guard

a1	attemptNo < 10
b1	roundNo < 6

```
(root)
└── a1  attemptNo < 10
    └── b1  roundNo < 6
```

เส้นทาง

```
a1 -> b1
```

Event

รหัส	event	guard
a1	sendcheckRound_1	
b1	receivecheckRound_1	
c1	sendscheduleAnswer_2	
d1	receivescheduleAnswer_2	
e1	sendackRound_3	a1
f1	receiveackRound_3	a1
g1	sendfetchAnswer_4	
g2	checkloop_ackRound_3_loop1	a1
h1	receivefetchAnswer_4	
i1	sendqueryScore_5	
j1	receivequeryScore_5	
k1	sendissueAnswer_6	b1
l1	receiveissueAnswer_6	b1
m1	sendapproveAnswer_7	b1
n1	receiveapproveAnswer_7	b1
o1	sendassignScore_8	b1
p1	receiveassignScore_8	b1
q1	sendcancelScore_9	b1
r1	receivecancelScore_9	b1
s1	sendrequestScore_10	b1
t1	receiverequestScore_10	b1
u1	sendreportScore_11	b1
v1	receivereportScore_11	b1
w1	sendreserveRound_12	
w2	checkloop_issueAnswer_6_loop2	b1
x1	receivereserveRound_12	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1 → m1 → n1 →
    │     o1 → p1 → q1 → r1 → s1 → t1 → u1 → v1
    │   ├── w1 → x1
    │   └── w2 ↺ k1
    └── g2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 ->
v1 -> w1 -> x1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 ->
v1 -> w2 ↺ k1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 ↺ e1
```

model_136 ParentalControlRouter · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendreleaseRule_1	
b1	receivereleaseRule_1	
c1	sendreserveReport_2	
d1	receivereserveReport_2	
e1	sendcancelRule_3	
f1	receivecancelRule_3	
g1	sendupdateRule_4	
h1	receiveupdateRule_4	
i1	sendnotifyRule_5	
j1	receivenotifyRule_5	
k1	sendconfirmBlock_6	
l1	receiveconfirmBlock_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1
```

model_145 PodcastPublishing · fail

Guard

a1	retryNo < 3
b1	retryNo mod 3 = 0
b2	retryNo mod 3 ≠ 0

```
(root)
└── a1  retryNo < 3
    ├── b1  retryNo mod 3 = 0
    └── b2  retryNo mod 3 ≠ 0
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

รหัส	event	guard
a1	sendvalidateFeed_1	
b1	receivevalidateFeed_1	
c1	sendassignAudio_2	
d1	receiveassignAudio_2	
e1	sendsendEpisode_3	a1
f1	receivesendEpisode_3	a1
g1	sendreserveEpisode_4	a1 b1
g2	sendcheckFeed_6	a1 b2
h1	receivereserveEpisode_4	a1 b1
h2	receivecheckFeed_6	a1 b2
i1	sendissueAudio_5	a1 b1
j1	receiveissueAudio_5	a1 b1
k1	sendsendAudio_7	a1
l1	receivesendAudio_7	a1
m1	sendlogAudio_8	
m2	checkloop_sendEpisode_3_loop1	a1
n1	receivelogAudio_8	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1
    │   ├── m1 → n1
    │   └── m2 ↺ e1
    └── g2 → h2 → k1 → l1
        ├── m1 → n1
        └── m2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m2 ↺ e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1 -> m1
-> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1 -> m2 ↺
e1
```

model_147 PrintFleetMonitor · fail

Guard

a1	isVip = 1
a2	isVip = 0

```
(root)
├── a1  isVip = 1
└── a2  isVip = 0
```

เส้นทาง

a1
a2

Event

รหัส	event	guard
a1	sendvalidateJob_1	
b1	receivevalidateJob_1	
c1	sendreserveAlert_2	
d1	receivereserveAlert_2	
e1	sendqueryToner_3	a1
e2	sendconfirmJob_4	a2
f1	receivequeryToner_3	a1
f2	receiveconfirmJob_4	a2
g1	sendfetchToner_5	
h1	receivefetchToner_5	
i1	sendscheduleToner_6	
j1	receivescheduleToner_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1
    └── e2 → f2 → g1 → h1 → i1 → j1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g1 -> h1 -> i1 -> j1

model_149 PublicWifiHotspot · fail

Guard

a1	optIn = 1
b1	load < 10
b2	load ≥ 10

```
(root)
└── a1  optIn = 1
    ├── b1  load < 10
    └── b2  load ≥ 10
```

เส้นทาง

a1 -> b1
a1 -> b2

Event

รหัส	event	guard
a1	sendscheduleQuota_1	
b1	receivescheduleQuota_1	
c1	sendscheduleVoucher_2	
d1	receivescheduleVoucher_2	
e1	sendvalidateQuota_3	
f1	receivevalidateQuota_3	
g1	sendupdateSession_4	
h1	receiveupdateSession_4	
i1	sendvalidateSession_5	
j1	receivevalidateSession_5	
k1	sendreserveVoucher_6	
l1	receivereserveVoucher_6	
m1	sendupdateQuota_7_opt1	a1
n1	receiveupdateQuota_7_opt1	a1
o1	sendapproveQuota_8_opt1	a1 b1
o2	sendsendSession_10_opt1	a1 b2
p1	receiveapproveQuota_8_opt1	a1 b1
p2	receivesendSession_10_opt1	a1 b2
q1	sendqueryVoucher_9_opt1	a1 b1
r1	receivequeryVoucher_9_opt1	a1 b1
s1	sendissueSession_11_opt1	a1
t1	receiveissueSession_11_opt1	a1
u1	sendissueQuota_12	
v1	receiveissueQuota_12	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
    ├── o1 → p1 → q1 → r1 → s1 → t1 → u1 → v1
    └── o2 → p2 → s1 → t1 → u1 → v1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o2 -> p2 -> s1 -> t1 -> u1 -> v1

model_159 SatelliteGroundStation · fail

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendupdateTelemetry_1	
b1	receiveupdateTelemetry_1	
c1	sendlogCommand_2	
d1	receivelogCommand_2	
e1	sendrequestCommand_3	
f1	receiverequestCommand_3	
g1	sendassignTelemetry_4	
h1	receiveassignTelemetry_4	
i1	sendcheckCommand_5	
j1	receivecheckCommand_5	
k1	sendreleasePass_6	
l1	receivereleasePass_6	
m1	sendreportPass_7	
n1	receivereportPass_7	
o1	sendupdateCommand_8	
p1	receiveupdateCommand_8	
q1	sendschedulePass_9	
r1	receiveschedulePass_9	
s1	sendsendTelemetry_10	
t1	receivesendTelemetry_10	
u1	sendscheduleTelemetry_11	
v1	receivescheduleTelemetry_11	
w1	sendvalidatePass_12	
x1	receivevalidatePass_12	
y1	sendconfirmCommand_13	
z1	receiveconfirmCommand_13	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1 →
      q1 → r1 → s1 → t1 → u1 → v1 → w1 → x1 →
      y1 → z1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1 -> w1 -> x1 -> y1 -> z1

model_194 VideoStreaming · fail

Guard

a1	stepNo < 7
b1	stepNo mod 2 = 0
b2	stepNo mod 2 ≠ 0

```
(root)
└── a1  stepNo < 7
    ├── b1  stepNo mod 2 = 0
    └── b2  stepNo mod 2 ≠ 0
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

รหัส	event	guard
a1	sendreserveLicense_1	
b1	receivereserveLicense_1	
c1	sendreportChunk_2	
d1	receivereportChunk_2	
e1	sendreportStream_3	a1
f1	receivereportStream_3	a1
g1	sendvalidateStream_4	a1 b1
g2	sendsendChunk_5	a1 b2
h1	receivevalidateStream_4	a1 b1
h2	receivesendChunk_5	a1 b2
i1	sendackChunk_7	a1
j1	receiveackChunk_7	a1
k1	sendscheduleLicense_8	
k2	checkloop_reportStream_3_loop1	a1
l1	receivescheduleLicense_8	
m1	sendfetchStream_9	
n1	receivefetchStream_9	
o1	sendscheduleStream_10	
p1	receivescheduleStream_10	
q1	sendupdateStream_11	
r1	receiveupdateStream_11	
s1	sendreleaseChunk_12	
t1	receivereleaseChunk_12	
u1	sendrequestStream_13	
v1	receiverequestStream_13	
w1	sendapproveStream_6	a1 b2
x1	receiveapproveStream_6	a1 b2

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1
    │   ├── k1 → l1 → m1 → n1 → o1 → p1 → q1 → r1 →
    │   │     s1 → t1 → u1 → v1
    │   └── k2 ↺ e1
    └── g2 → h2 → w1 → x1 → i1 → j1
        ├── k1 → l1 → m1 → n1 → o1 → p1 → q1 → r1 →
        │     s1 → t1 → u1 → v1
        └── k2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 ->
v1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 ↺
e1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> w1 -> x1 -> i1
-> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 ->
t1 -> u1 -> v1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> w1 -> x1 -> i1
-> j1 -> k2 ↺ e1
```

เกษตรและสิ่งแวดล้อม · Agriculture & Environment (15)

model_038 SmartGreenhouse · pass

Guard

a1	cycleNo < 24
b1	humidity < 40
b2	humidity ≥ 40

```
(root)
└── a1  cycleNo < 24
    ├── b1  humidity < 40
    └── b2  humidity ≥ 40
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

รหัส	event	guard
a1	sendstartCycle_1	
b1	receivestartCycle_1	
c1	sendreadClimate_2	a1
d1	receivereadClimate_2	a1
e1	sendclimateData_3	a1
f1	receiveclimateData_3	a1
g1	sendstartSprinkler_4	a1 b1
g2	sendstartFan_6	a1 b2
h1	receivestartSprinkler_4	a1 b1
h2	receivestartFan_6	a1 b2
i1	sendsprinklerOn_5	a1 b1
j1	receivesprinklerOn_5	a1 b1
k1	sendendCycle_7	
k2	checkloop_readClimate_2_loop1	a1
l1	receiveendCycle_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1
    │   ├── k1 → l1
    │   └── k2 ↺ c1
    └── g2 → h2
        ├── k1 → l1
        └── k2 ↺ c1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 ↺
c1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> k2 ↺ c1
```

model_051 AirQualityMonitor · pass

Guard

a1	useCache = 1
b1	online = 1
b2	online = 0

```
(root)
└── a1  useCache = 1
    ├── b1  online = 1
    └── b2  online = 0
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

รหัส	event	guard
a1	sendcancelFan_1	
b1	receivecancelFan_1	
c1	sendreleaseAlert_2	
d1	receivereleaseAlert_2	
e1	sendsendAlert_3_opt1	a1
f1	receivesendAlert_3_opt1	a1
g1	sendfetchFan_4	
h1	receivefetchFan_4	
i1	sendlogFan_5	
j1	receivelogFan_5	
k1	sendqueryAlert_6	b1
k2	sendcancelAlert_13	b2
l1	receivequeryAlert_6	b1
l2	receivecancelAlert_13	b2
m1	sendnotifyAlert_7	b1
n1	receivenotifyAlert_7	b1
o1	sendreservePm_8	b1
p1	receivereservePm_8	b1
q1	sendapproveFan_9	b1
r1	receiveapproveFan_9	b1
s1	sendreportFan_10	b1
t1	receivereportFan_10	b1
u1	sendrequestPm_11	b1
v1	receiverequestPm_11	b1
w1	sendfetchPm_12	b1
x1	receivefetchPm_12	b1
y1	sendvalidateFan_14	
z1	receivevalidateFan_14	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1
    ├── k1 → l1 → m1 → n1 → o1 → p1 → q1 → r1 →
    │     s1 → t1 → u1 → v1 → w1 → x1 → y1 → z1
    └── k2 → l2 → y1 → z1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 ->
v1 -> w1 -> x1 -> y1 -> z1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2
-> l2 -> y1 -> z1
```

model_070 CenterPivotIrrigation · fail

Guard

a1	cycleNo < 9
b1	cycleNo mod 3 = 0

```
(root)
└── a1  cycleNo < 9
    └── b1  cycleNo mod 3 = 0
```

เส้นทาง

```
a1 -> b1
```

Event

รหัส	event	guard
a1	sendcancelWater_1	
b1	receivecancelWater_1	
c1	sendvalidateWater_2	
d1	receivevalidateWater_2	
e1	sendreserveWater_3	a1
f1	receivereserveWater_3	a1
g1	sendreleaseSector_4_opt1	a1 b1
h1	receivereleaseSector_4_opt1	a1 b1
i1	sendupdateWater_5_opt1	a1 b1
j1	receiveupdateWater_5_opt1	a1 b1
k1	sendcancelSector_6	a1
l1	receivecancelSector_6	a1
m1	sendscheduleWater_7	a1
n1	receivescheduleWater_7	a1
o1	sendqueryRain_8	
o2	checkloop_reserveWater_3_loop1	a1
p1	receivequeryRain_8	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
    ├── o1 → p1
    └── o2 ↺ e1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o2 ↺ e1
```

model_080 CropDroneSpraying · fail

Guard

a1	t_obs ≤ 11
a2	t_obs > 11

```
(root)
├── a1  t_obs ≤ 11
└── a2  t_obs > 11
```

เส้นทาง

```
a1
a2
```

Event

รหัส	event	guard
a1	sendapproveArea_1	
b1	receiveapproveArea_1	
c1	sendcheckMission_2	
d1	receivecheckMission_2	
e1	sendsyncSpray_3	a1
e2	sendassignSpray_4	a2
f1	receivesyncSpray_3	a1
f2	receiveassignSpray_4	a2
g1	sendscheduleSpray_5	
h1	receivescheduleSpray_5	
i1	sendassignMission_6	
j1	receiveassignMission_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1
    └── e2 → f2 → g1 → h1 → i1 → j1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g1 -> h1 -> i1 -> j1
```

model_092 FarmMilkingRobot · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendreserveYield_1	
b1	receivereserveYield_1	
c1	sendlogYield_2	
d1	receivelogYield_2	
e1	sendreportYield_3	
f1	receivereportYield_3	
g1	sendreserveMilk_4	
h1	receivereserveMilk_4	
i1	sendqueryHealth_5	
j1	receivequeryHealth_5	
k1	sendissueMilk_6	
l1	receiveissueMilk_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1

model_098 ForestFireSensor · fail

Guard

a1	grade = MANUAL
a2	grade = AUTO
b1	t_obs ≤ 3
c1	retryNo < 11
b2	t_obs > 3

```
(root)
├── a1  grade = MANUAL
│   ├── b1  t_obs ≤ 3
│   │   └── c1  retryNo < 11
│   └── b2  t_obs > 3
└── a2  grade = AUTO
    ├── b1  t_obs ≤ 3
    │   └── c1  retryNo < 11
    └── b2  t_obs > 3
```

เส้นทาง

```
a1 -> b1 -> c1
a1 -> b2
a2 -> b1 -> c1
a2 -> b2
```

Event

รหัส	event	guard
a1	sendreleaseAlert_1	
b1	receivereleaseAlert_1	
c1	sendschedulePatrol_2	
d1	receiveschedulePatrol_2	
e1	sendsyncPatrol_3	a1
e2	sendcancelAlert_4	a2
f1	receivesyncPatrol_3	a1
f2	receivecancelAlert_4	a2
g1	sendissueSmoke_5	
h1	receiveissueSmoke_5	
i1	sendnotifyAlert_6	
j1	receivenotifyAlert_6	
k1	sendreserveAlert_7	b1
k2	sendscheduleSmoke_11	b2
l1	receivereserveAlert_7	b1
l2	receivescheduleSmoke_11	b2
m1	sendvalidateSmoke_8	b1 c1
n1	receivevalidateSmoke_8	b1 c1
o1	sendcheckAlert_9	b1 c1
p1	receivecheckAlert_9	b1 c1
q1	sendassignAlert_10	b1
q2	checkloop_validateSmoke_8_loop1	b1 c1
r1	receiveassignAlert_10	b1
s1	sendqueryPatrol_12	
t1	receivequeryPatrol_12	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1
    │   ├── k1 → l1 → m1 → n1 → o1 → p1
    │   │   ├── q1 → r1 → s1 → t1
    │   │   └── q2 ↺ m1
    │   └── k2 → l2 → s1 → t1
    └── e2 → f2 → g1 → h1 → i1 → j1
        ├── k1 → l1 → m1 → n1 → o1 → p1
        │   ├── q1 → r1 → s1 → t1
        │   └── q2 ↺ m1
        └── k2 → l2 → s1 → t1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q2 ↺ m1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2
-> l2 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q2 ↺ m1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g1 -> h1 -> i1 -> j1 -> k2
-> l2 -> s1 -> t1
```

model_103 GreenhouseLighting · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendapproveEnergy_1	
b1	receiveapproveEnergy_1	
c1	sendackLight_2	
d1	receiveackLight_2	
e1	sendcheckLux_3	
f1	receivecheckLux_3	
g1	sendreserveLux_4	
h1	receivereserveLux_4	
i1	sendreleaseLight_5	
j1	receivereleaseLight_5	
k1	sendnotifyEnergy_6	
l1	receivenotifyEnergy_6	
m1	sendqueryEnergy_7	
n1	receivequeryEnergy_7	
o1	sendrequestLight_8	
p1	receiverequestLight_8	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1 → o1 → p1
```

เส้นทาง

a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1

model_116 IrrigationZoneControl · pass

Guard

a1	attemptNo < 12

```
(root)
└── a1  attemptNo < 12
```

เส้นทาง
a1

Event

รหัส	event	guard
a1	sendqueryZone_1	
b1	receivequeryZone_1	
c1	sendackSchedule_2	
d1	receiveackSchedule_2	
e1	sendrequestZone_3	a1
f1	receiverequestZone_3	a1
g1	sendcheckMoisture_4	a1
h1	receivecheckMoisture_4	a1
i1	sendqueryMoisture_5	a1
j1	receivequeryMoisture_5	a1
k1	sendcancelZone_6	
k2	checkloop_requestZone_3_loop1	a1
l1	receivecancelZone_6	
m1	sendrequestSchedule_7	
n1	receiverequestSchedule_7	
o1	sendreleaseMoisture_8	
p1	receivereleaseMoisture_8	
q1	sendreserveMoisture_9	
r1	receivereserveMoisture_9	
s1	sendscheduleSchedule_10	
t1	receivescheduleSchedule_10	
u1	sendsendSchedule_11	
v1	receivesendSchedule_11	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1
    ├── k1 → l1 → m1 → n1 → o1 → p1 → q1 → r1 →
    │     s1 → t1 → u1 → v1
    └── k2 ↺ e1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1 -> u1 -> v1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2 ↺ e1

model_121 LivestockFeeder · fail

Guard

a1	approved = 1
b1	express = 1
a2	approved = 0

```
(root)
├── a1  approved = 1
│   └── b1  express = 1
└── a2  approved = 0
```

เส้นทาง

```
a1 -> b1
a2
```

Event

รหัส	event	guard
a1	sendreportRation_1	
b1	receivereportRation_1	
c1	sendsyncFeed_2	
d1	receivesyncFeed_2	
e1	sendackFeed_3	a1
e2	sendcancelRation_7	a2
f1	receiveackFeed_3	a1
f2	receivecancelRation_7	a2
g1	sendupdateRation_4_opt1	a1 b1
h1	receiveupdateRation_4_opt1	a1 b1
i1	sendqueryRation_5_opt1	a1 b1
j1	receivequeryRation_5_opt1	a1 b1
k1	sendconfirmFeed_6	a1
l1	receiveconfirmFeed_6	a1
m1	sendconfirmRation_8	
n1	receiveconfirmRation_8	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1 → k1 → l1 →
    │     m1 → n1
    └── e2 → f2 → m1 → n1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> m1 -> n1
```

model_151 RecyclingKiosk · fail

Guard

a1	express = 1
b1	roundNo < 6
c1	urgent = 1
c2	urgent = 0

```
(root)
└── a1  express = 1
    └── b1  roundNo < 6
        ├── c1  urgent = 1
        └── c2  urgent = 0
```

เส้นทาง

```
a1 -> b1 -> c1
a1 -> b1 -> c2
```

Event

รหัส	event	guard
a1	sendconfirmCredit_1	
b1	receiveconfirmCredit_1	
c1	sendfetchVoucher_2	
d1	receivefetchVoucher_2	
e1	sendsyncCredit_3_opt1	a1
f1	receivesyncCredit_3_opt1	a1
g1	sendnotifyCredit_4_opt1	a1 b1
h1	receivenotifyCredit_4_opt1	a1 b1
i1	sendconfirmBottle_5_opt1	a1
i2	checkloop_notifyCredit_4_loop1	a1 b1
j1	receiveconfirmBottle_5_opt1	a1
k1	sendconfirmVoucher_6	
l1	receiveconfirmVoucher_6	
m1	sendackVoucher_7	
n1	receiveackVoucher_7	
o1	sendassignBottle_8	c1
o2	sendscheduleVoucher_9	c2
p1	receiveassignBottle_8	c1
p2	receivescheduleVoucher_9	c2
q1	sendqueryCredit_10	
r1	receivequeryCredit_10	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1
    ├── i1 → j1 → k1 → l1 → m1 → n1
    │   ├── o1 → p1 → q1 → r1
    │   └── o2 → p2 → q1 → r1
    └── i2 ↺ g1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o2 -> p2 -> q1 -> r1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i2 ↺ g1
```

model_166 SmartBeehive · fail

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendupdateHoney_1	
b1	receiveupdateHoney_1	
c1	sendreleaseWeight_2	
d1	receivereleaseWeight_2	
e1	sendlogWeight_3	
f1	receivelogWeight_3	
g1	sendissueWeight_4	
h1	receiveissueWeight_4	
i1	sendcheckWeight_5	
j1	receivecheckWeight_5	
k1	sendscheduleSwarm_6	
l1	receivescheduleSwarm_6	
m1	sendrequestHoney_7	
n1	receiverequestHoney_7	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1 → m1 → n1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1
```

model_174 SmartRainGauge · pass

Guard

a1	useCache = 1
b1	score < 50
b2	score ≥ 50

```
(root)
└── a1  useCache = 1
    ├── b1  score < 50
    └── b2  score ≥ 50
```

เส้นทาง

```
a1 -> b1
a1 -> b2
```

Event

รหัส	event	guard
a1	sendqueryWarning_1	
b1	receivequeryWarning_1	
c1	sendconfirmRainfall_2	
d1	receiveconfirmRainfall_2	
e1	sendreserveRainfall_3_opt1	a1
f1	receivereserveRainfall_3_opt1	a1
g1	sendcheckLevel_4_opt1	a1 b1
g2	sendscheduleLevel_9_opt1	a1 b2
h1	receivecheckLevel_4_opt1	a1 b1
h2	receivescheduleLevel_9_opt1	a1 b2
i1	sendapproveRainfall_5_opt1	a1 b1
j1	receiveapproveRainfall_5_opt1	a1 b1
k1	sendreserveLevel_6_opt1	a1 b1
l1	receivereserveLevel_6_opt1	a1 b1
m1	sendvalidateLevel_7_opt1	a1 b1
n1	receivevalidateLevel_7_opt1	a1 b1
o1	sendupdateWarning_8_opt1	a1 b1
p1	receiveupdateWarning_8_opt1	a1 b1
q1	sendsyncRainfall_10_opt1	a1
r1	receivesyncRainfall_10_opt1	a1
s1	sendcancelRainfall_11	
t1	receivecancelRainfall_11	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1 → m1 → n1 →
    │     o1 → p1 → q1 → r1 → s1 → t1
    └── g2 → h2 → q1 → r1 → s1 → t1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1 -> o1 -> p1 -> q1 -> r1 -> s1 -> t1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 -> h2 -> q1 -> r1 -> s1
-> t1
```

model_177 SmartTrashCompactor · pass

Guard

ไม่มีเงื่อนไขจากแผนภาพ

Event

รหัส	event	guard
a1	sendcheckRoute_1	
b1	receivecheckRoute_1	
c1	sendassignLevel_2	
d1	receiveassignLevel_2	
e1	sendscheduleRoute_3	
f1	receivescheduleRoute_3	
g1	sendscheduleLevel_4	
h1	receivescheduleLevel_4	
i1	sendnotifyLevel_5	
j1	receivenotifyLevel_5	
k1	sendackService_6	
l1	receiveackService_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1 → g1 → h1 →
      i1 → j1 → k1 → l1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1
```

model_179 SmartWasteBin · pass

Guard

a1	hasKey = 1
a2	hasKey = 0
b1	phase = GOLD
b2	phase = SILVER
b3	phase = BRONZE

```
(root)
├── a1  hasKey = 1
│   ├── b1  phase = GOLD
│   ├── b2  phase = SILVER
│   └── b3  phase = BRONZE
└── a2  hasKey = 0
    ├── b1  phase = GOLD
    ├── b2  phase = SILVER
    └── b3  phase = BRONZE
```

เส้นทาง

```
a1 -> b1
a1 -> b2
a1 -> b3
a2 -> b1
a2 -> b2
a2 -> b3
```

Event

รหัส	event	guard
a1	sendreserveRoute_1	
b1	receivereserveRoute_1	
c1	sendsendRoute_2	
d1	receivesendRoute_2	
e1	sendassignRoute_3	a1
e2	sendreservePickup_4	a2
f1	receiveassignRoute_3	a1
f2	receivereservePickup_4	a2
g1	sendscheduleFill_5	
h1	receivescheduleFill_5	
i1	sendupdatePickup_6	
j1	receiveupdatePickup_6	
k1	sendlogPickup_7	b1
k2	sendissueRoute_8	b2
k3	sendsyncPickup_9	b3
l1	receivelogPickup_7	b1
l2	receiveissueRoute_8	b2
l3	receivesyncPickup_9	b3
m1	sendreleaseRoute_10	
n1	receivereleaseRoute_10	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1
    ├── e1 → f1 → g1 → h1 → i1 → j1
    │   ├── k1 → l1 → m1 → n1
    │   ├── k2 → l2 → m1 → n1
    │   └── k3 → l3 → m1 → n1
    └── e2 → f2 → g1 → h1 → i1 → j1
        ├── k1 → l1 → m1 → n1
        ├── k2 → l2 → m1 → n1
        └── k3 → l3 → m1 → n1
```

เส้นทาง

```
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k2
-> l2 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k3
-> l3 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g1 -> h1 -> i1 -> j1 -> k1
-> l1 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g1 -> h1 -> i1 -> j1 -> k2
-> l2 -> m1 -> n1
a1 -> b1 -> c1 -> d1 -> e2 -> f2 -> g1 -> h1 -> i1 -> j1 -> k3
-> l3 -> m1 -> n1
```

model_180 SoilTestingLab · pass

Guard

a1	roundNo < 12

```
(root)
└── a1  roundNo < 12
```

เส้นทาง
a1

Event

รหัส	event	guard
a1	sendassignReport_1	
b1	receiveassignReport_1	
c1	sendsendNutrient_2	
d1	receivesendNutrient_2	
e1	sendreleaseReport_3	a1
f1	receivereleaseReport_3	a1
g1	sendcancelNutrient_4	
g2	checkloop_releaseReport_3_loop1	a1
h1	receivecancelNutrient_4	
i1	sendquerySample_5	
j1	receivequerySample_5	
k1	sendnotifyReport_6	
l1	receivenotifyReport_6	

```
(root) INITIALISATION
└── a1 → b1 → c1 → d1 → e1 → f1
    ├── g1 → h1 → i1 → j1 → k1 → l1
    └── g2 ↺ e1
```

เส้นทาง
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g1 -> h1 -> i1 -> j1 -> k1 -> l1
a1 -> b1 -> c1 -> d1 -> e1 -> f1 -> g2 ↺ e1